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Stratton et al.

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(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SYSTEM FOR PROCESSING BONE STOCK INCLUDING A BONE CLEANING HEAD, A BONE MILLING HEAD AND BASE THAT POWERS BOTH THE CLEANING HEAD AND THE MILLING HEAD**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 528 days.

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(22) Filed: **Sep. 26, 2022**

(65) **Prior Publication Data**

US 2023/0010256 A1 Jan. 12, 2023

Related U.S. Application Data

(60) Continuation of application No. 16/777,277, filed on Jan. 30, 2020, now Pat. No. 11,452,621, which is a
(Continued)

(51) **Int. Cl.**
A61F 2/46 (2006.01)

(52) **U.S. Cl.**
CPC **A61F 2/4644** (2013.01); **A61F 2002/4645** (2013.01); **A61F 2002/4646** (2013.01)

(58) **Field of Classification Search**
CPC A61F 2/4644; A61F 2002/4645; A61F 2002/4646

See application file for complete search history.

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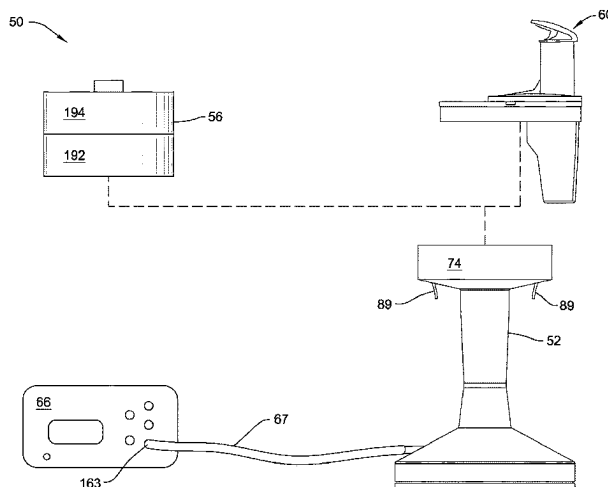
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(74) *Attorney, Agent, or Firm* — Howard & Howard
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(57) **ABSTRACT**

A system for cleaning bone that includes a base unit with a motor, a cleaning head with a cleaning element and a mill head with a mill element. Both the cleaning head and the mill head are designed to be coupled to the base unit. Both the cleaning element and mill element have features that facilitate their coupling to the motor. When the cleaning head is attached to the base unit, a motor in the base unit rotates the cleaning element to remove soft tissue from the bone so as to clean the bone. The mill element is placed on the base unit and the cleaned bone placed in the mill head. The actuation of the base unit motor results in the actuations of the mill element. The actuation of the mill element converts the cleaned bone into bone chips.

20 Claims, 47 Drawing Sheets



Related U.S. Application Data

continuation of application No. 15/915,562, filed on Mar. 8, 2018, now Pat. No. 10,588,757, which is a continuation of application No. 15/174,281, filed on Jun. 6, 2016, now Pat. No. 10,045,863, which is a division of application No. 14/157,975, filed on Jan. 17, 2014, now Pat. No. 9,370,436, which is a division of application No. 13/462,120, filed on May 2, 2012, now Pat. No. 8,672,942, which is a continuation of application No. PCT/US2010/055646, filed on Nov. 5, 2010.

- (60) Provisional application No. 61/258,667, filed on Nov. 6, 2009.

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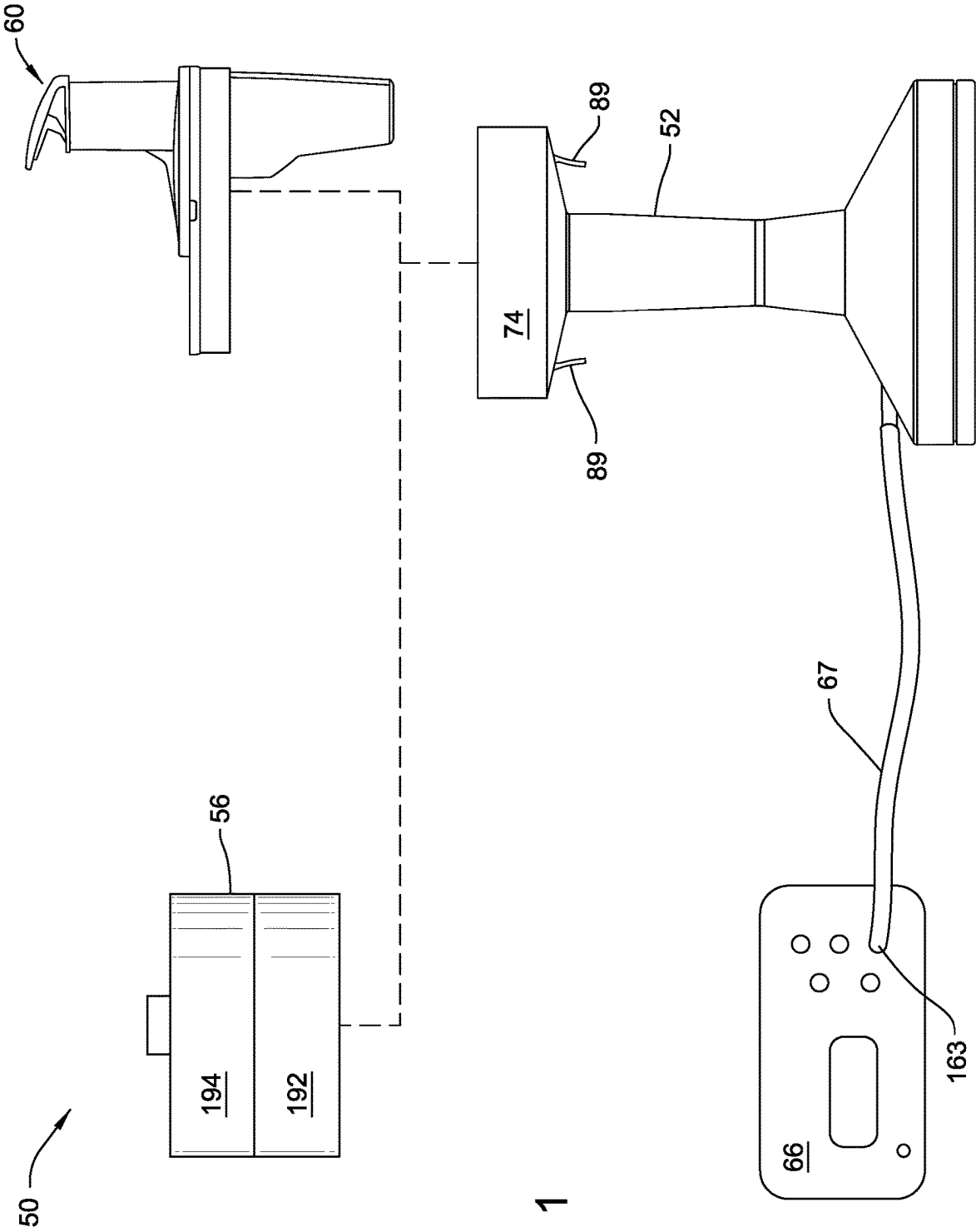


FIG. 1

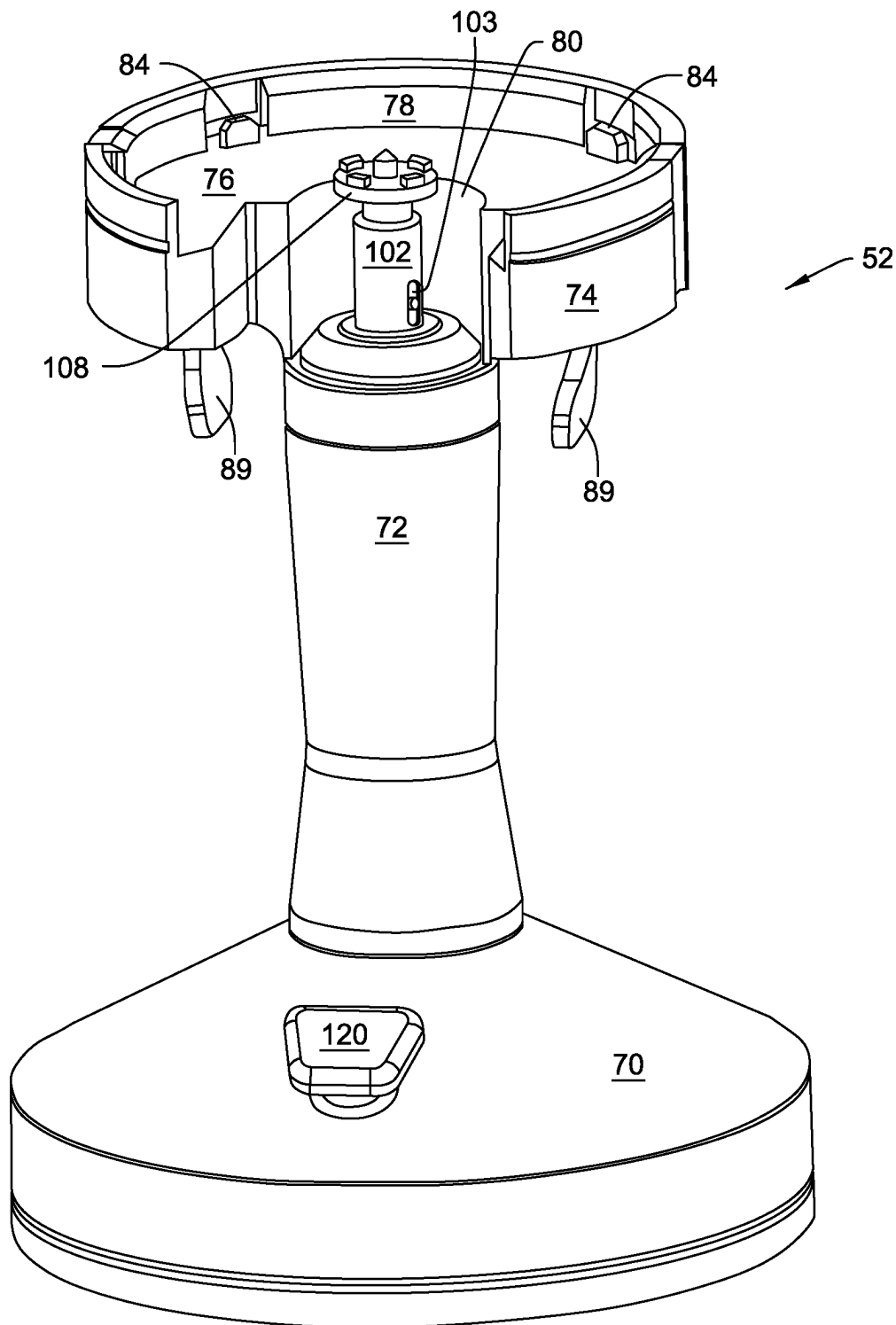


FIG. 2

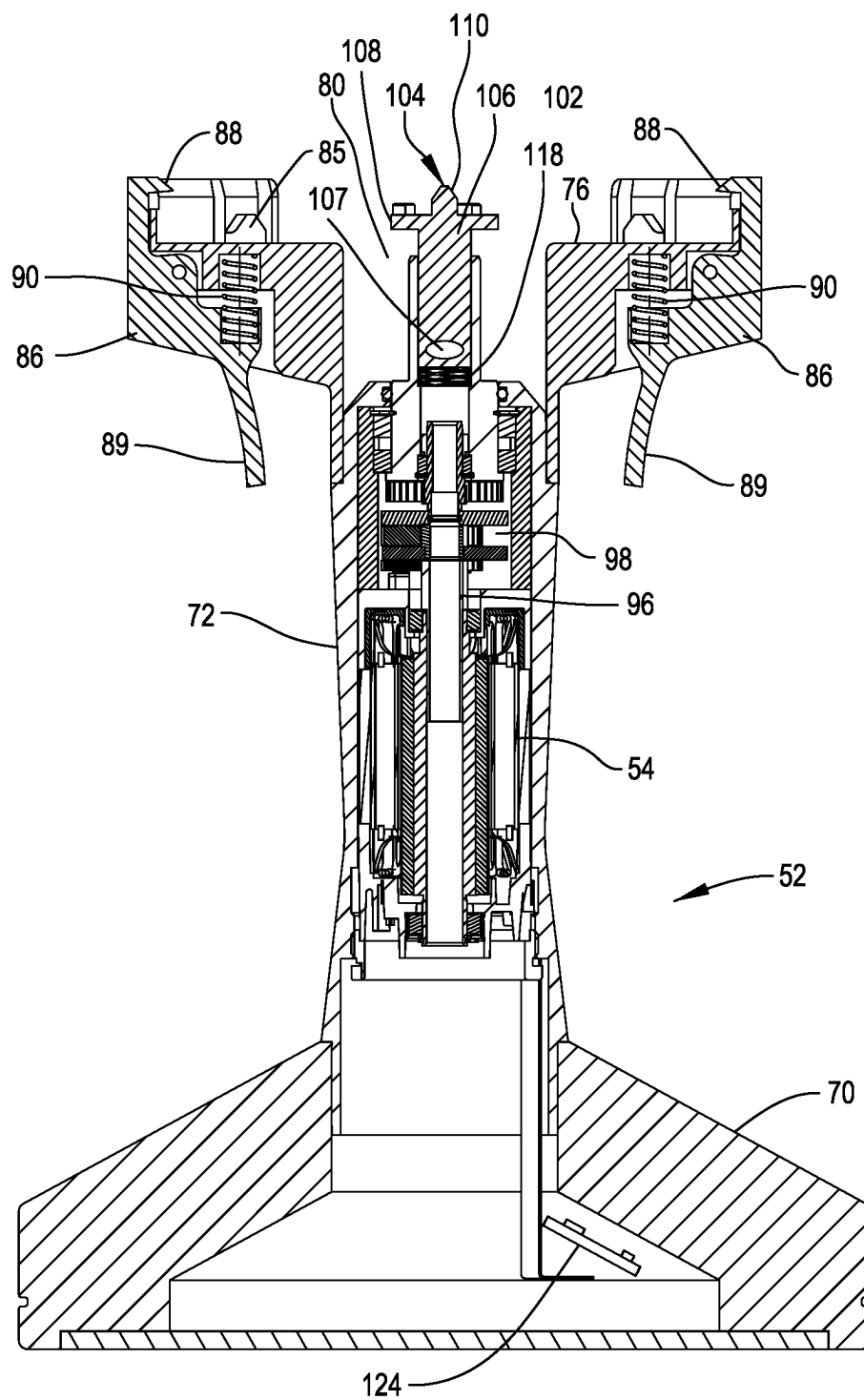


FIG. 3

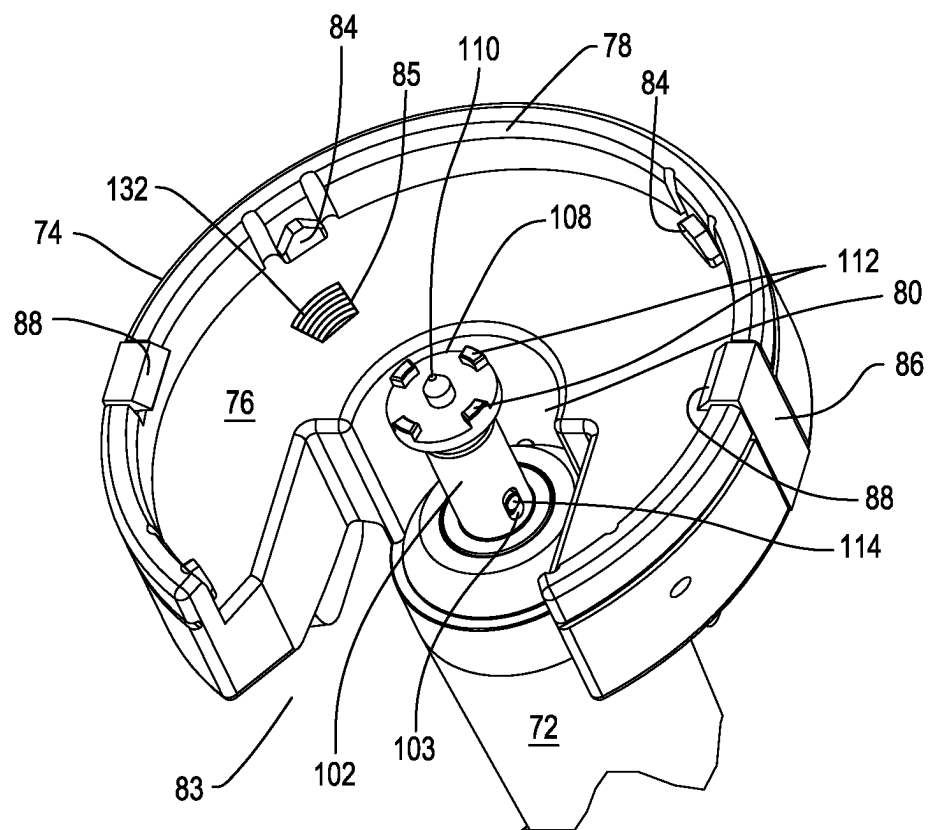


FIG. 4

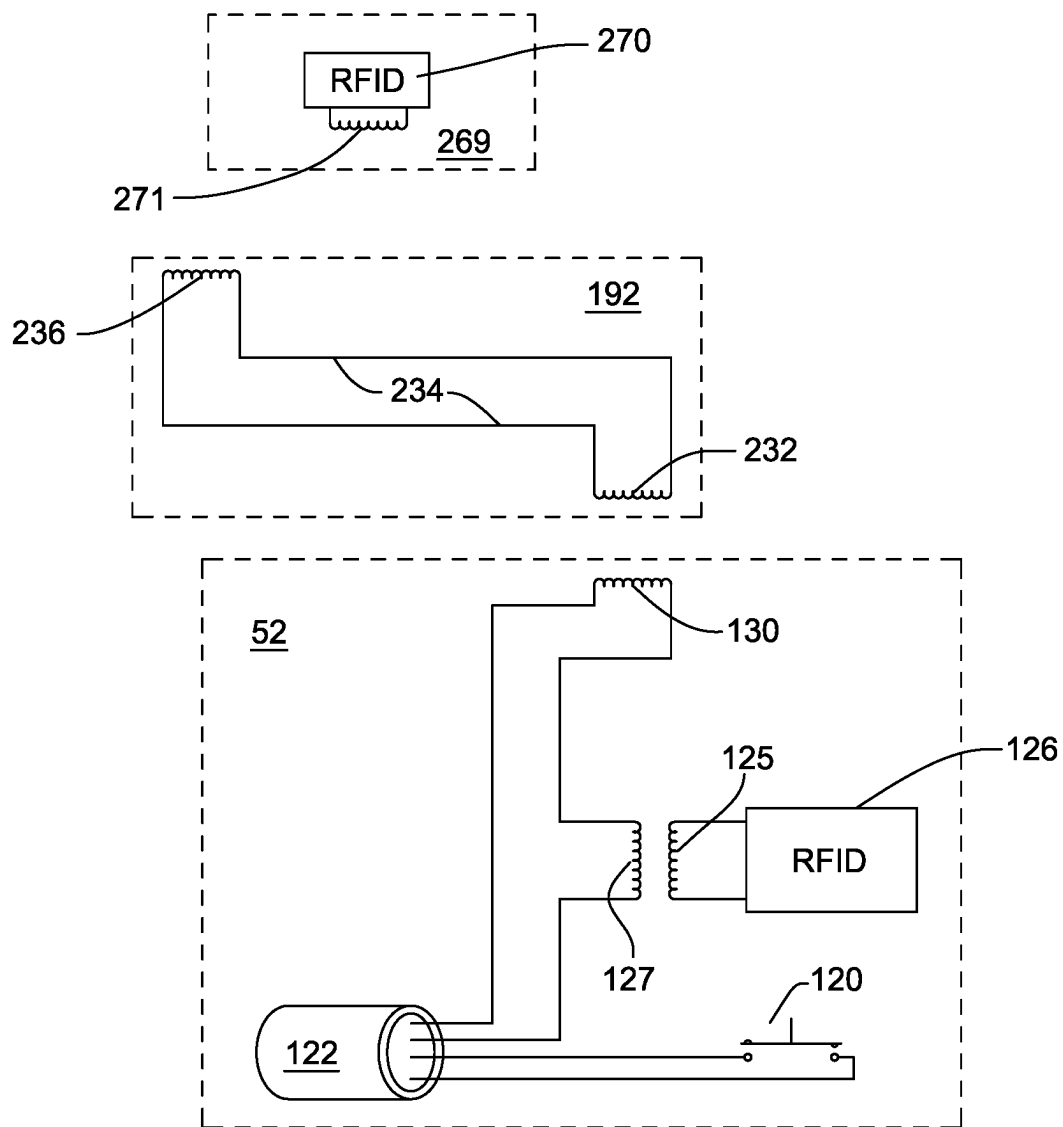


FIG. 5

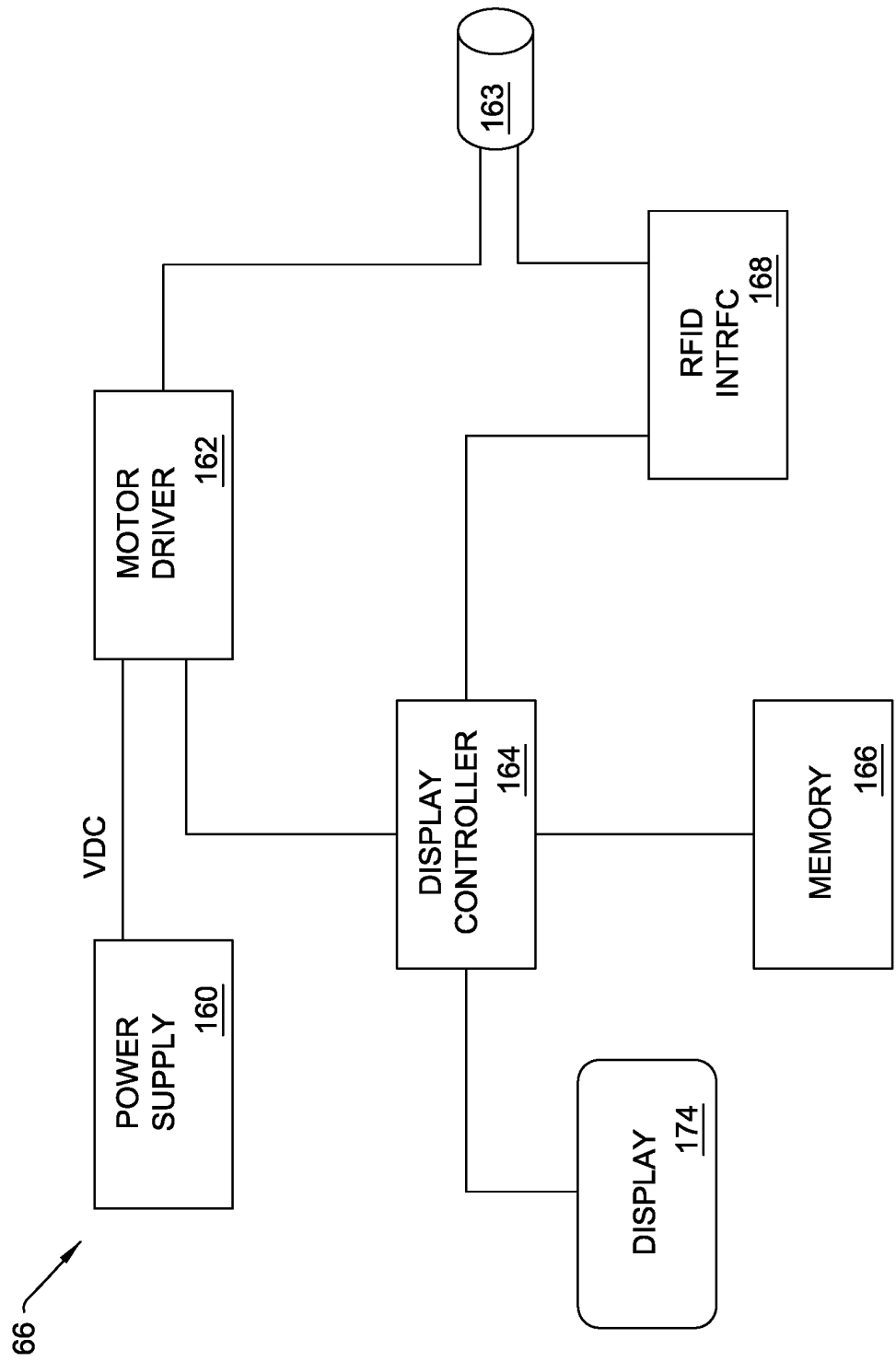
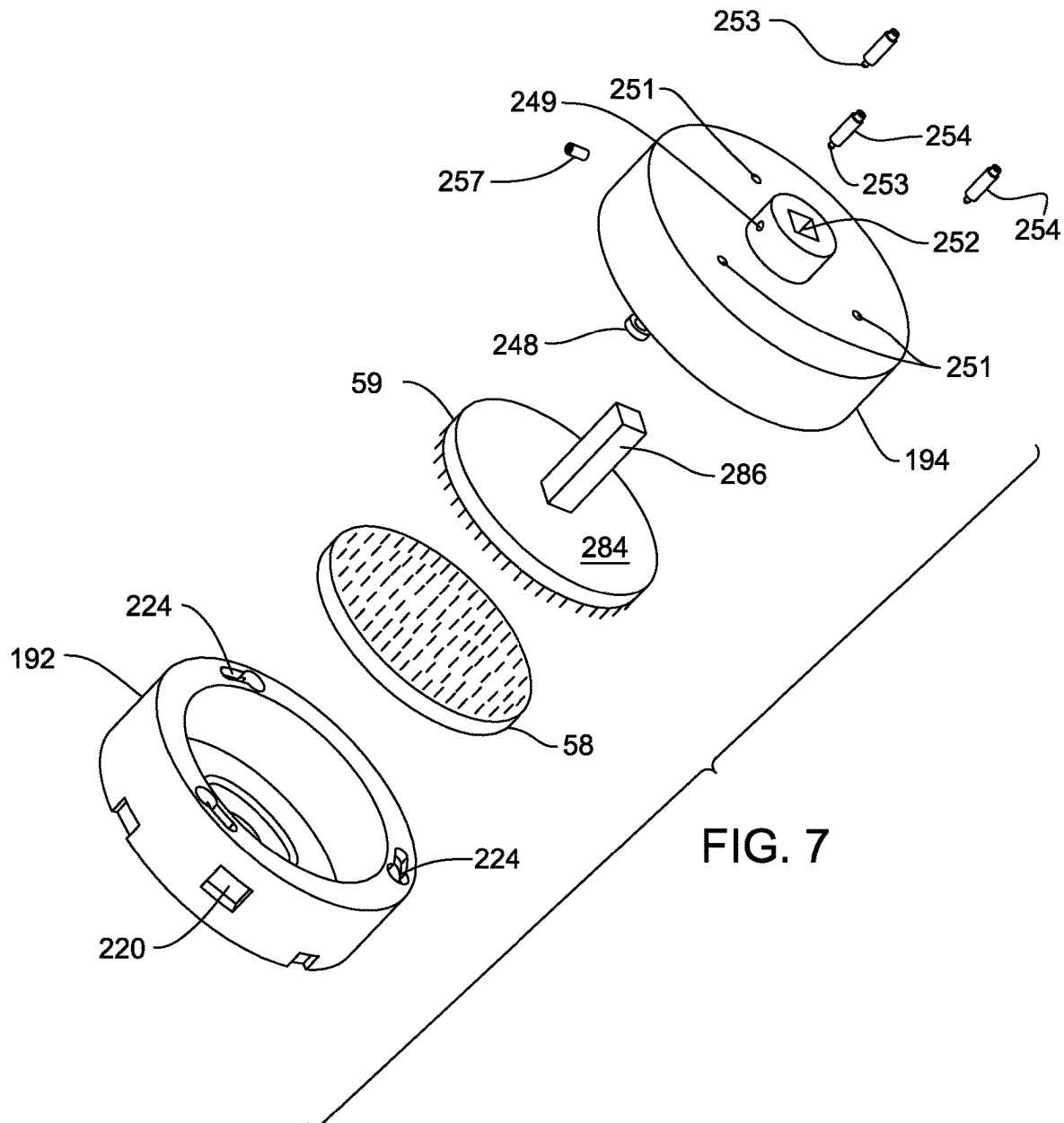


FIG. 6



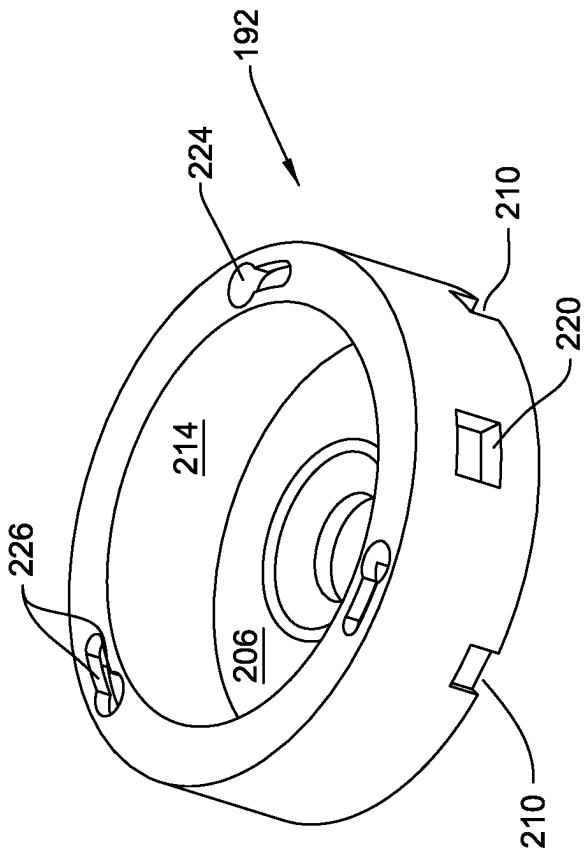


FIG. 8

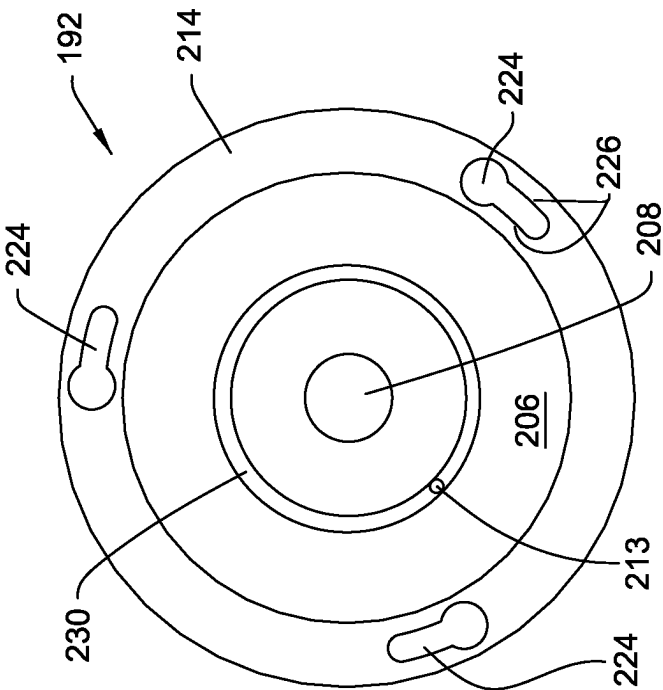


FIG. 9

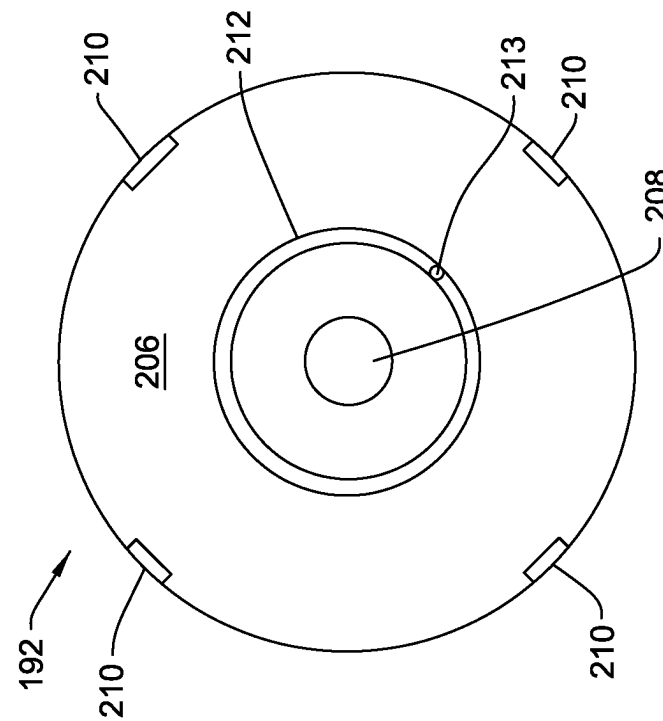


FIG. 10

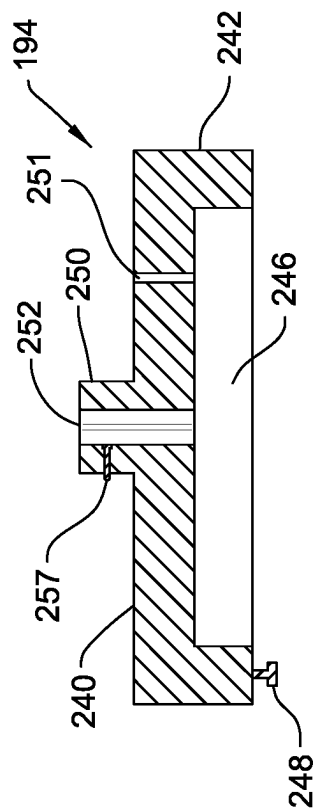


FIG. 11

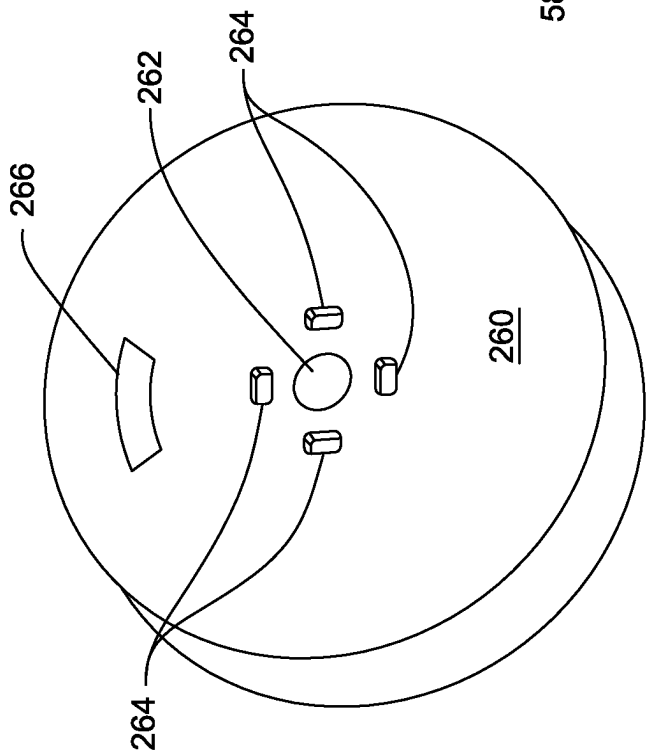


FIG. 12

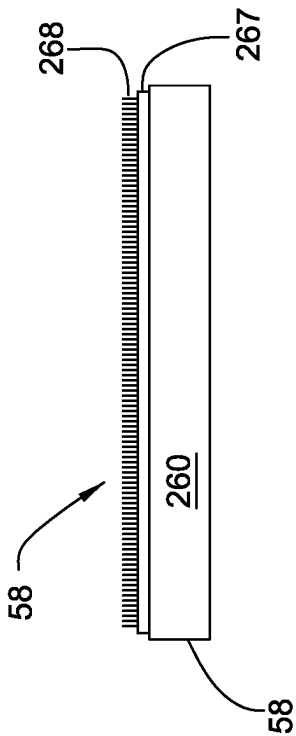


FIG. 13

272



DEVICE IDENTIFICATION	<u>273</u>
USE Y/N	<u>274</u>
MINIMUM SPEED	<u>275</u>
DEFAULT SPEED	<u>276</u>
MAXIMUM SPEED	<u>277</u>
DIRECTION	<u>278</u>

FIG. 14

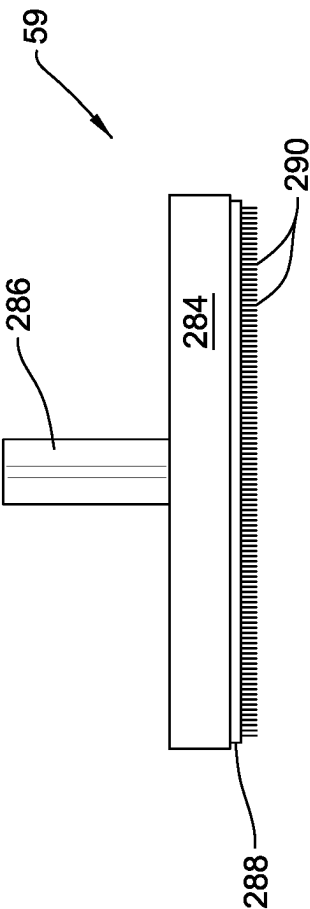


FIG. 15

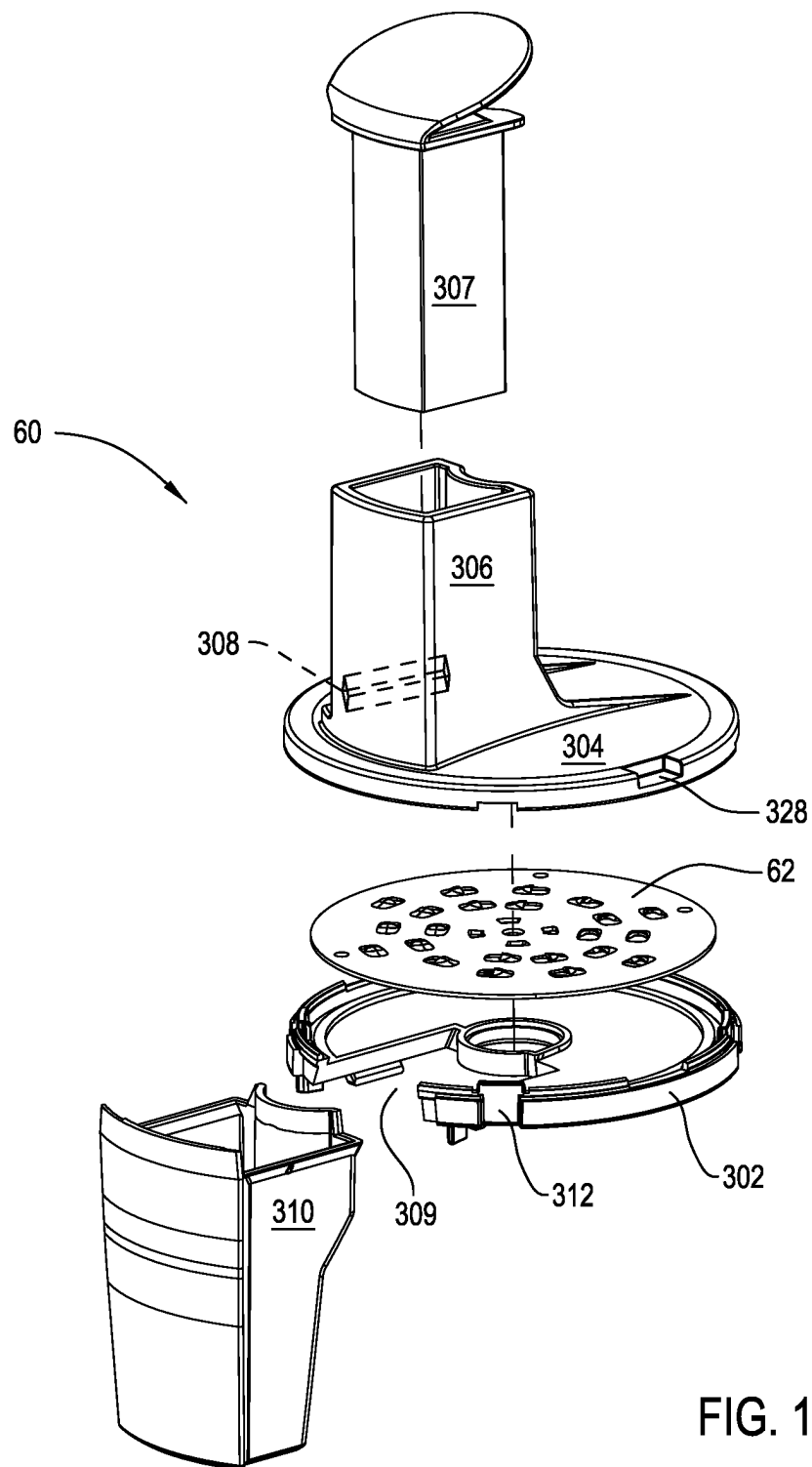


FIG. 16

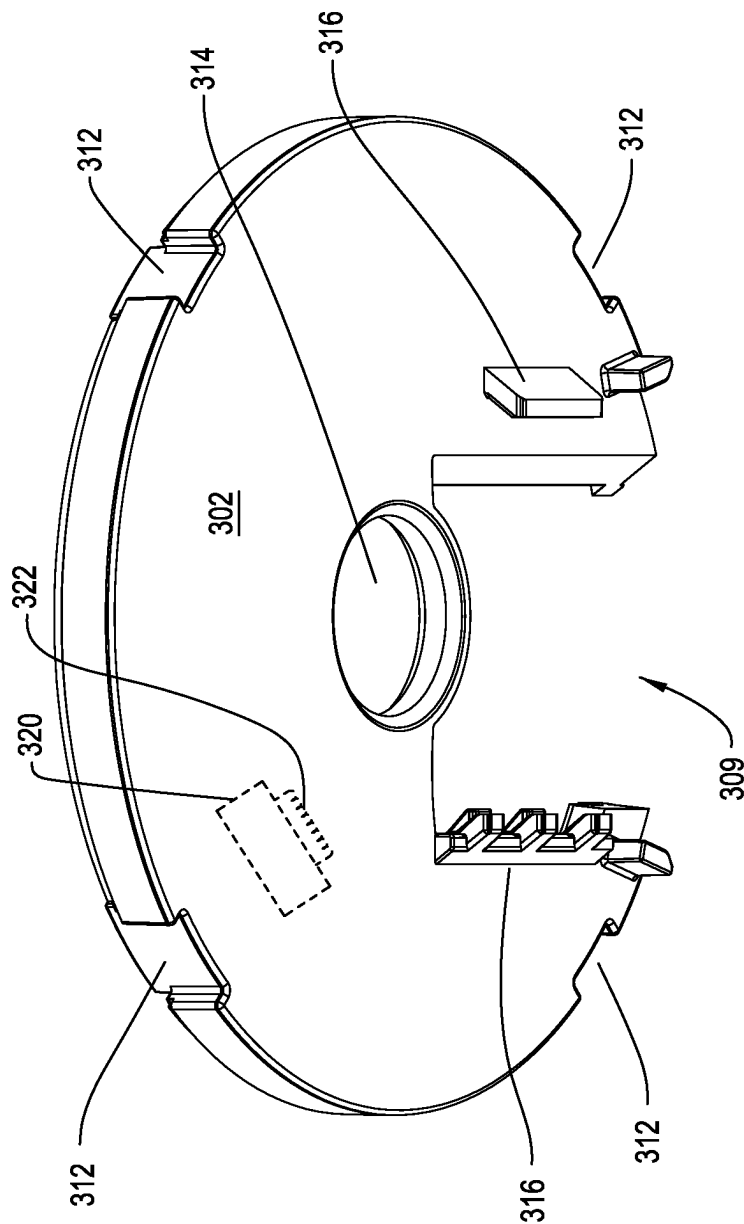


FIG. 17

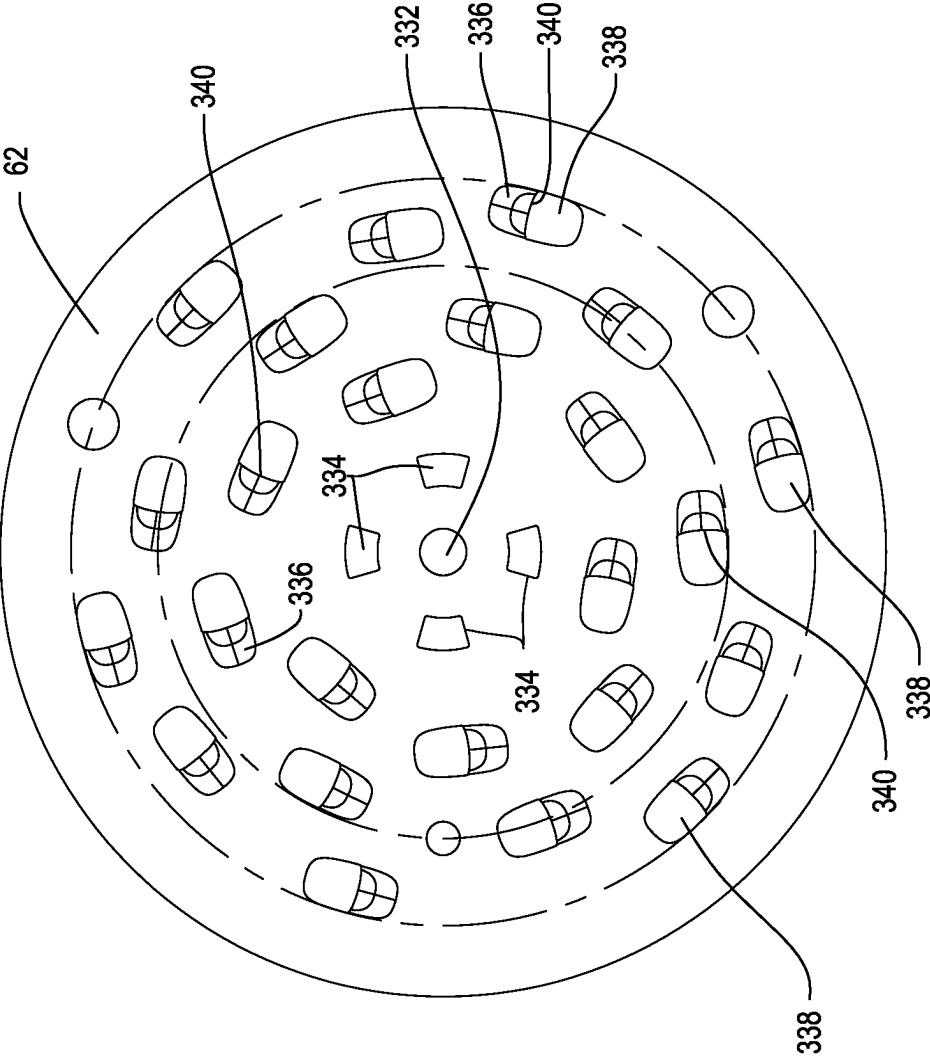


FIG. 18

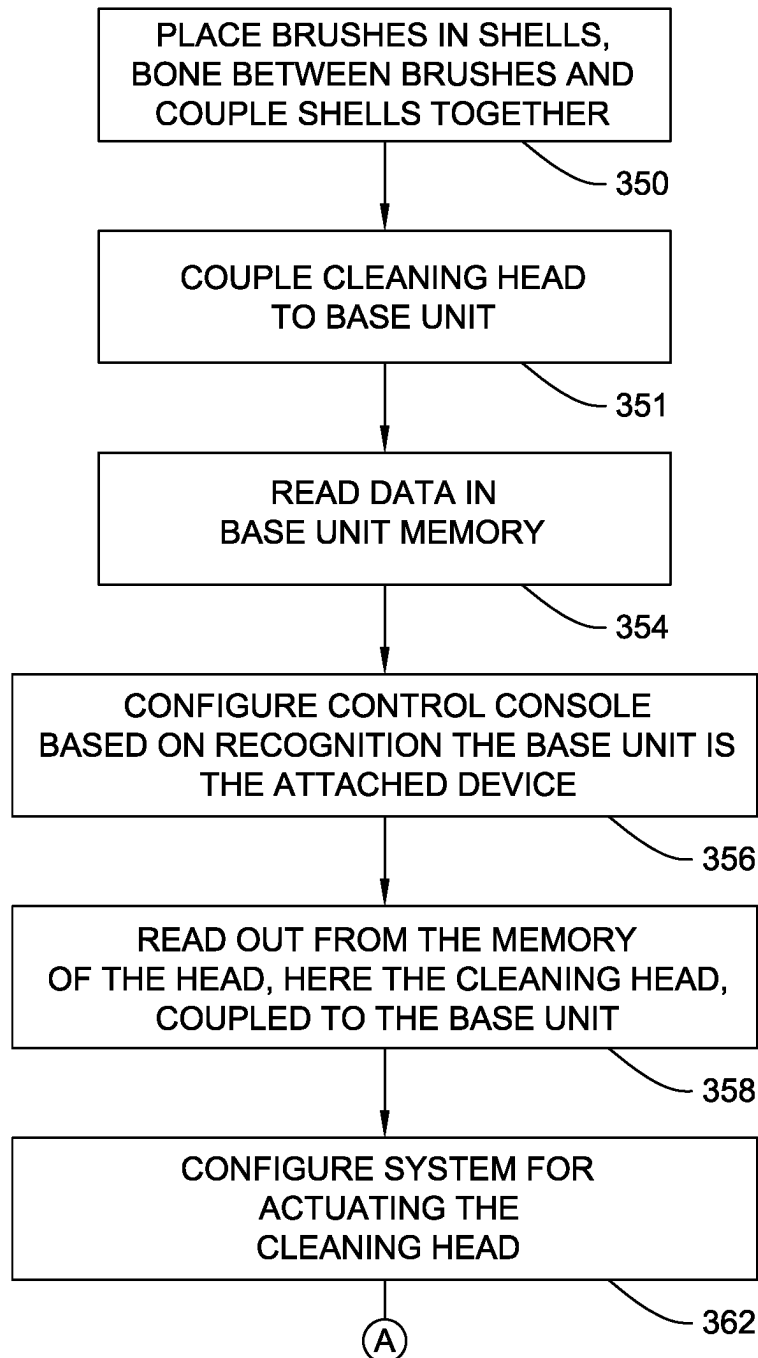


FIG. 19A

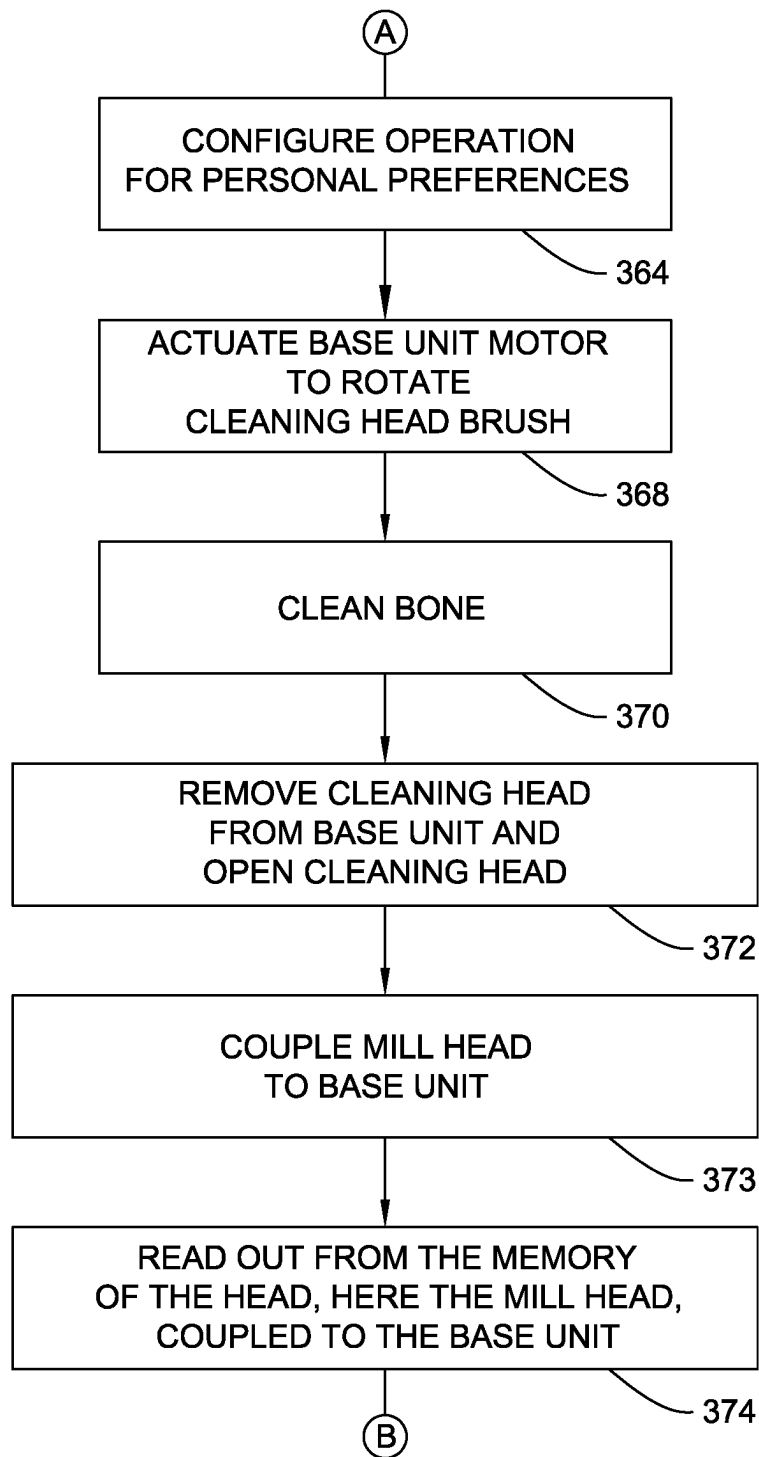


FIG. 19B

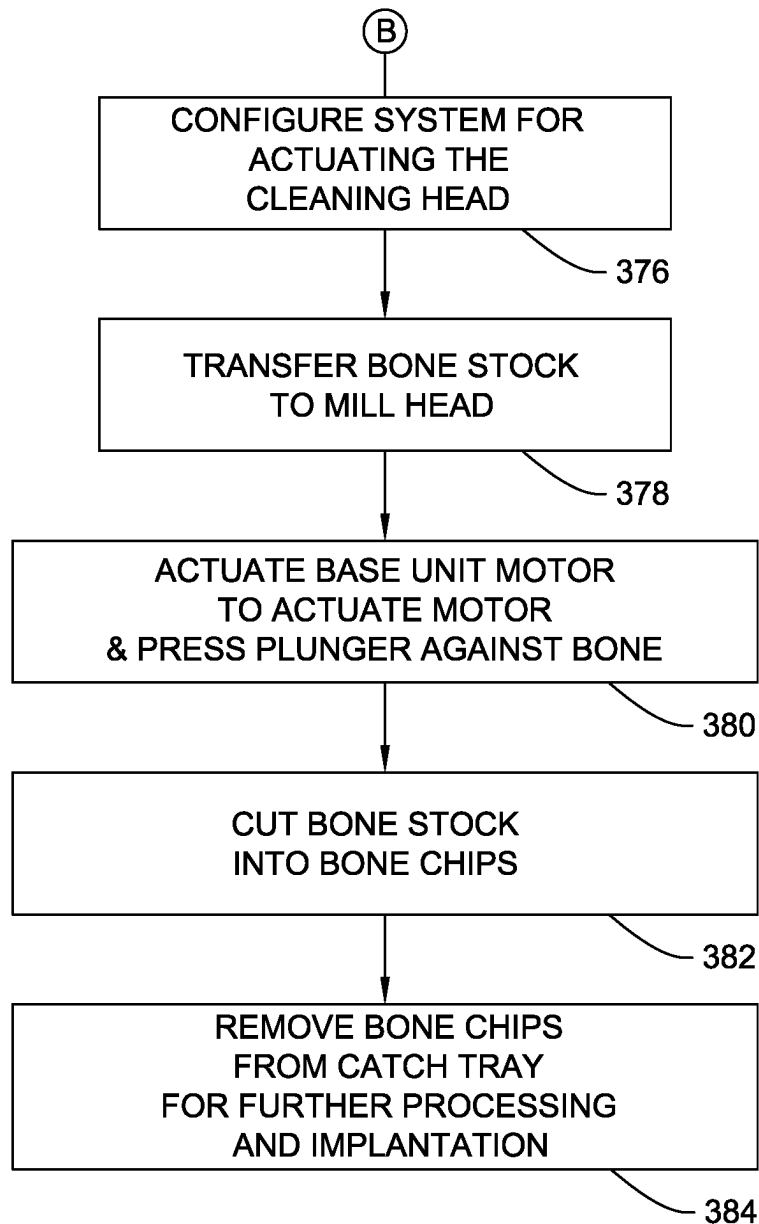
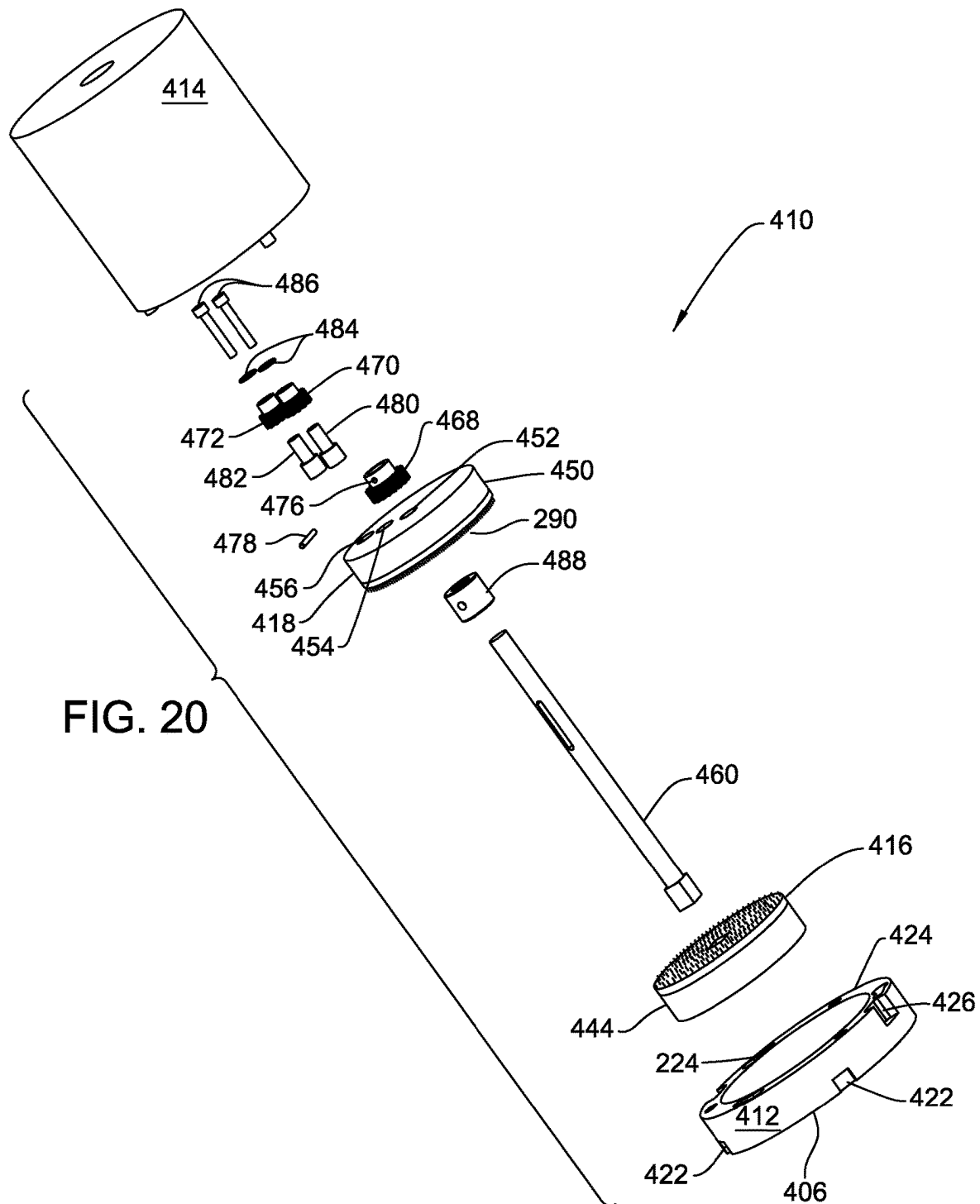


FIG. 19C



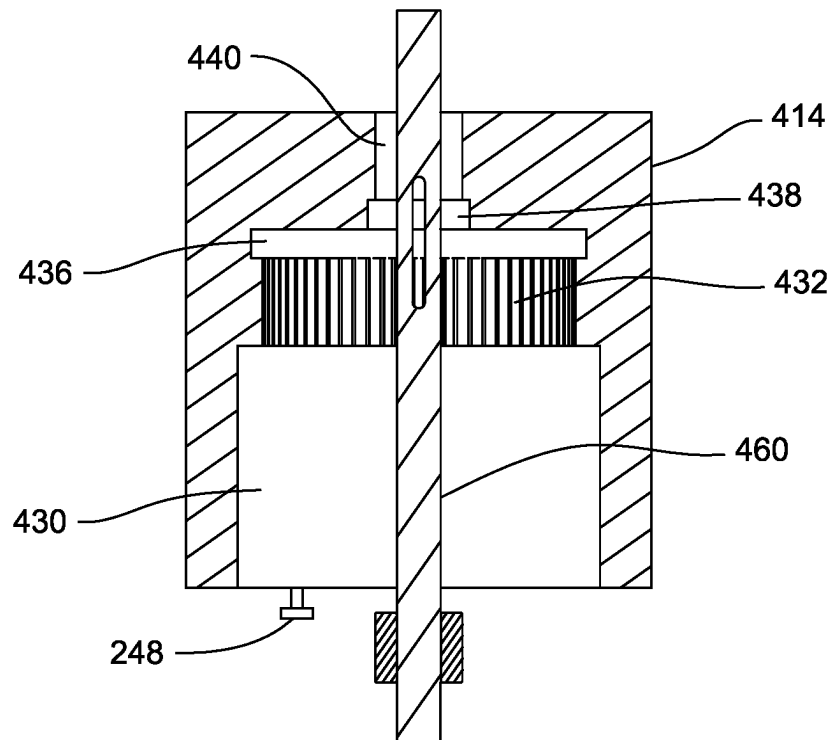


FIG. 21

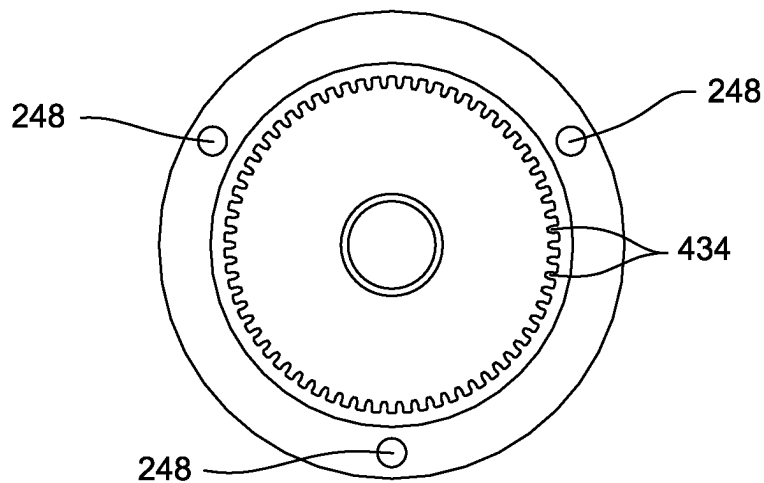


FIG. 22

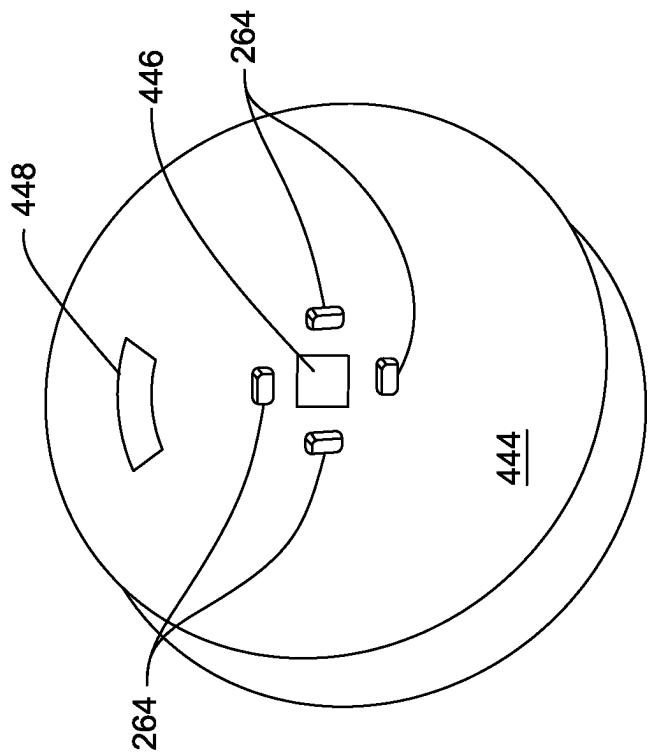


FIG. 23

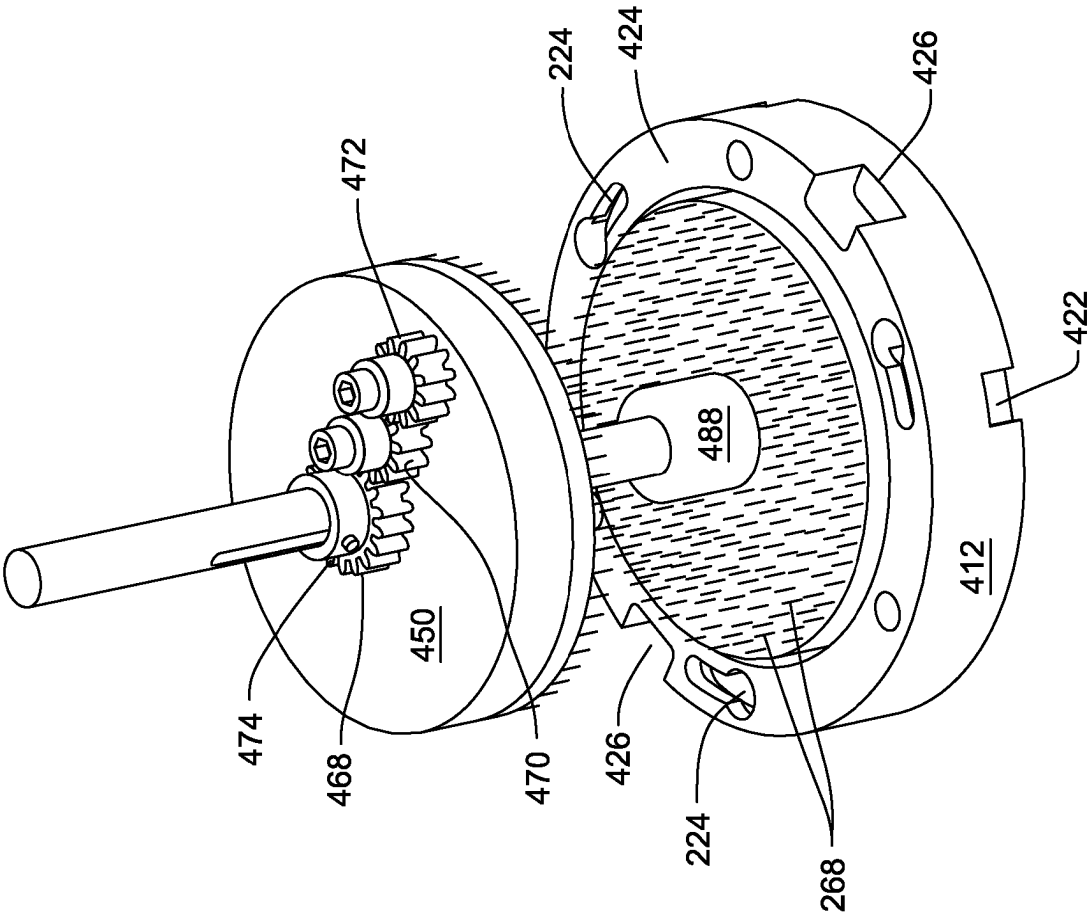


FIG. 24

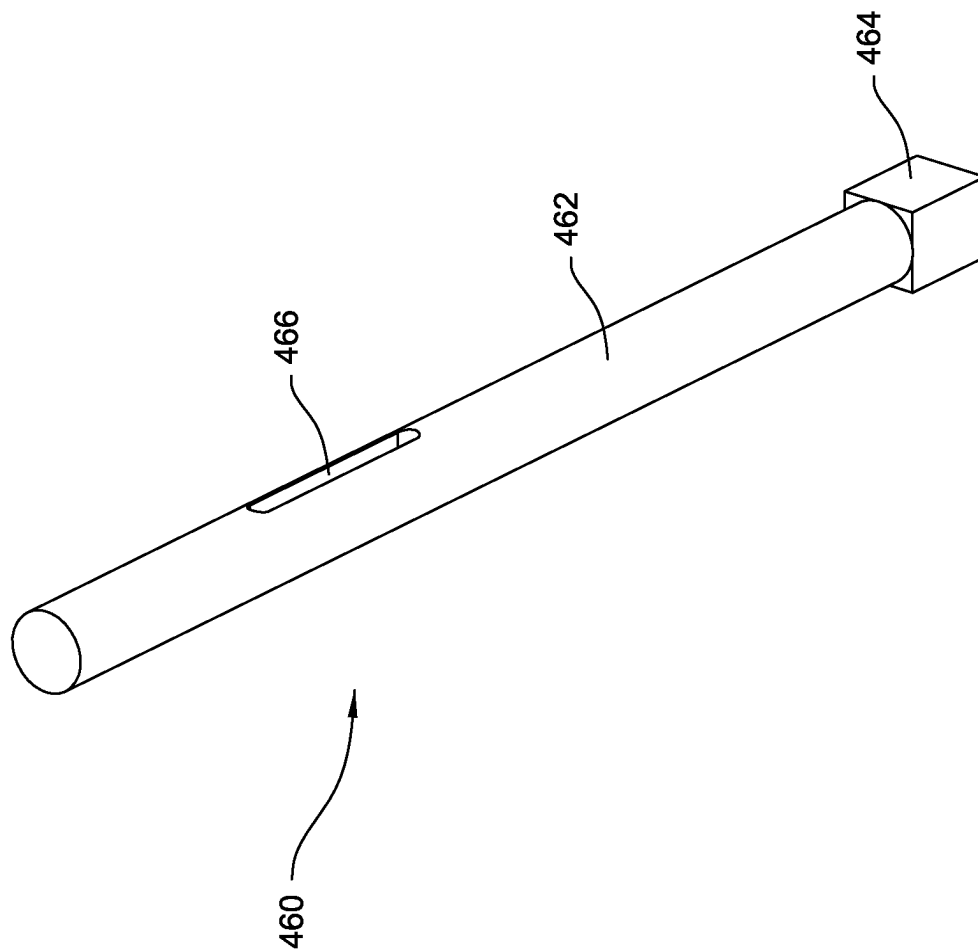
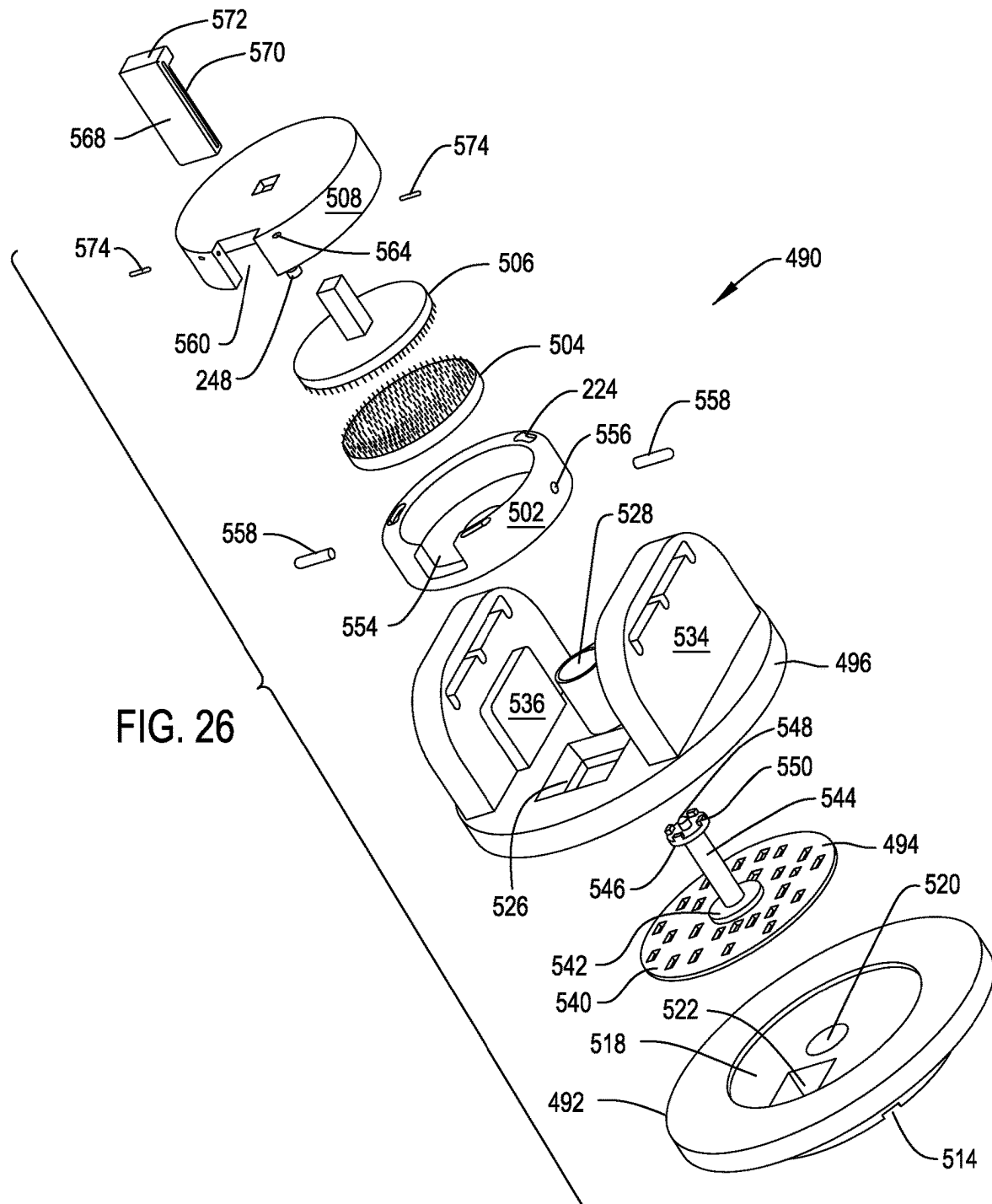


FIG. 25



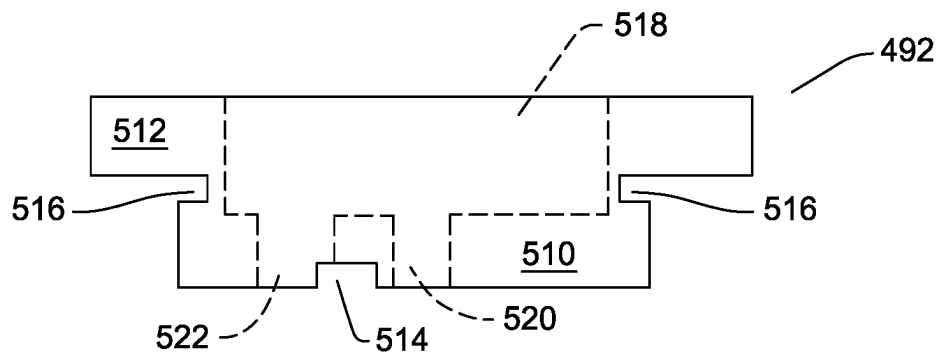


FIG. 27

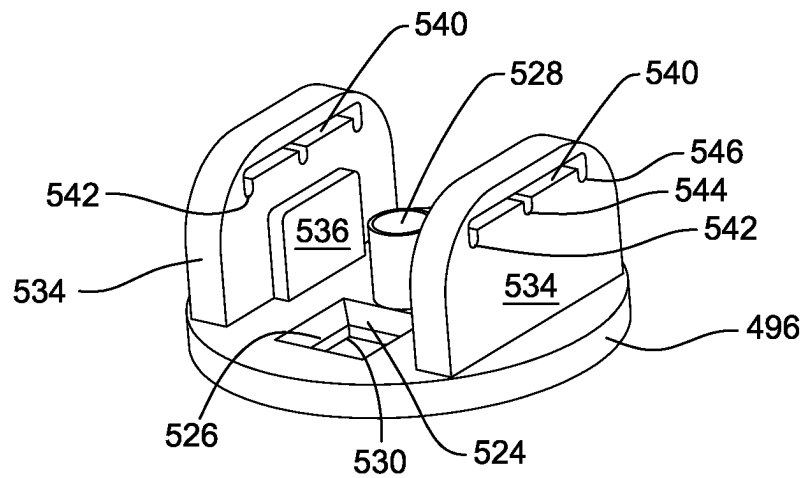


FIG. 28

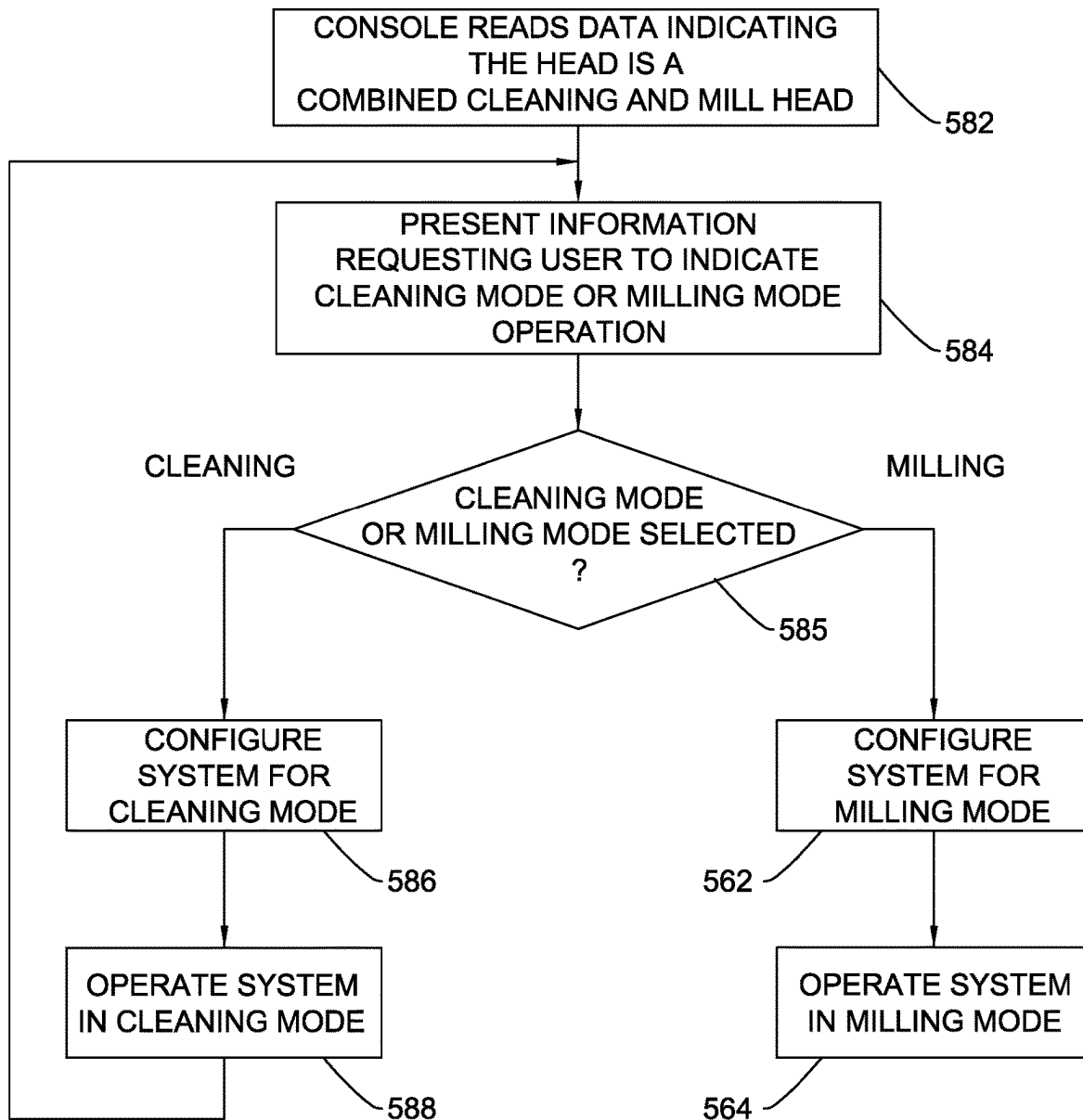


FIG. 29

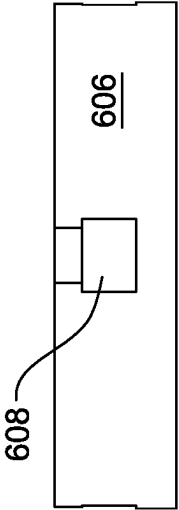
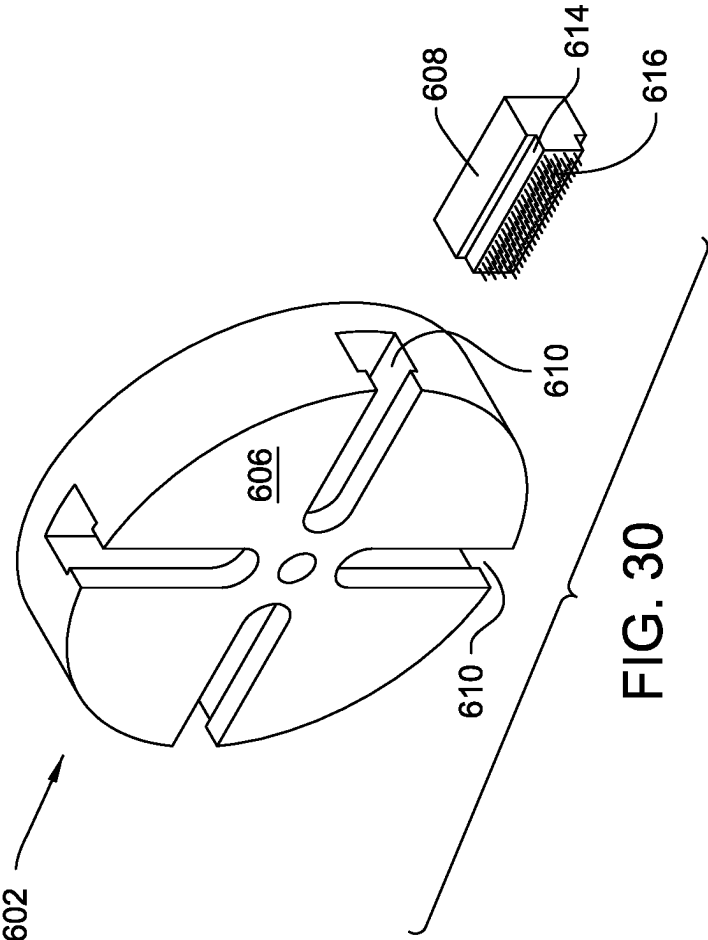


FIG. 31

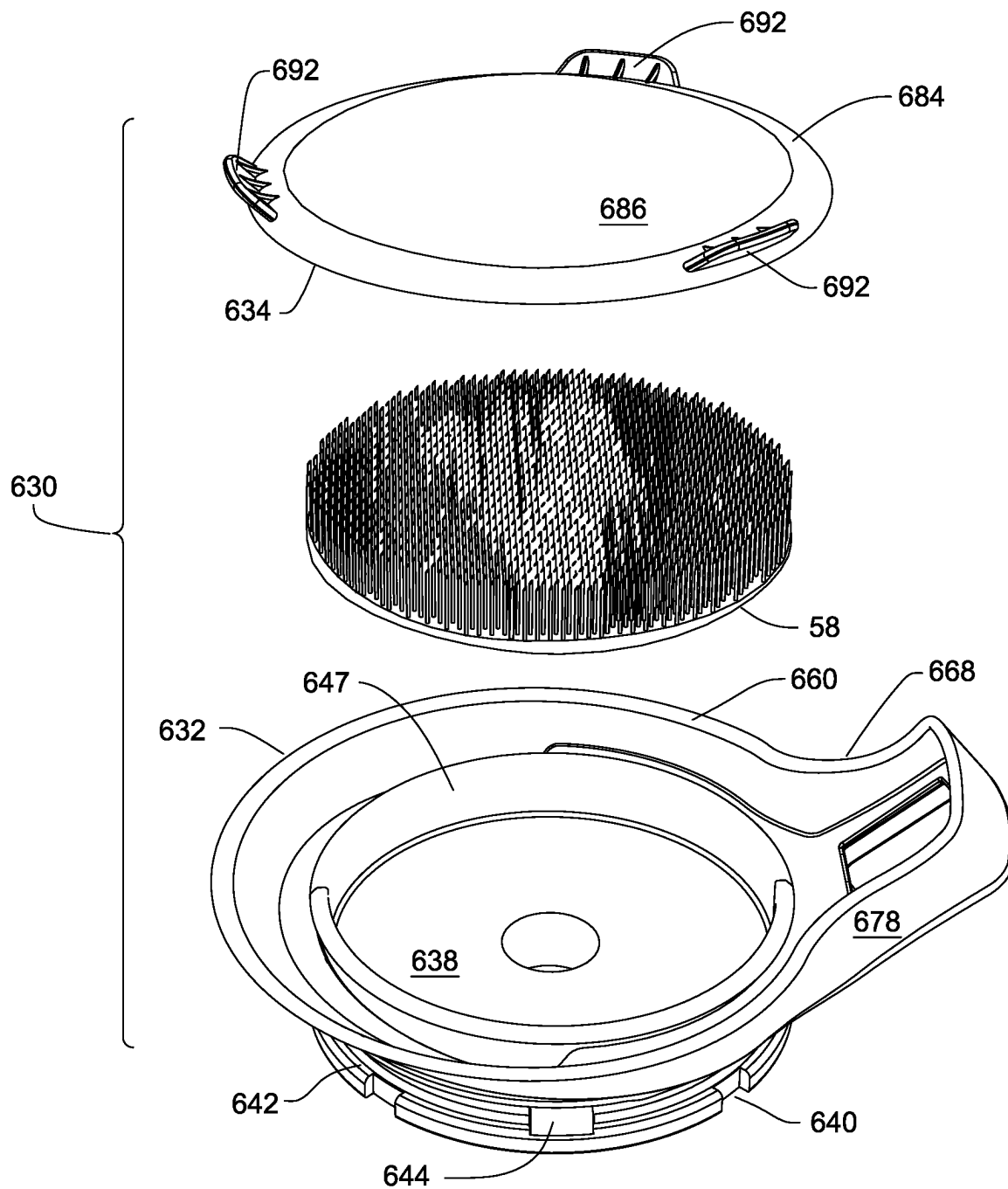


FIG. 32

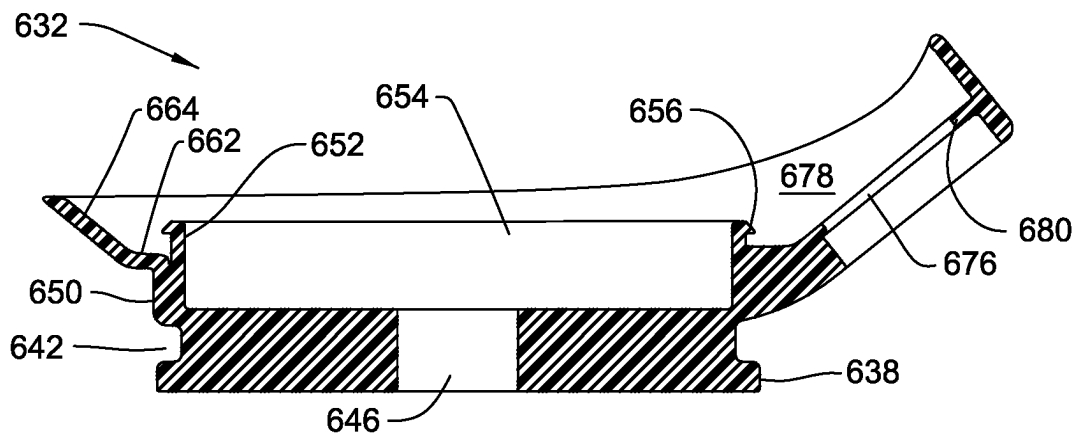


FIG. 34

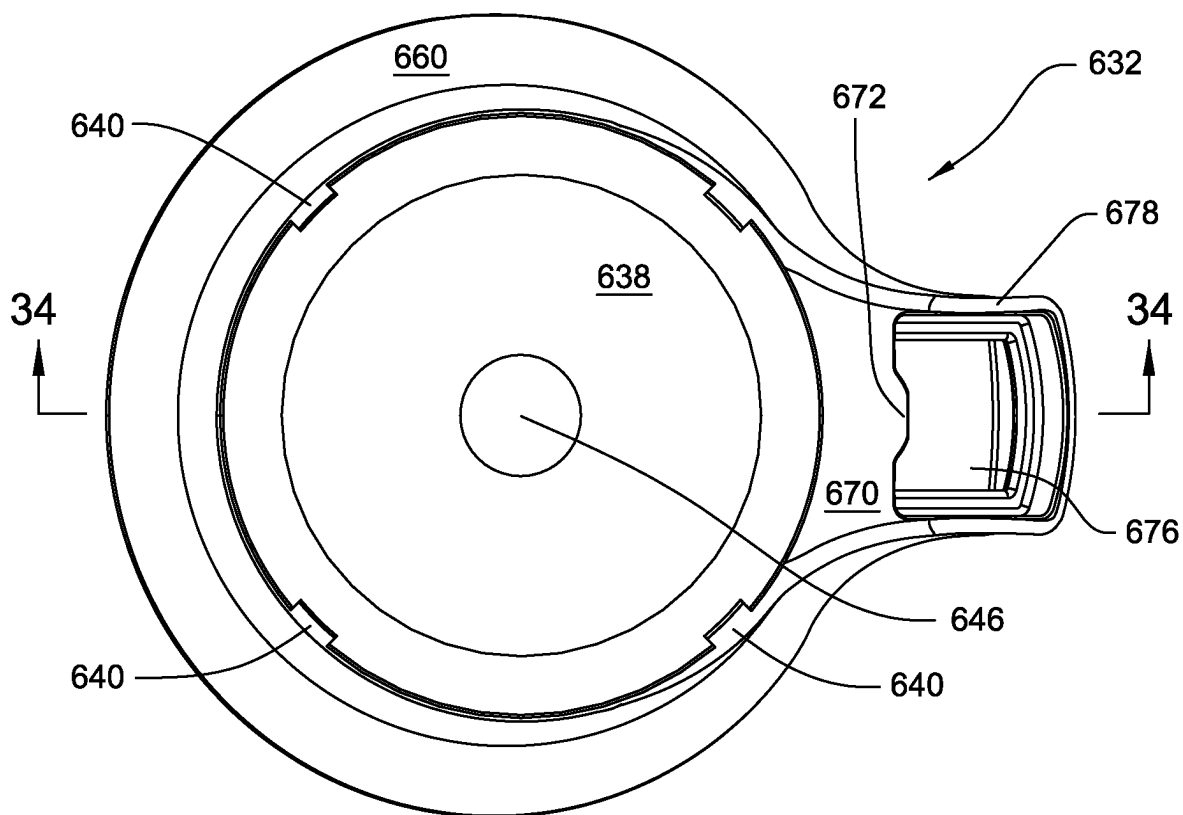


FIG. 33

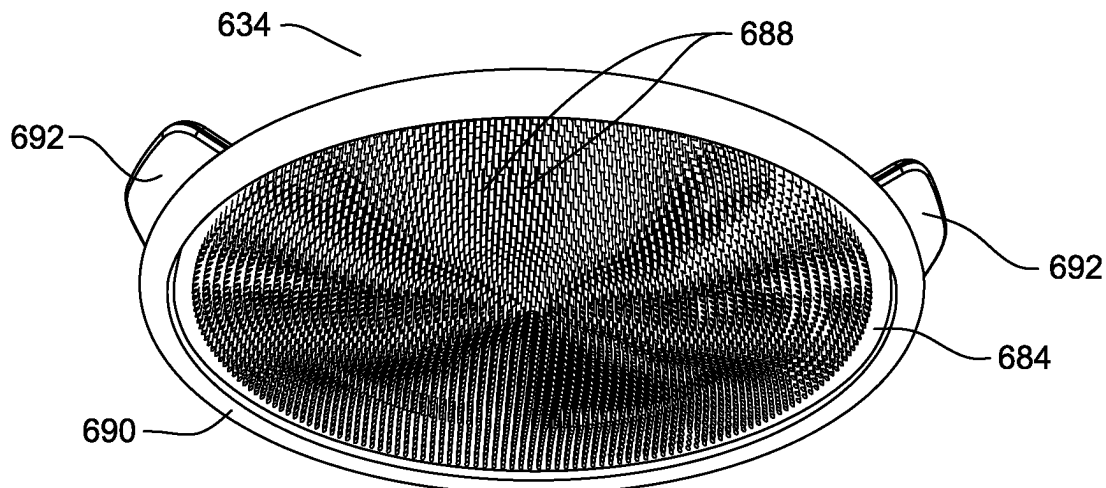


FIG. 36

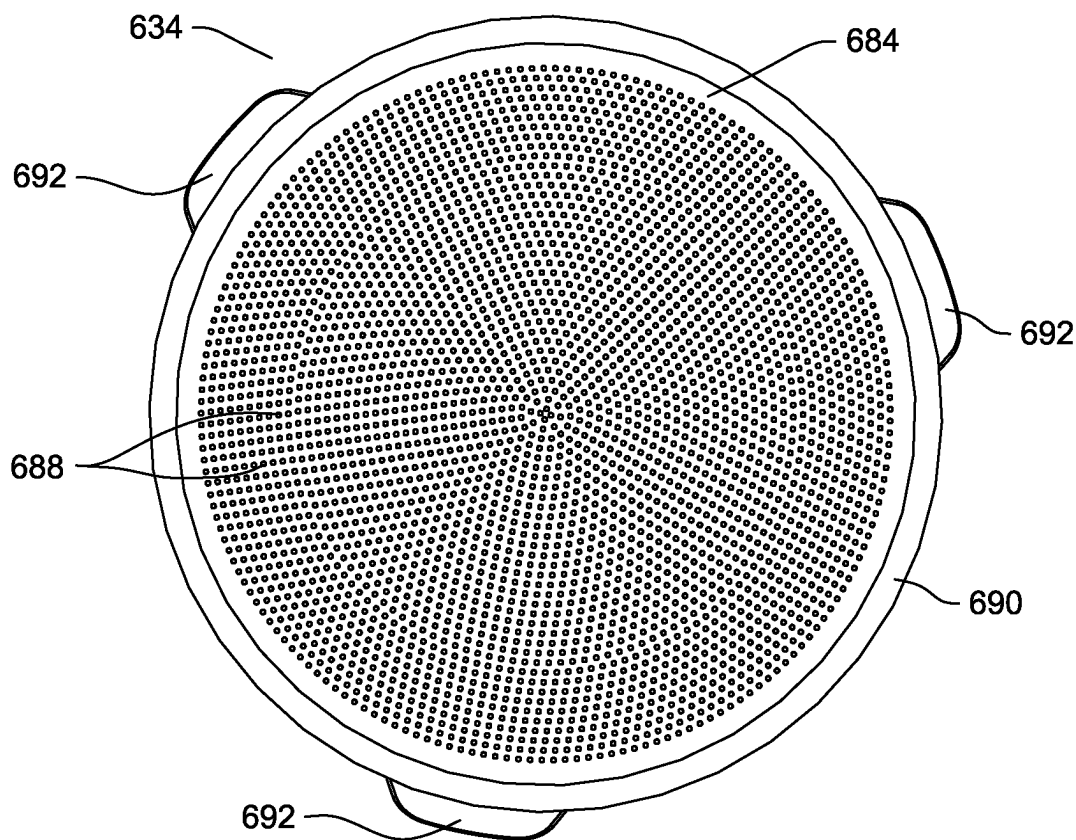


FIG. 35

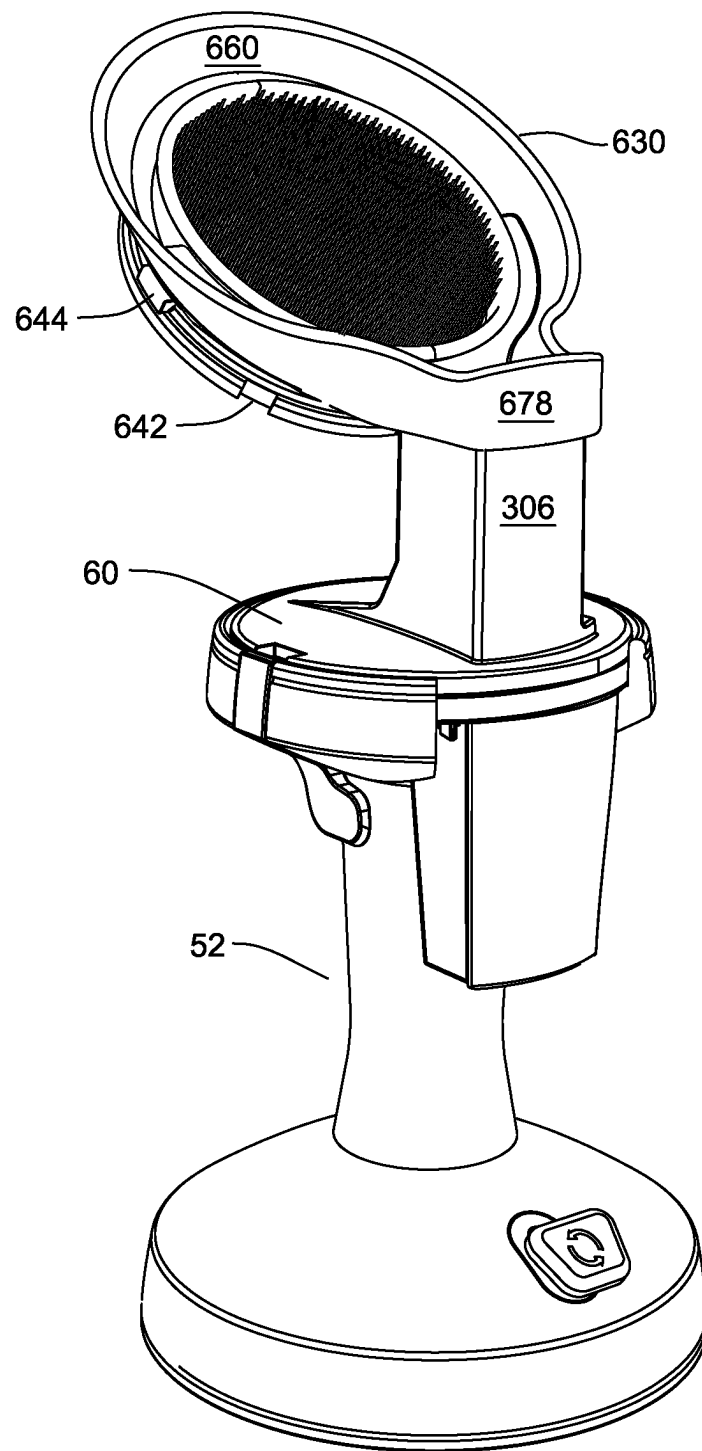
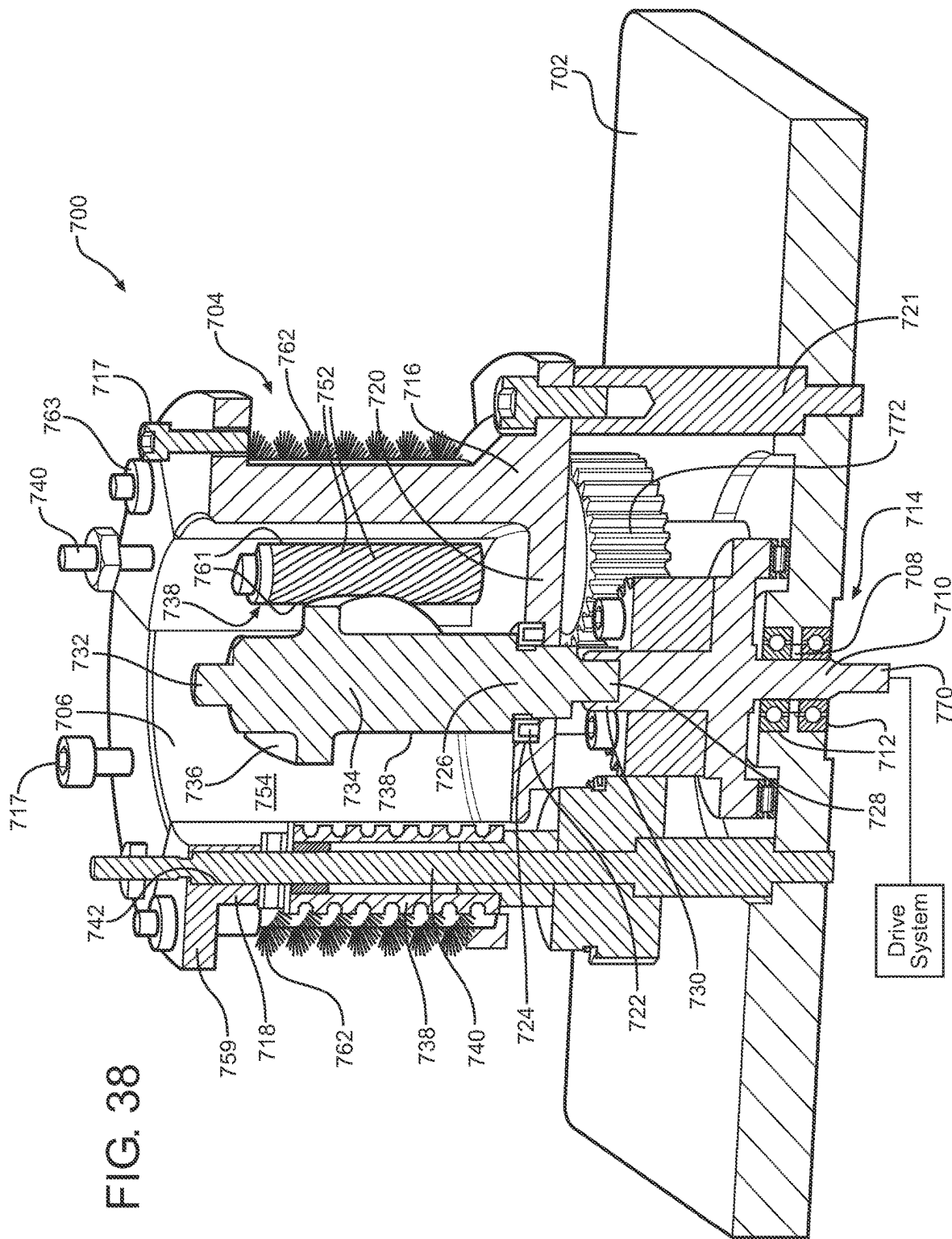


FIG. 37



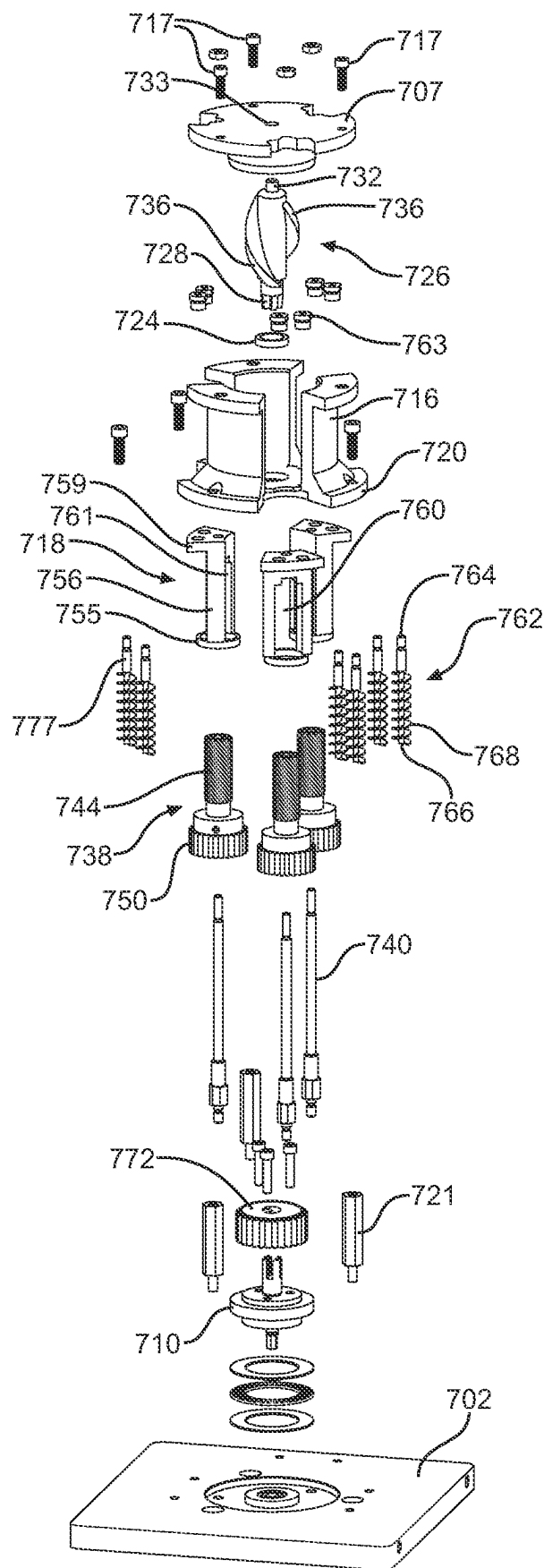


FIG. 39

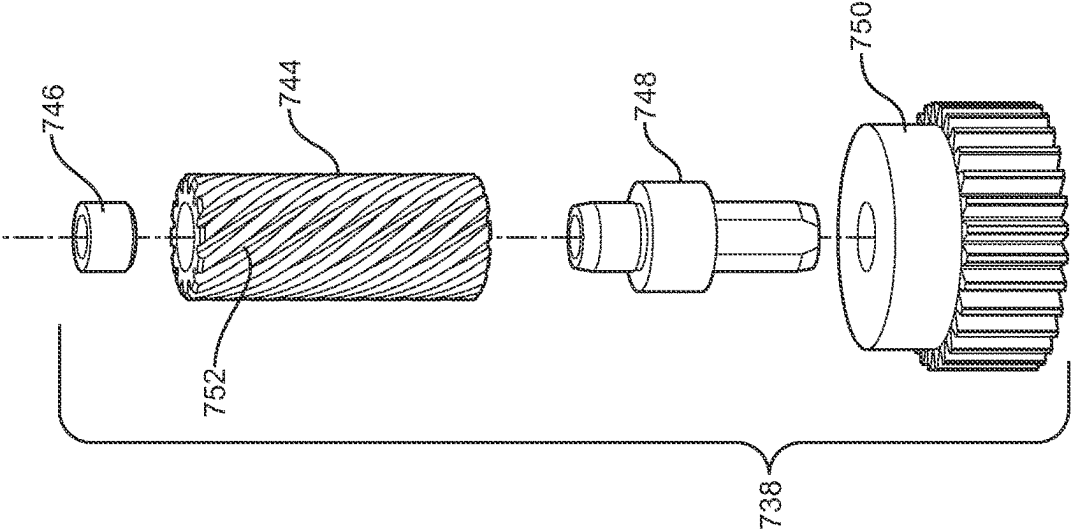


FIG. 41

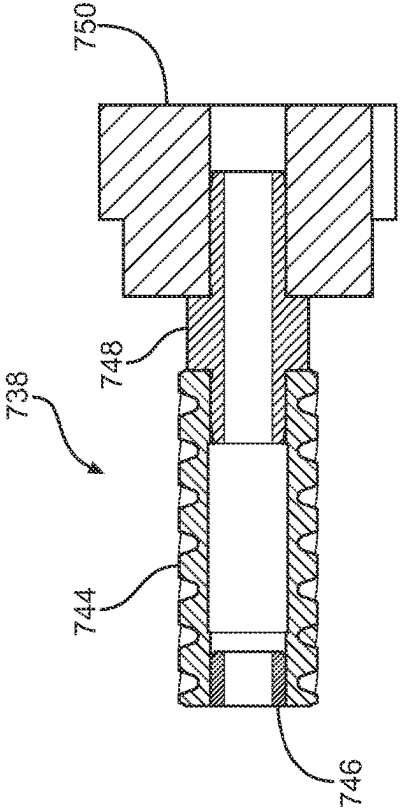


FIG. 40

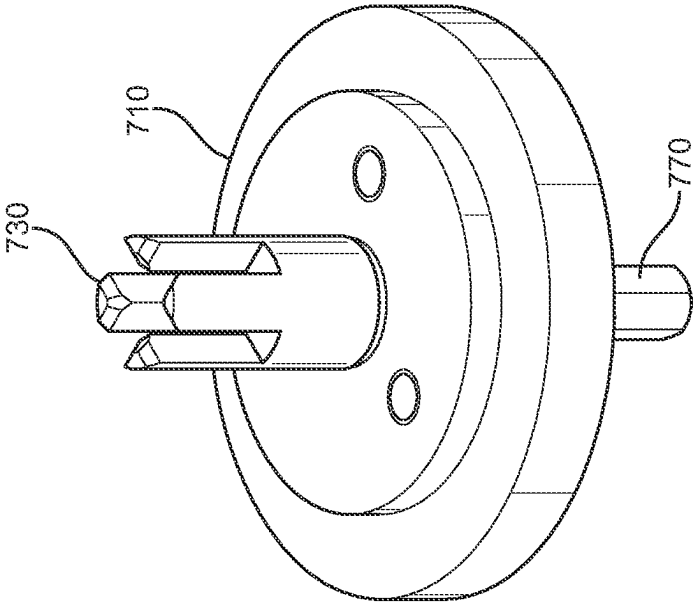


FIG. 42

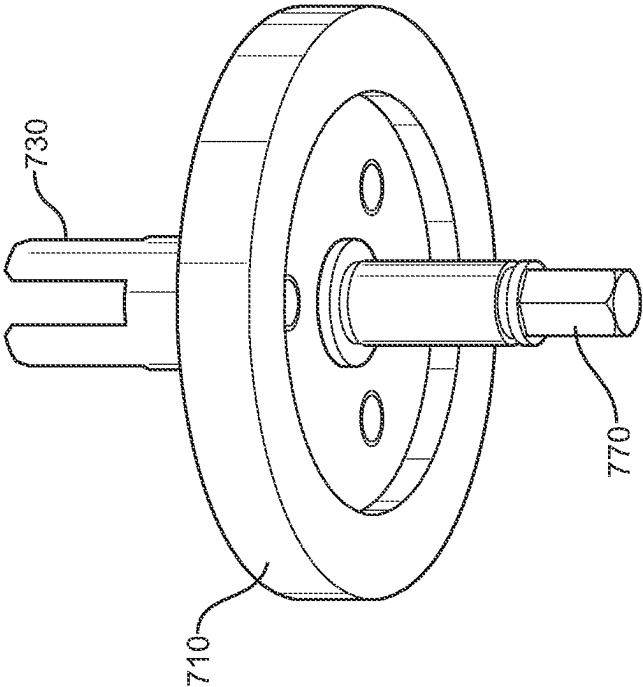
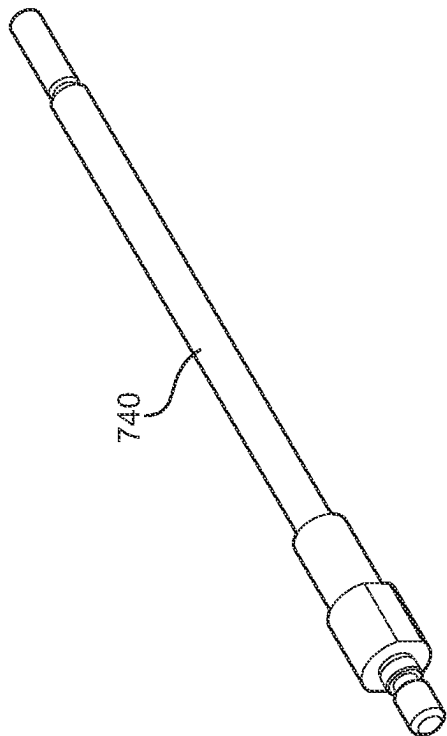
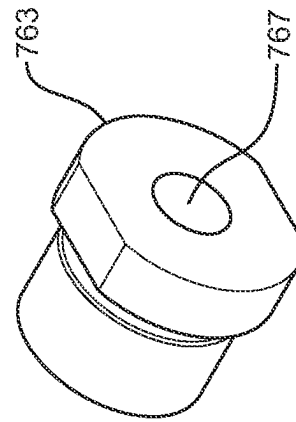
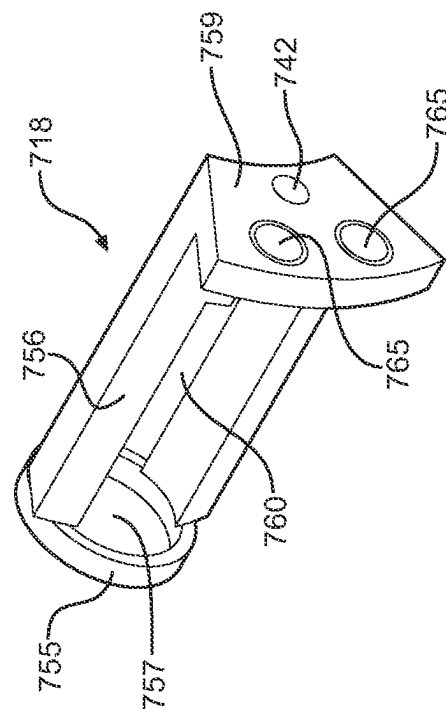


FIG. 43



44
45²⁴
46
47

64
G
F

545

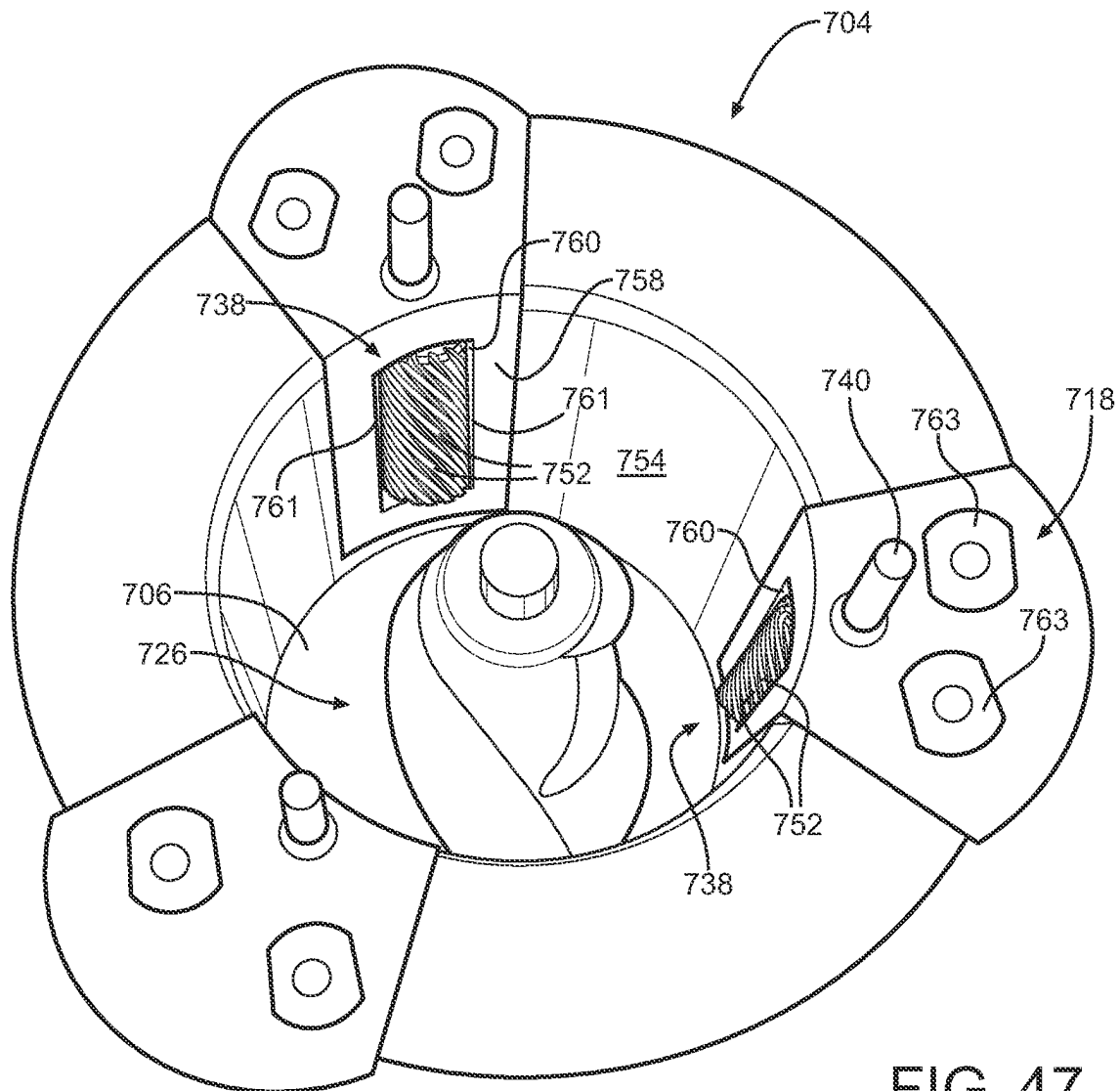


FIG. 47

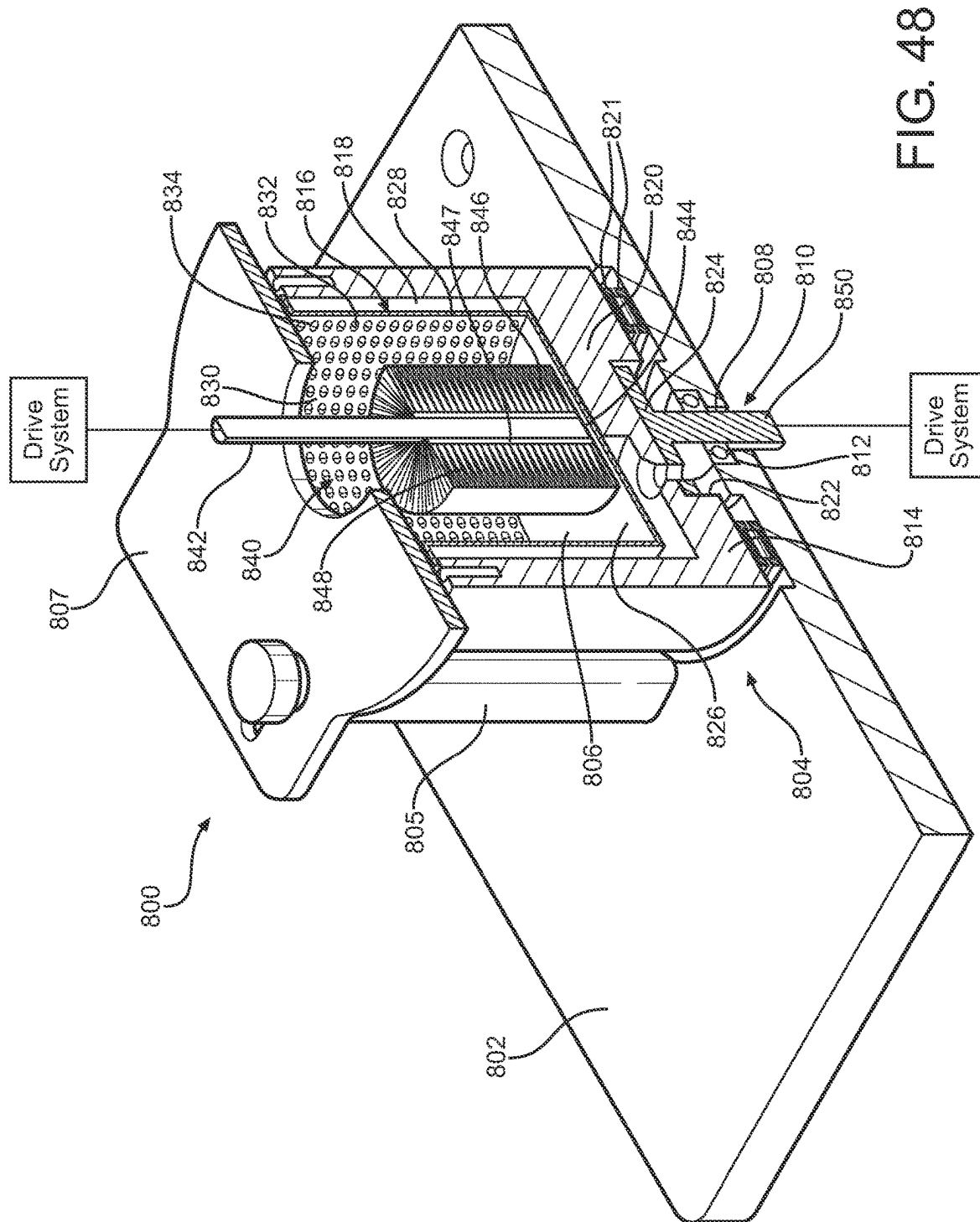
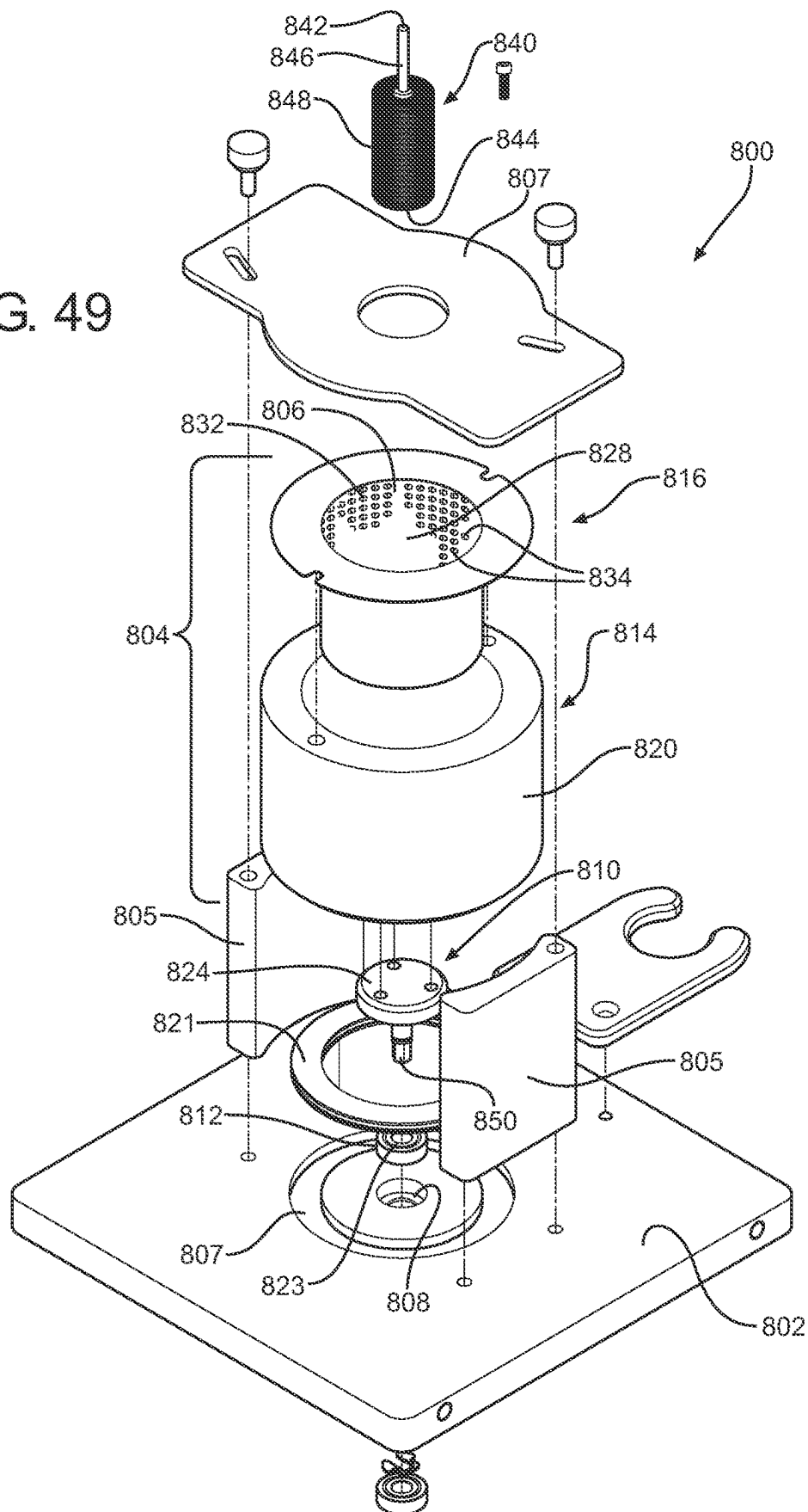


FIG. 49



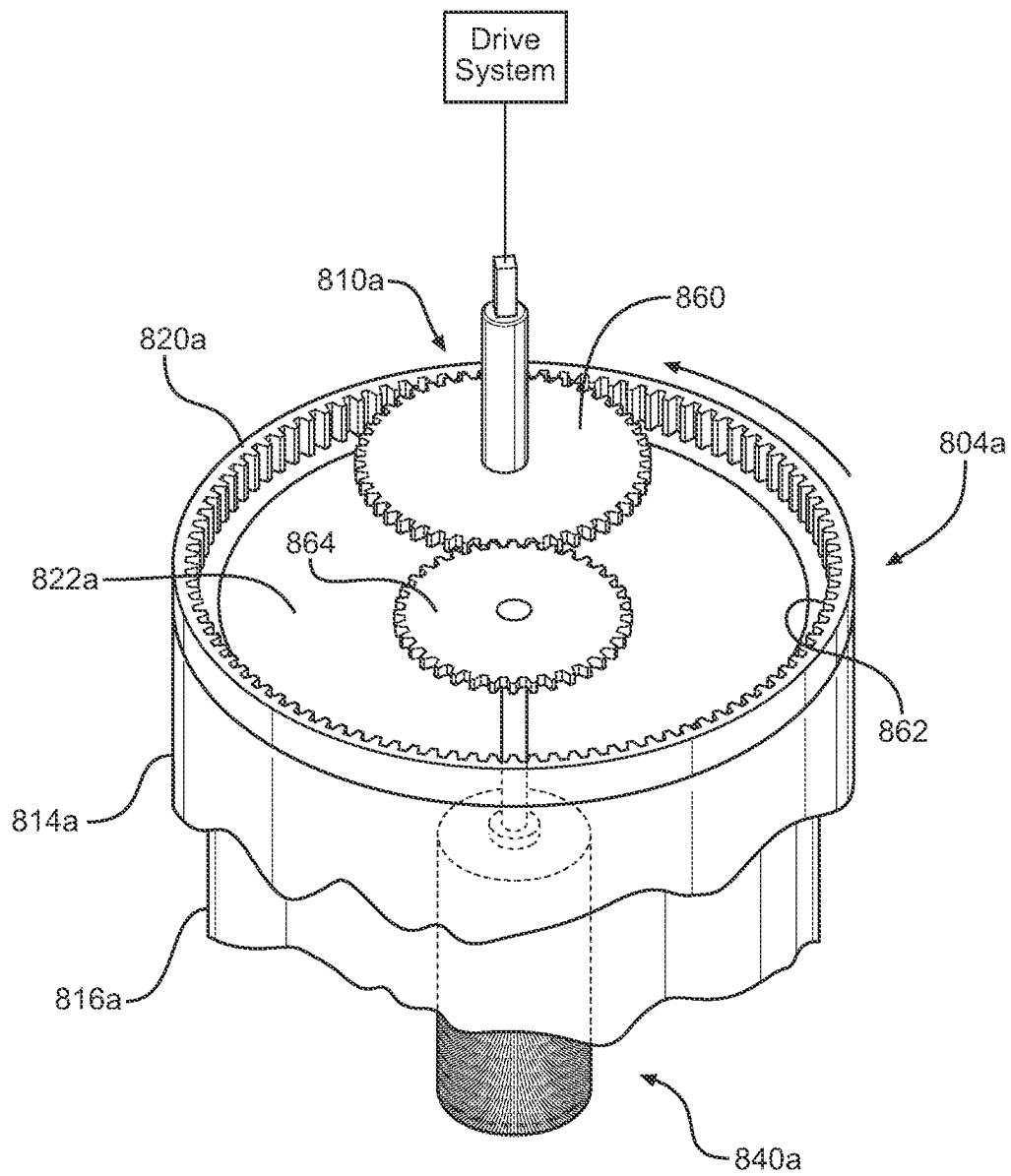


FIG. 49A

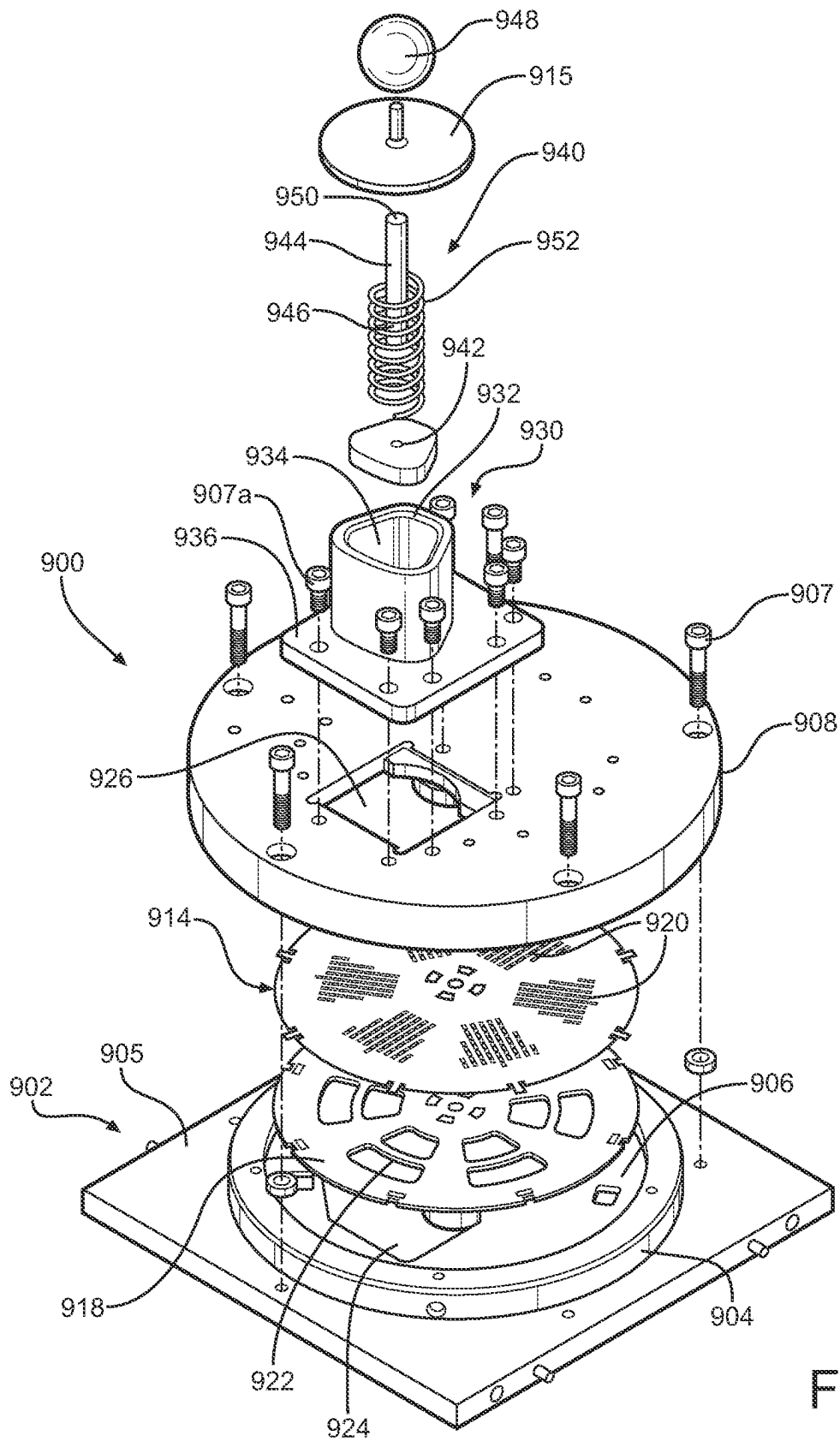
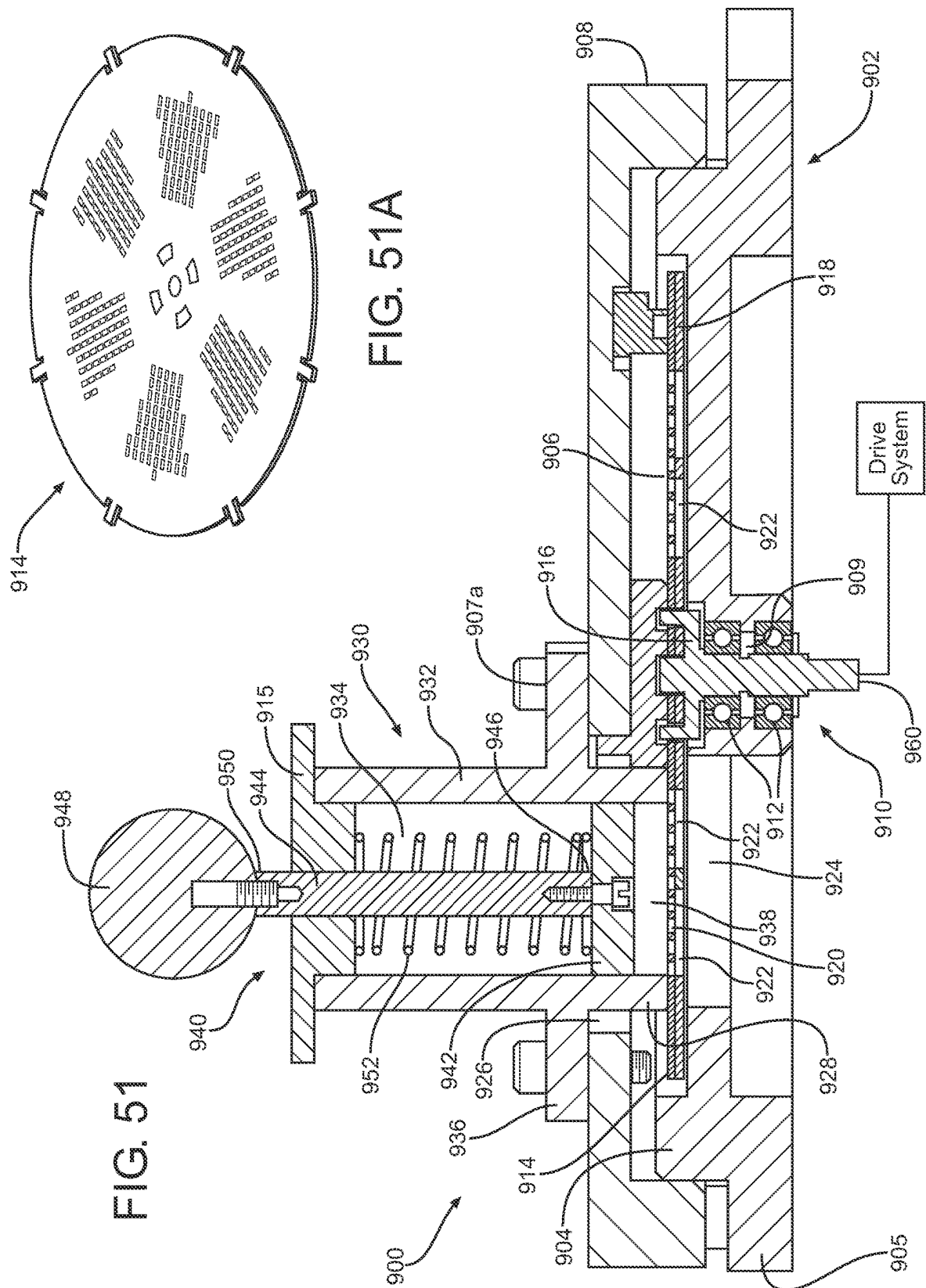


FIG. 50



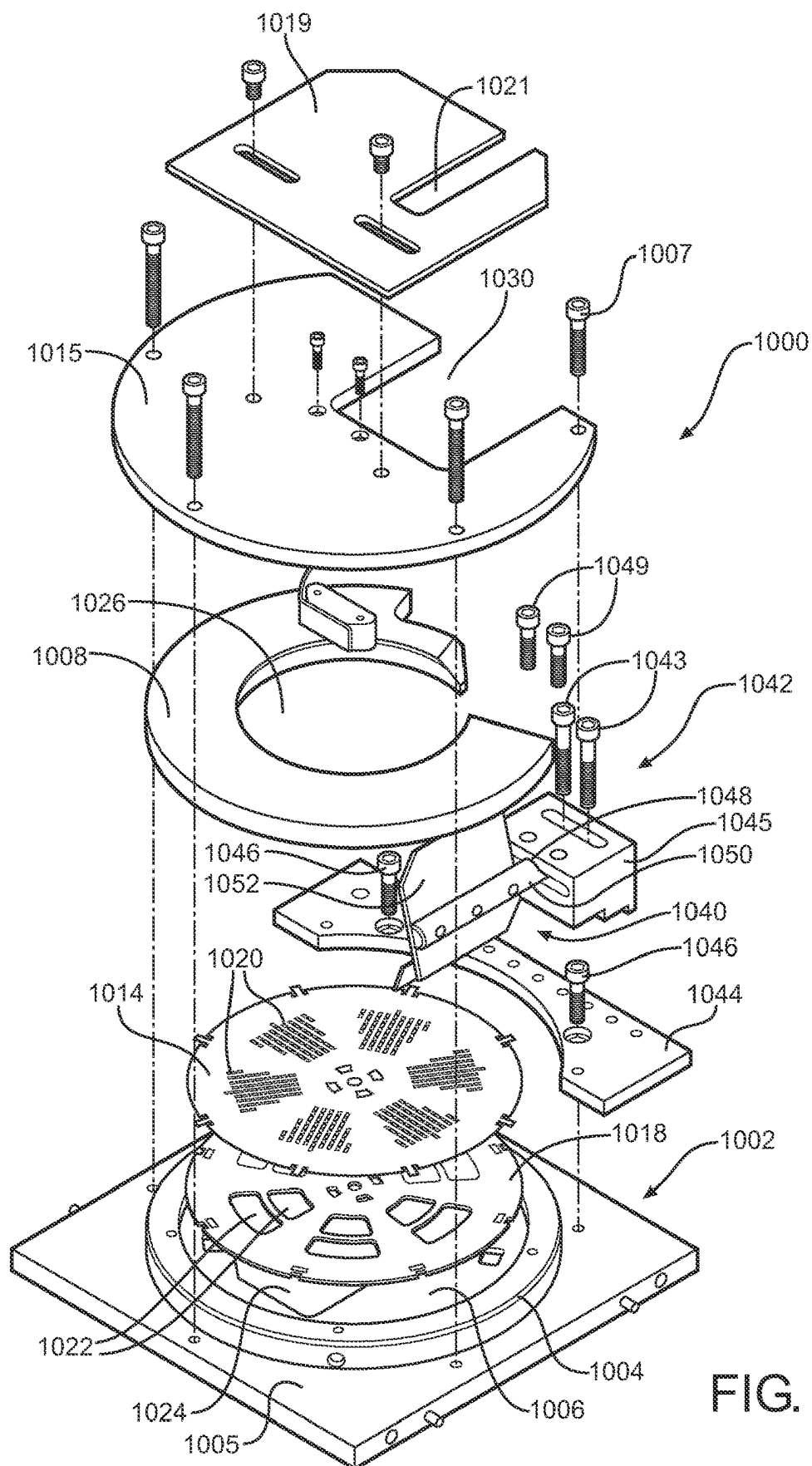
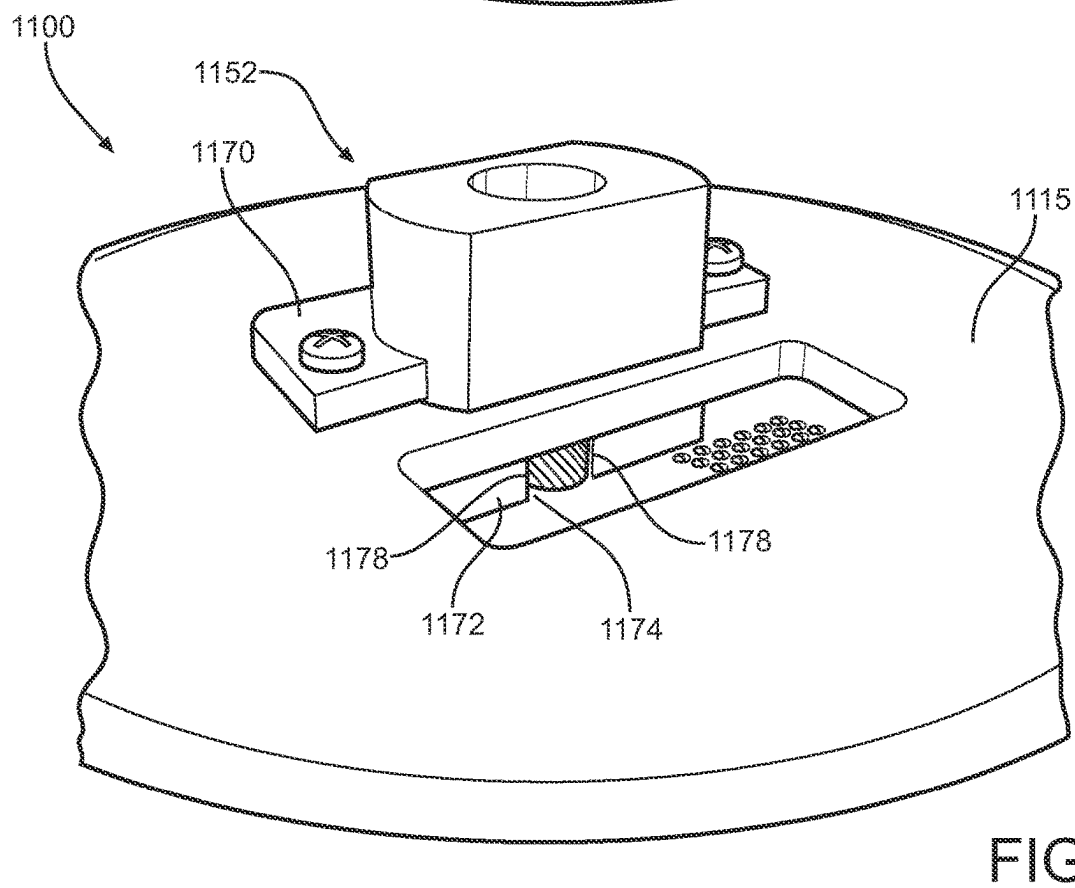
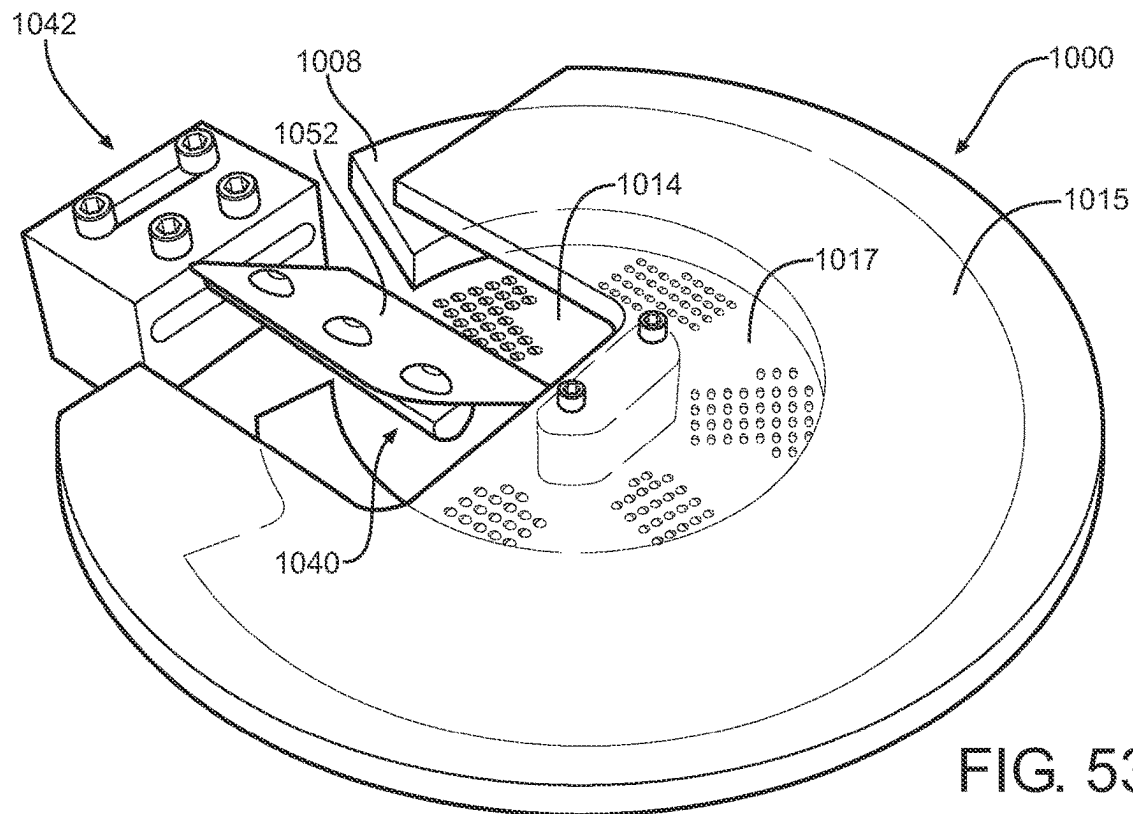
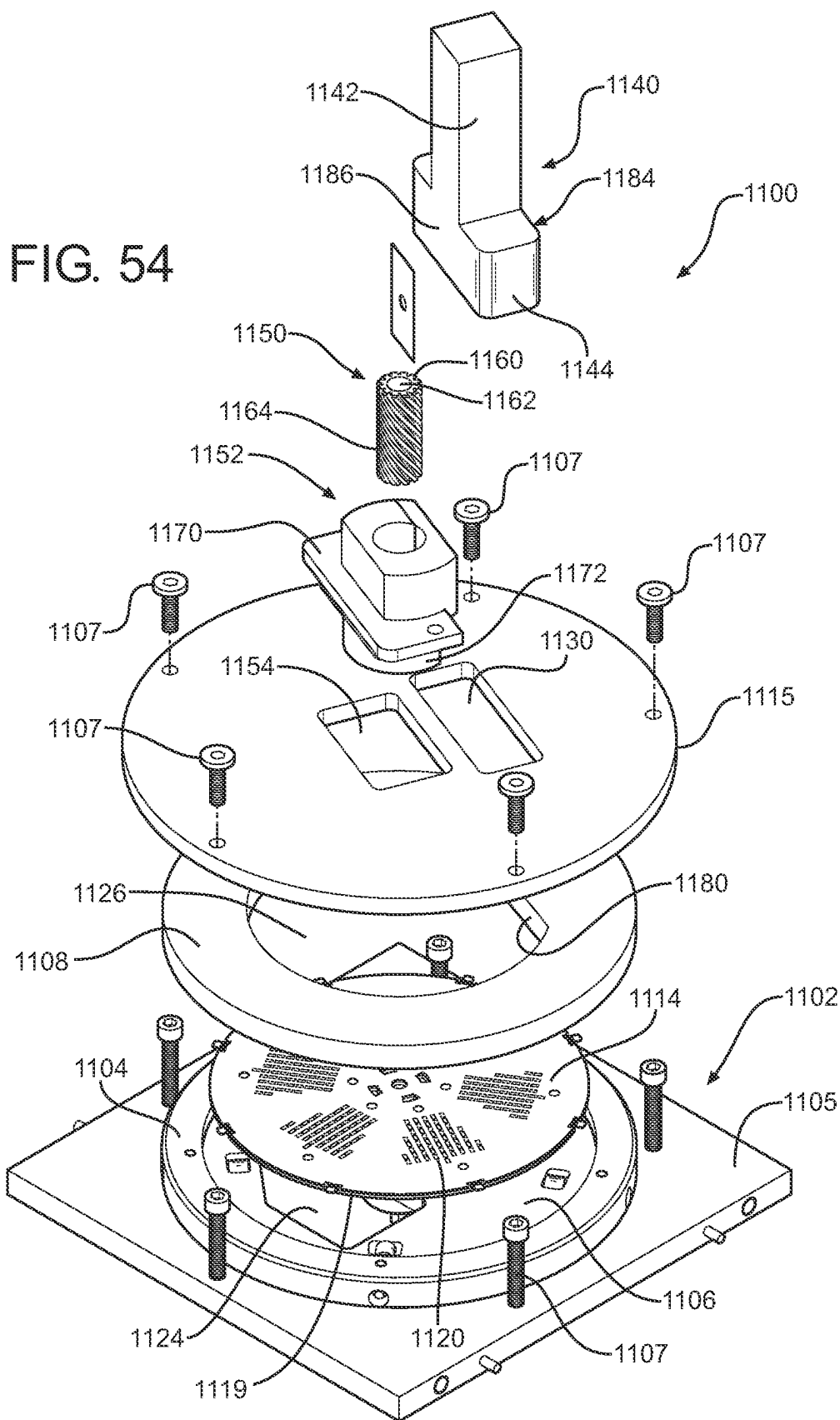
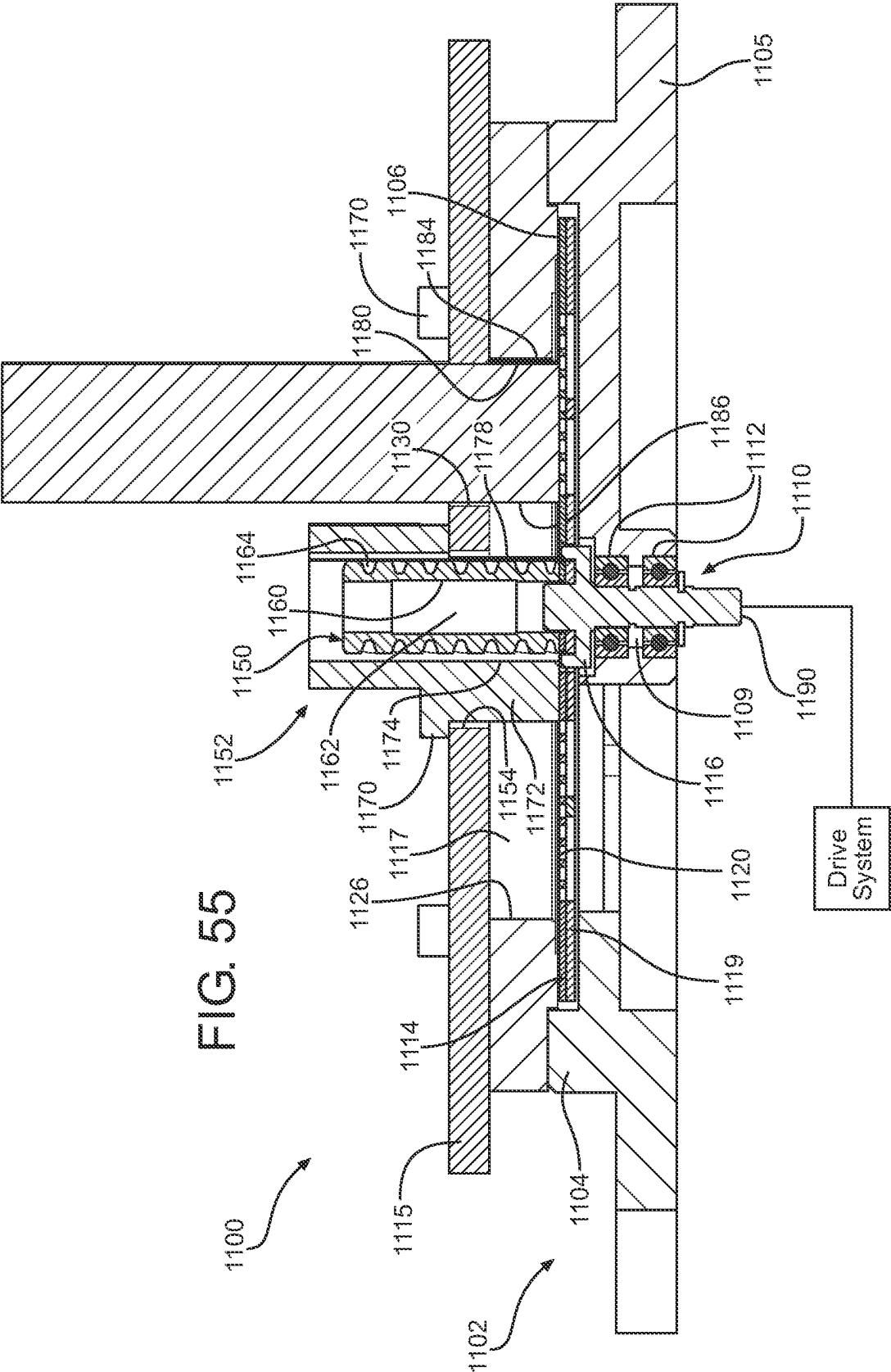


FIG. 52







1

**SYSTEM FOR PROCESSING BONE STOCK
INCLUDING A BONE CLEANING HEAD, A
BONE MILLING HEAD AND BASE THAT
POWERS BOTH THE CLEANING HEAD
AND THE MILLING HEAD**

RELATIONSHIP TO EARLIER FILED
APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/777,277, filed 30 Jan. 2020. U.S. patent application Ser. No. 16/777,277 is a continuation of U.S. patent application Ser. No. 15/915,562 filed 8 Mar. 2018, now U.S. Pat. No. 10,588,757. U.S. patent application Ser. No. 15/915,562 is a continuation of U.S. patent application Ser. No. 15/174,281 filed 6 Jun. 2016, now U.S. Pat. No. 10,045,863. U.S. patent application Ser. No. 15/174,281 is a divisional of U.S. patent application Ser. No. 14/157,975 filed 17 Jan. 2014, now U.S. Pat. No. 9,370,436. U.S. patent application Ser. No. 14/157,975 is a divisional of U.S. patent application Ser. No. 13/462,120 filed 2 May 2012, now U.S. Pat. No. 8,672,942. U.S. patent application Ser. No. 13/462,120 is a continuation of PCT Application No. PCT/US2010/055646 filed 5 Nov. 2010. PCT Application No. PCT/US2010/055646 is a non-provisional of U.S. Provisional App. No. 61/258,667 filed 6 Nov. 2009. The contents of the above-listed priority applications are explicitly incorporated herein by reference.

FIELD OF THE INVENTION

This invention is directed to a system that first cleans bone stock and, once the bone stock is cleaned, mills the bone to form bone chips.

BACKGROUND OF THE INVENTION

There are a number of different surgical procedures in which chip-sized bone is used as filler adjacent other sections of bone. For example, in a spinal fusion procedure, it is a known practice to place a compound formed out of milled bone around the rods used to hold adjacent vertebra in alignment. This compound serves as a lattice upon which the tissues forming the vertebra grow so as to form a foundation of bone around the rods. This foundation distributes the load imposed on the rods. Bone chips are also used as filler and/or growth formation lattice in orthopedic surgical procedures and other procedures such as maxillo-facial procedures.

Bone chips are used as a filler/growth formation lattice in these procedures because the material, the proteins from which the bone is formed, serves as make-up material from which the blast cells of the adjacent living bone cells form new bone.

The ideal source of stock for bone chips is the patient into which the bone chips are to be packed. This is because the patient's own bone, own tissue, is less likely to be rejected by the patient's immune system than donor bone. Accordingly, in a procedure in which bone chips are required, the bone stock often harvested from one of the patient's bones afford to lose a small section of bone, typically between 0.25 and 3 cm³. Bone that is removed from the patient for transplant into another part of the patient is referred to as autograft bone.

Once the bone is harvested, it is cleaned. After cleaning, the bone is milled to form chips. The Applicant's Assignee's U.S. Patent Pub. No. US 2009/011735 A1/PCT Pub. No.

2

WO 2009/061728 A1, BONE MILL INCLUDING A BASE AND A MILL HEAD SEPARATE FROM THE BASE, THE MILL HEAD INCLUDING A REMOVABLE CATCH TRAY, the contents of which are explicitly incorporated herein by reference, discloses a bone mill capable of converting bone stock into bone chips. This bone mill includes a base with a motor. A mill head, that contains the bone milling components, is removably attached to the base. When the head is attached to the base, the motor engages at least one of the milling components. Actuation of the motor results in a like actuation of the milling component. This results of conversion of bone stock into bone chips.

The bone mill of the incorporated by reference publication is understood to perform a more than adequate job of milling bone stock into bone chips. Nevertheless, prior to this process, it is still necessary to clean the bone to remove ligaments and other tissue that are not suitable stock for forming bone chips. Presently, surgical personnel perform this task manually using curettes, rongeurs, brushes and/or cobbs. It may take 15 minutes or more for surgical personnel to perform this process.

Moreover, to perform the cleaning process, the surgical personnel may need to firmly grasp the bone. Exerting such force on the bone may cause tearing of the gloves worn by the surgical personnel. Such tearing could result in the possibility that skin of the surgical personnel may come into direct contact with the bone. This contact can result in contamination of the bone.

SUMMARY OF THE INVENTION

This invention is related to a new and useful system for first cleaning bone and, once the bone is cleaned, milling the bone to form bone chips.

One version of the system of this invention includes: a base unit; a cleaning head; and a mill head. Internal to the base unit is a motor. The base unit also includes components for releasably holding first the cleaning head and then the mill head. Internal to the cleaning head is at least one cleaning element. In some embodiments, the cleaning element is a moveable brush, a rotating grater, and/or a rotating fluted screw. Attached to the cleaning element are features that releasably couple the cleaning element to the base unit motor. Internal to the mill head is a moveable mill element. The mill element is designed to, when actuated, mill bone stock into bone chips. The mill element includes features for releasably coupling the mill element to the base unit motor.

The system of this invention is employed to convert harvested bone stock into bone chips by first coupling the cleaning head to the base. The harvested bone stock is placed in the cleaning head. The base unit motor is actuated to cause a like actuation of the cleaning head cleaning element. The movement of the cleaning element against the bone stock removes the ligaments, muscle, connective tissue and other debris material from the surface of the bone stock.

Once the bone stock is cleaned, the cleaning head is removed from the base unit. The mill head is fitted to the base unit. The bone stock is placed in the mill head. The base unit motor is actuated cause a like actuation of the mill element. Actuation of the mill element converts the bone stock into bone chips suitable for implantation into the patient.

The integrated system of this invention includes components for both cleaning and milling the bone stock. By having the mechanized cleaning head perform the cleaning process, the need for operating room personnel to perform

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this task is eliminated. In many situations, the cleaning head cleans bone stock in less time than it takes an individual to perform the same task.

Still another advantage of the integrated system of this invention is that a single unit, the base unit, provides the motive power needed to actuate the cleaning element internal to the cleaning head and the mill element internal to the mill head. The need to provide essentially duplicative power units, one for each head, is eliminated.

In an alternative version of the invention, a single head is attached to the base. The head has a module with a brush for cleaning the bone. Below the cleaning module, the head has a mill module. Internal to the mill module are components for milling the bone stock into chips. The mill module is dimensioned to be removably coupled to the base unit. The modules have components that, when the head is attached to the base unit, couple the brush and moving mill element to the motor internal to base unit. In one embodiment of this version of the invention, the cleaning module is moveable relative to the mill module.

The alternative version of the invention is prepared for use by coupling the head to the base. The harvested bone stock is placed in the cleaning head. The base unit motor is actuated. The motor drives the brush internal to the head so as to clean the bone. Once the bone is cleaned, the cleaning module is moved to a position in which it is directed to feed port integral with the mill module. The cleaned bone is discharged from the cleaning module into the mill module. Once the bone is so transferred to the mill module, the motor is again actuated. As a result of this actuation of the motor, the consequential actuation of the mill element converts the previously cleaned bone stock into bone chips.

An assembly for cleaning bone stock includes a base. A shell is supported by the base for defining a void space for receiving the bone stock to be cleaned. At least one cleaning element is disposed in the void space. A drive assembly is coupled to the at least one cleaning element to actuate the at least one cleaning element to clean the bone stock. In some embodiments, the cleaning element is a rotating brush, a rotating grater, and/or a rotating fluted screw.

A further advantage of the above alternative versions of the invention is that once the bone stock is placed in the cleaning head, the need for surgical personnel to handle bone is substantially eliminated.

BRIEF DESCRIPTION OF THE DRAWINGS

The invention is pointed out with particularity in the claims. The above and further features and advantages of this invention are better understood by the following detailed description taken in conjunction with the following drawings in which:

FIG. 1 depicts the basic components of the integrated system for cleaning and milling bone of this invention;

FIG. 2 is a perspective view of the base unit;

FIG. 3 is a cross sectional view of components internal to the base unit.

FIG. 4 is a perspective view of the top of the base unit;

FIG. 5 is a schematic and block diagram of the memories internal to the base unit and cleaning head and the components over which data are read from and written to these memories;

FIG. 6 is a simplified block diagram of the circuits internal to the control console;

FIG. 7 is an exploded view of the components forming the cleaning head;

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FIG. 8 is a perspective view of the lower shell of the cleaning head;

FIG. 9 is a view of the bottom surface of the lower shell of the cleaning head;

FIG. 10 is a plan view, looking down, of the cleaning head lower shell;

FIG. 11 is a cross sectional view of the cleaning head upper shell;

FIG. 12 is a perspective view of the bottom surface of the substrate of the cleaning head lower brush;

FIG. 13 is a side view of the cleaning head lower brush;

FIG. 14 depicts data fields in the memory of the RFID fitted to the cleaning head lower brush substrate;

FIG. 15 is a side view of the cleaning head upper brush;

FIG. 16 is an exploded view of the system mill head;

FIG. 17 is a perspective view of the bottom shell of the mill head;

FIG. 18 is a plan view of the mill element, here a cutting disc, internal to the mill head;

FIGS. 19A through 19C, when assembled together, form a flow chart of the steps performed to both clean bone stock and convert the bone stock into chips using the system of this invention;

FIG. 20 is an exploded view of an alternative cleaning head of this invention;

FIG. 21 is a cross sectional view of the upper shell and shaft of the mill head of FIG. 20;

FIG. 22 is a plan view looking upwardly, of the bottom of the upper shell, shaft and spacer ring of the mill head of FIG. 20;

FIG. 23 is a perspective view of the underside of the lower brush substrate of the mill head of FIG. 20;

FIG. 24 is a perspective view of the drive assembly of the mill head of FIG. 20;

FIG. 25 is a perspective view of the drive shaft of the mill head of FIG. 20;

FIG. 26 is an exploded view of the integrated cleaning and mill head of this invention;

FIG. 27 is a side plan view of the lower plate of the head of FIG. 26;

FIG. 28 is a perspective view of the upper plate of the head of FIG. 26;

FIG. 29 are a flow chart of the process steps executed during the operation of the head of FIG. 26;

FIG. 30 is an exploded view of an alternative brush of the system of this invention;

FIG. 31 is a side plan view of the substrate/superstrate of the brush of FIG. 30.

FIG. 32 is a exploded view of a second alternative cleaning head of this invention;

FIG. 33 is a top plan view of the base of the alternative cleaning head of FIG. 32;

FIG. 34 is a cross sectional view along line 34-34 of the base of FIG. 33;

FIG. 35 is a bottom plan view of the cap of the cleaning head of FIG. 32;

FIG. 36 is a perspective view of the bottom of the cap of the cleaning head of FIG. 32;

FIG. 37 depicts how the cleaning head of FIG. 32 is mated to the complementary mill head to facilitate the transfer of cleaned bone stock from the cleaning head to the mill head;

FIG. 38 is a cross-sectional perspective view of a third alternative cleaning head of this invention comprising a plurality of rotating fluted screws and a central agitator;

FIG. 39 is an exploded view of the alternative cleaning head of FIG. 38;

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FIG. 40 is cross-sectional view of one of the rotating fluted screws of the alternative cleaning head of FIG. 39;

FIG. 41 is an exploded view of the rotating fluted screw of FIG. 40;

FIGS. 42 and 43 are top and bottom perspective views of the spindle used to drive the rotating fluted screws of the alternative cleaning head of FIG. 38;

FIG. 44 is a perspective view of an axle on which the rotating fluted screws rotate in the cleaning head of FIG. 38;

FIG. 45 is a perspective view of shaving block of the alternative cleaning head of FIG. 38;

FIG. 46 is a perspective view of a plug for the alternative cleaning head of FIG. 38;

FIG. 47 is a partial perspective view illustrating the void space and shaving block with rotating flute screw of the alternative cleaning head of FIG. 38;

FIG. 48 is a cross-sectional perspective view of a fourth alternative cleaning head of this invention comprising a rotating inner basket and a rotating brush;

FIG. 49 is an exploded view of the alternative cleaning head of FIG. 48;

FIG. 49A is a partial perspective view an alternative gear train for the alternative cleaning head of FIG. 48;

FIG. 50 is an exploded view of a fifth alternative cleaning head with a rotating grater and a plunger;

FIG. 51 is a cross-sectional view of the alternative cleaning head of FIG. 50;

FIG. 51A is a top perspective view of the rotating grater of FIG. 50;

FIG. 52 is an exploded view of a sixth alternative cleaning head of this invention comprising a rotating grater and an impingement plate;

FIG. 53 is a top perspective view of the alternative cleaning head of FIG. 52 illustrating the rotating grater and the impingement plate;

FIG. 54 is an exploded view of a seventh alternative cleaning head of this invention comprising a rotating grater, rotating fluted screw, and plunger;

FIG. 55 is a cross-sectional view of the alternative cleaning head of FIG. 54; and

FIG. 56 is a top perspective view of the alternative cleaning head of FIG. 54 illustrating the rotating grater and rotating fluted screw.

DETAILED DESCRIPTION

I. Overview

FIG. 1 illustrates the basic components of integrated system 50 of this invention for cleaning and milling bone stock. System 50 includes a base unit 52. Internal to the base unit 52 is a motor 54, (FIG. 3). A cleaning head 56 is removably attached to the base unit 52. Internal to the cleaning head are brushes 58 and 59 (FIG. 7). Cleaning head 56 is configured so that, when the cleaning head 56 is attached to the base unit 52, brush 58 is connected to the motor 54 so as to be actuated by the motor 54. The system 50 includes a mill head 60 that, like cleaning head 56, is configured to be removably attached to the base unit 52. A mill element 62 (FIG. 16), sometimes referred to as a cutting device, is moveably mounted inside, the mill head 60. Mill element 62 includes features that, when the mill head 60 is mounted to the base unit 52 couple the mill element to the motor 54.

Also part of system 50 is a control console 66. Control console 66 supplies the energization signals to the motor 54 that actuate the motor 54. Cable 67 connected between the

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base unit 52 and console 66 contains the conductors (not illustrated) over which energization signals are supplied from the console 66 to the motor 54.

System 50 of this invention is used by coupling the cleaning head 56 to the base 52. Harvested bone stock is placed in the cleaning head 56. The motor 54 is actuated so as to result in a like actuation of brush 58. The action of the brush 58 against the bone stock strips the soft tissue and other debris from the bone stock. The cleaning head 56 is removed from the base unit 52 and the mill head 60 is fitted to the base unit. The cleaned bone stock is placed in the mill head 60. Base unit motor 54 is again actuated so as to result in a like actuation of the mill element 62. As a consequence of the actuation of the mill element 62, the cleaned bone stock is milled into bone chips suitable for implantation into the patient.

II. Base Unit and Control Console

The base unit 52, now described by reference to FIGS. 2 and 3, includes a circular foot 70. A leg 72 having a circular cross section extends upwardly from foot 70. Leg 72 is tubular in shape. A pedestal 74 is disposed on top of leg 72. The pedestal 74 tapers outwardly from the leg 72. Pedestal 74 has a generally circular top surface 76. The pedestal is further formed to have a lip 78 that extends upwardly around the outer perimeter of the top surface. The outer circumference of lip 78, which is the outer circumference of the pedestal 74, is less than that of the circumference of foot 70 and larger than that of leg 72. Pedestal 74 is further formed so as to have an opening 80 in the center of top surface 76.

While pedestal 74 is generally circular in shape, as best seen in FIG. 4, a notch 83 extends inwardly from the outer perimeter. Notch 83 thus forms a break in lip 78. In the illustrated version of the invention notch 83 extends to center opening 80. The pedestal is further formed to include a number of arcuately spaced apart teeth 84. Each tooth 84 extends upwardly from the outer perimeter of the pedestal top surface 76 adjacent lip 78. Pedestal 74 is further formed to have a rectangular opening 85 that is spaced away from both center opening 80 and notch 84.

Two retention arms 86, seen best in FIGS. 3 and 4, are pivotally mounted to the pedestal 74. Retention arms 86 are mounted to the pedestal in cutouts formed in the lip 78, (cutouts not identified). Each retention arm has a finger 88 that, when the arm is at rest, extends over a portion of the pedestal top surface 76. When the arms 86 are so positioned, the arms are in the "locked" state. Each retention arm 86 has a tab 89 located below the pedestal 74. By depressing the tab 89 inwardly, towards the underside of the pedestal 74, the arm 86 is pivoted outwardly so as to pivot the associated finger 88 away from its position over the pedestal top surface 76. When the arms 86 are so positioned, the arms can be considered in the "release" state. A spring 90 disposed between inner surface of the pedestal 74 and each arm 86, normally holds each arm in the locked state.

Referring to FIGS. 3 and 4, motor 54 includes a shaft 96 also disposed in the center hollow of leg 72. Shaft 96 extends upwardly toward pedestal center opening 80. A gear head at the top of shaft 96 (gear head not identified) engages a gear train 98 disposed in leg 72 above the motor 54. Gear train 98 steps down the rotational of speed of the rotational moment output by motor shaft 96. The gear assembly 98 has an output shaft 102 disposed in the pedestal center opening 80 below the top surface 76. Output shaft 102 is tubular in shape. Shaft 102 is formed to have two diametrically

opposed oval openings **103** (one shown in FIG. 2) that extend longitudinally along the shaft.

In some versions of the invention, motor **54** and gear train **98** are collectively provided so that the gear train output shaft **102** can rotate at speeds between 100 and 500 RPM. These speeds are the under load speed when bone stock is disposed in either the cleaning head **56** or mill head **60**. For reasons apparent below, the motor **54** and gear train **98** are designed to drive the output shaft **102** in an oscillatory pattern.

A drive spindle **104** is disposed in output shaft **102**. The drive spindle **104** includes a stem **106**. Above stem **106**, spindle **104** is shaped to have a disc-shaped head **108**. A number of different components extend upwardly from the top surface of the spindle head **108**. One of these components is an alignment pin **110**. The alignment pin **110** is coaxial with the longitudinal axis of the spindle **104** and extends upwardly from the center of the head **108**. Pin **110** is shaped so that the lower portion, the portion that extends upwardly from the spindle head **108**, has a cylindrical shape. The top portion of alignment pin has a shape of a cone with a flattened tip. (The individual sections of alignment pin **110** are not identified.)

Four equiangularly spaced apart alignment teeth **112** also extend upwardly from the top surface of the spindle head **108**. Teeth **112** are located around the outer perimeter of the spindle head **108**. The arcuate outer surfaces of teeth **112** are flush with the outer surface of the spindle head **108**. Each tooth **112** has a pair of inwardly tapered side surfaces and an arcuate inner surface. (Surfaces not identified.) Teeth **112** do not extend as far above the spindle head **108** as does alignment pin **110**.

Spindle **104** is dimensioned and positioned so that stem **106** is slidably mounted in the bore that extends through the gear assembly output shaft **102**, (bore not identified). A pin **114** extends through a bore **107** in the spindle stem **106**. The opposed ends of the pin **114** are seated in diametrically opposed openings **103** formed in shaft **102**. Pin **114** holds the drive spindle **104** to the shaft **102** so that the spindle rotates in unison with the shaft and is able to move longitudinally relative to the gear assembly **98**.

A spring **118** is disposed in the output shaft **102** below spindle stem **106**. Spring **118** is a wave spring. One end of spring **118** is seated on the annular step internal to the output shaft **102** (step not identified). The opposed end of spring **118** is disposed against the bottom end of spindle stem **106**. Spring **118** exerts an upward bias on spindle stem **106**. This force, which can be overcome by the application of manual force, normally displaces the spindle **104** so that the head **108** is urged away from the pedestal top surface **76**.

A spring biased, normal open press button switch **120** is mounted to the base unit foot **70** (FIG. 2). A socket **122**, shown symbolically in FIG. 5, receives cable **67** from control console **66**. Internal to foot **70** is a circuit board **124** (FIG. 3). Mounted to the circuit board **124** are components that function as the interface between switch **122** and the conductors that extend to socket **122**. Also disposed on circuit board **124** are components that function as interfaces between the power conductors internal to cable **67** and the conductors that extend to the windings of the motor **54**. The specific structure and configuration of these components as well as of the conductors that extend to motor **54**, switch **120** and socket **122** are neither illustrated or part of this invention.

Also disposed on circuit board **124** is a non-volatile memory **126** seen in FIG. 5. Memory **126** contains data describing the base unit **52**. These data include data identi-

fying the type of device; here, that the device is a bone cleaner/bone mill base unit. These data also contain data useful for supplying energization signals to the motor **54**. These latter data include data indicating the speed range of the motor and the currents the motor should draw. A more complete list of the data that may be contained in memory **126** can be found in the Applicants' Assignee's U.S. Pat. No. 6,017,354, INTEGRATED SYSTEM FOR POWERED SURGICAL TOOLS, the contents of which is explicitly incorporated herein by reference. In one version of the invention, memory **126** is a radio frequency identification device (RFID) and is identified as such in FIG. 5. An antenna (coil) **125** is connected to the memory **126**. A coil **127** is connected to the conductors **123** that extend from the socket **122**. Signals between the control console **66** and the memory **126** are inductively exchanged between memory coil **125** and base unit coil **127**.

Base unit **52** also includes an assembly for reading non volatile memories integral with the cleaning head **56** and mill head **60**. This assembly includes a coil **130**. Coil **130** is disposed in pedestal opening **85**. Coil **130** is encased in a block **132** (FIG. 4) disposed in opening **85**. Coil **130** is series connected to coil **127**. Block **132** is formed from material permeable to RF energy and that can withstand the rigors of autoclave sterilization.

FIG. 6 is a block diagram of some of the basic assemblies internal to the control console **66**. One of these assemblies is the power supply **160**. The power supply **160** converts the line signal into a DC voltage suitable for application to the windings internal to the base unit motor **54**. Power supply **160** also produces AC and DC voltages by the other components internal to the control console and internal to the control console **66** and base unit **52**. For reasons of simplicity, the only the connection shown out of the power supply **160** is the VDC that is applied to the motor windings. This VDC signal is applied to a motor driver **162**. Motor driver **162** selectively ties the individual windings of the motor to either the VDC signal or ground. This is the commutation the current flowed through the motor windings (windings not illustrated). The motor driver **162** is connected to cable **67** by a socket **163** integral with the console **66**.

Motor driver **162** selectively ties the motor windings to the VDC signal or ground based on both feedback signals from the windings and command signals from a display controller **164**. The display controller **164** generates command signals that indicate the speed at which the motor should run, the maximum currents the windings should draw and the sequence in which the voltages are applied across the windings. These last data are used to regulate the direction in which the motor shaft rotates. Display controller **164** generates these command signals based on both user entered data and stored data that indicates the characteristics of the energization signals that are to be applied to the motor.

The data indicating the characteristics of the energization signals that are to be applied to the base unit motor **54** are retrieved from different sources. These data may be stored in memory **166** internal to the control console **66**. These data may be retrieved from the memory **126** internal to the base unit **52**. Alternatively, these data may be retrieved from a memory **270** internal to the cleaning head **56** (FIG. 5) or a memory **320** (FIG. 17) internal to the mill head **60**.

To read the data in the base unit memory **126**, cleaning head memory **270** and mill head memory **320** control console **66** includes a RFID interface **168**. RFID interface **168** is connected to the display controller **164**. In response to command signals from the display controller **164**, the RFID interface sends read request signals to the comple-

mentary memories. In response the read request, the memory **126**, **270** or **320** writes out the stored data. Interface **168** converts these data signals into digital signals that are interpreted by the display controller **164**.

Control console **66** also includes a touch screen display **174**. Display controller **164** generates both data images and images of command buttons for presentation on the display **174**. The display controller **164** receives the signals when an individual presses the display buttons. In response to the depression of the buttons, display controller **164** generates the appropriate commands to cause the user-requested operation of the base unit motor **54**.

A more detailed understanding of the structure of the control console **66** can be found in the Applicants' Assignee's U.S. Pat. No. 7,422,582, CONTROL CONSOLE TO WHICH POWERED SURGICAL HANDPIECES ARE CONNECTED, THE CONSOLE CONFIGURED TO SIMULTANEOUSLY ENERGIZE MORE THAN ONE AND LESS THAN ALL OF THE HANDPIECES, the contents of which are explicitly incorporated herein by reference.

III. Cleaning Head

As seen by reference to FIG. 7, the cleaning head includes opposed lower and upper shells **192** and **194**, respectively, that are releasably coupled together. A lower brush **58** is rotatably and removably disposed in the lower shell **192**. An upper brush **59** is disposed in the upper shell **194**. While the upper brush **59** is removably and, to a limited degree moveably mounted with regard to the upper shell, the upper brush does not rotate. A cap, (not illustrated) may be removably fitted over the exposed top of upper shell **194**.

Lower shell **192** is formed from aluminum or other material that can withstand repetitive autoclave sterilization. As seen best in FIGS. 8, 9 and 10, the lower shell is shaped to have a disc shaped base **206**. Base **206** has an outer diameter that allows the shell **192** to be slip fitted in the void space immediately above pedestal top surface **76** within lip **78**. Shell base **206** has a center opening **208**. Center opening **208** has a diameter that is approximately 2 mm larger than the diameter of spindle head **108**. Four equiangularly spaced apart notches **210** extend inwardly and upwardly from the downwardly directed face of the shell base. Notches **210** are dimensioned so that when the cleaning head **56** is fitted to base unit **52**, pedestal teeth **84** are able to seat in the notches.

Located radially outwardly from opening **208**, base **206** is formed to have an annularly extending groove **212**. Groove **212** is positioned so that, when the cleaning head **56** is seated in the pedestal void space **79**, groove **212** extends over the space above coil **130**. A hole **213** extends upwardly from the base of groove **212**.

A ring **214** is integrally formed with and extends upwardly from the outer perimeter of shell base **206**. The outer diameter of ring **214** is coincident with the outer diameter of base **206**. Ring **214** defines a cylindrical void space (not identified) within the lower shell **192**. The outwardly directed face of the shell base **206** functions as the base of this void space. Lower shell **192** is further formed to have two additional notches **220** that are diametrically opposed from each other. (Only one notch **220** seen in FIG. 8.) Notches **220** extend inwardly from the outer cylindrical surface of the shell **192** at a location above the bottom of the shell. More particularly, shell **192** is formed so that when the shell is seated in pedestal void space **79** and teeth **84** are in notches **210**, notches **220** are positioned so the fingers **88**

integral with the pedestal retention arms **86** can seat against the shell ring surfaces that define the bases of the notches **220**.

Lower shell **192** is further formed to have three equiangularly spaced apart slots **224** in the exposed circular, outwardly directed face of ring **214**. Each slot **224** has what can be generally described as a keyhole shape. That is, each slot **224** has a circular section and a section that resembles a section of a curve that extends away from the circular section (individual sections not identified). The width across each curved segment section is less than the diameter of the circular section. More particularly, lower shell **192** is formed so that opposed ledges **226** extend over the slot curved sections to provide the appearance of these sections having a width less than diameter of the slot circular sections. Below the ledges **226** the common widths of slots is constant.

Shell base **206** is further formed to have a circular groove **230** in the outwardly directed face of the base. Hole **213** opens into the surface of the base **206** that forms the base of the groove **230**.

A coil **232**, seen in FIG. 5, is disposed in base groove **212**. Not seen is the ring formed of PEEK, polyetherimide or other sterilizable plastic in which coil **232** is embedded. Accordingly, both the coil **232** and the ring are seated in groove **212** so as to be flush with the downwardly directed face of shell base **206**. Conductors **234** extend from coil **232** through shell base hole **213**. Conductors **234** extend to a coil **236** seated in shell base groove **230**. Coil **236** is embedded in the same type of ring (not illustrated) in which coil **232** is embedded. Coil **236** and its complementary ring are seated in groove **230** so as to be flush with the adjacent outwardly directed face of shell base **206**.

The cleaning head upper shell **194**, now described by reference to FIGS. 7 and 11, is formed from the same material from which lower shell **192** is formed. Upper shell **194** includes a circular plate **240**. Plate **240** has the same outer diameter of the lower shell **192** and functions as the superstructure of the upper shell **194**. An annular skirt **242** formed integrally with the plate **240**, extends downwardly from the outer perimeter of the plate. The bottom surface of plate **240** and skirt **242** thus define a void space **246** internal to the upper shell **194** that extends upwardly from the outer face of the skirt to the downwardly directed surface of plate **240**.

Three equiangularly spaced apart pins **248** extend downwardly from the downwardly directed face of skirt **242** (Only one pin **248** seen in both FIGS. 7 and 11.) Each pin **248** has a narrow diameter stem (not identified) and a large diameter head (not identified). The pin heads are dimensioned to pass through large diameter opening of one the lower shell slots **224** but not the narrow diameter curved section of the slot. Upper shell pins **248** are thus the fastening members that complement the lower shell slots **224** so as to removably hold the shells **192** and **194** together without the aid of other fastening components. A cylindrical boss **250** formed integrally with the plate **240** extends upwardly from the exposed upper surface of the plate. Boss **250** is centered on the longitudinal axis of the plate **240**. A bore **252** extends through the plate **240** and overlying boss **250**. Bore **252** is centered on the longitudinal axis of the plate **240** and boss **250**. Bore **252** has a non-circular cross sectional shape. In the illustrated version of the invention, the bore **252** has a cross-sectional shape that is square. A threaded bore **249** extends laterally from the side of boss **250** into bore **252**.

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Three load pins **253** are slidably mounted in through holes **251** formed in plate **240**. Holes **251** are equiangularly spaced apart from each other and located outwardly of shell boss **250**. Each pin **254** extends into void space **246**. A helical spring **254** is disposed around the section pin **253** disposed in the shell void space **246**. Springs **254** urge the pins **253** toward the lower shell **192**. Not seen are the heads of the pins against which the springs **254** abut and the feet of the pins that prevent the pins from falling out of the holes **251**.

FIGS. **12** and **13** illustrate the features of the cleaning head lower brush **58**. Brush **58** is formed to have disc-shaped substrate **260** formed of metal or a sterilizable plastic such as a glass-filled nylon. Substrate **260** has an outer diameter that is generally at least 0.5 mm less than the diameter of the void space defined by the inner wall of lower shell ring **214**. Thus brush **58** is able to float, laterally shift position, within the lower shell **192**. While the opposed upper and lower faces of substrate **260** are generally planer and parallel, a number of indentations extend upwardly from the inner face lower face of substrate **260**. One of these indentations is a closed-end bore **262** that is centered over the longitudinal axis of the disc. Bore **262** has a diameter that allows the base unit drive spindle alignment pin **110** to seat therein. Four equiangularly spaced apart notches **264** are the other indentations that extend upwardly from the bottom-directed face of lower brush substrate **260**. Notches **264** are positioned so that when spindle pin **110** is seated in substrate bore **262**, spindle teeth **112** seat in the notches. Substrate **260** is also formed to have a single notch **266**. Notch **266** is positioned so that when lower brush **58** is seated in the lower shell **192**, the notch **266** is disposed over coil **236**.

Lower brush **58** includes a number of bristles **268** that extend upwardly from the upwardly directed face of substrate **260**. The bristles **268** are formed from a stainless steel. Bristles **268** are attached to substrate by an adhesive **267** such as an epoxy adhesive. In manufacture, adhesive **267** is initially applied over upwardly directed face of substrate **260**. Before the adhesive cures, the bristles **268** are planted in the adhesive **267**.

An RFID chip **270**, illustrated in FIG. **5**, is disposed in substrate notch **266**. RFID chip **270** functions as the memory for the lower brush **194**. Attached to RFID chip **270** is a coil **271**. RFID chip **270** and coil **271** are encapsulated in a plastic block **269** (shown in phantom in FIG. **5**). Block **269** is formed from a plastic that allows inductive signal exchanged between shell coil **236** and brush coil **271**.

FIG. **14** illustrates different data stored in the actual memory **272** internal to the RFID chip **270**. The data in memory **272** include a device identification field **273**. The data in field **273** identifies that the associated device is a bone cleaning brush. If there are a number of different bone cleaning brushes, the data in field **273** identifies the specific kind of brush. A use field **274** contains data indicating whether or not the brush is useable. It is anticipated that, owing to the expenses associated with post-use sterilization, each brush is a use once brush. Accordingly, field **274** may be a single bit field that is initially set to contain data indicating that the brush **58** can be used.

Memory **272** also contains minimum, default and maximum motor speeds fields **275**, **276** and **277**, respectively. The data in the minimum and maximum motor speed fields **275** and **277**, respectively, indicate, respectively, the preferred minimum and maximum speeds at which the base unit motor **54** should be driven to rotate the lower brush **58**. The default speed field **276** indicates that speed at which the motor is to be driven in the event the personnel operating the system do not set any other speed.

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A direction field **278** indicates the direction in which base unit motor **54** should be driven. Typically, the data in field **278** indicates if the motor is to be driven in a single direction or in an oscillatory mode. If the motor, actually, the lower brush **58**, is to be driven in a single direction the direction of rotation is irrelevant. If the lower brush **58** is to be oscillated back and forth, field **278** may include additional data indication through which how many degrees the brush should rotate before the direction of rotation is reversed.

In some versions of the invention, field **278** may contain data indicating a sequence of rotation in which the lower brush **58** upon actuation. One sequence may include an initial rotation of the brush through 10 rotations in one direction followed by rotation in oscillatory pattern wherein in each phase of the oscillation the brush rotates 2 rotations (720°) in one direction before the direction of rotation is reversed.

The upper brush **59**, now described with reference to FIG. **15**, includes a disc shaped superstrate **284**. Superstrate **284** is formed from the same material from which lower brush substrate **260** is formed. The superstrate **284** has an outer diameter equal to the lower brush substrate **260**. A post **286** extends upwardly from the longitudinal center axis of superstrate **284**. Post **286** is dimensioned to slidably fit in upper shell bore **252**.

Bristles **290** extend downwardly from the bottom facing surface of superstrate **284**. A layer of adhesive **288** holds the bristles **290** to superstrate **284**.

A ball plunger **257** (FIG. **7**) is fitted in upper shell bore **249**. Ball plunger **257** is set to press against post **286** to prevent upper brush **59** from falling out of upper shell **194**. In some versions of the invention, brush post **286** is formed with a detent (not illustrated) for receiving the plunger head.

IV. Mill Head

FIG. **16** illustrates the basic components of the mill head **60**. There are bottom and top shells **302** and **304**, respectively. When assembled together, shells **302** and **304** form the housing of the mill head **60**. Mill element **62** is a disc shaped member sandwiched between shells **302** and **304**. Top shell **304** is formed to have an open ended feed sleeve **306**. The center space defined by sleeve **306** opens into the space in which mill element **62** is seated. A plunger **308** is slidably fitted in feed sleeve **306**.

Fitted to top shell **304** is an impingement plate **308** (shown in phantom as a rectangular bar). Impingement plate **308** is mounted to the top shell so as to be near the base of the feed sleeve and immediately above the mill element **62**.

Bottom shell **302** is formed to have an opening **309** immediately bellow the feed sleeve **306**. A removably catch tray **310** is slidably fitted to the bottom shell **302**. Catch tray **310** is fitted to bottom sleeve **306** to receive bone chips discharged from mill head **60** through opening **309**.

Mill head bottom shell **302**, seen in FIG. **17**, is generally circularly shaped. More particularly, shell **302** is dimensioned to seat within the base unit void space **79**. Shell **302** is formed to have four equiangularly spaced apart notches **312** that extend inwardly from the outer perimeter of the shell. Notches **312** are positioned so that when mill head **60** is seated on the base unit pedestal **74**, pedestal teeth **84** seat in the notches **312**. Shell **302** has a center located opening **314** dimensioned to receive base unit drive spindle **104**.

Opening **309** extends inwardly from the perimeter of the bottom shell **302**. The bottom shell **302** is further formed to have two rails **316** located on the opposed sides of opening **309**. Rails **316** are dimensioned to allow the catch tray to be

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slidably held to the shell 302 below opening 309. For reasons not relevant to the present invention, the rails 316 are shaped differently from each other.

An RFID chip 320, shown as a phantom rectangle in FIG. 17, is embedded in bottom shell 302. A complementary coil 322, also shown in phantom is attached to RFID chip 320 and also embedded in shell 320. Coil 322 is located within the shell 302 so that when the mill head 60 is disposed on the base unit 52, the coil 322 is disposed over base unit coil 130. The memory of RFID 320 contains data similar to that contained in cleaning head RFID 270. Specifically, there is a data field in which data identifying the type of device, a mill head 56, are stored. There is also at least one data field indicating the speed at which the base unit motor 54 should operate when the mill head 56 is attached. Typically, the mill element 62 is intended to be driven at a single speed. Accordingly, in a number of versions of the invention, RFID 320 only contains a single data field containing data indicating the speed at which mill element 62 is to be rotated. Also, the mill element 62 is typically only rotated in a single direction. Accordingly, RFID 320 likewise typically does not contain any data containing instructions regarding a forward/reverse sequence for driving the mill element 62.

Returning to FIG. 16, it can be seen that top shell 304, like bottom shell 302 is generally disc shaped. The top shell 304 is further shaped to have the same outer diameter as the bottom shell 302. Collectively, shells 302 and 304 are shaped so that when the mill head 60 is seated in base unit void space, the outer face of the top shell 304 is slightly above the pedestal lip 78. The top shell 304 is formed so as to have two diametrically opposed notches 328; one notch seen in FIG. 16. Notches 328 are positioned so that when the mill head 60 is fitted to the base unit pedestal 74, the base unit retention arm fingers 88 are able to seat in the notches 328.

Mill element 62, seen best in FIG. 18, may be formed from metal such as 410 Stainless Steel. In the illustrated version of the invention, the mill element 62 is generally in the form of a planar disc. In the illustrated version of the invention, mill element 62 is sometimes referred to as a cutting disc. Collectively, the components forming the mill head 60 are dimensioned so that the mill element 62 can engage in lateral movement, as well as some up-and-down movement, within the mill head housing.

The mill element 62 is further shaped to have a center-located hole 332. Hole 332 is dimensioned to receive the alignment pin 110 integral with the base unit drive spindle 104. Located around hole 332, mill element 62 is formed to have four equiangularly shaped apart openings 334. Each opening 334 is shaped to receive a separate one of the teeth 112 integral with spindle 104. Accordingly, openings 334 are arcuately shaped. The circle defined by the outer circumference of openings 334 is less than bottom shell opening 314.

The mill element 62 is further formed to have a number of cutting scallops 336. Integral with and longitudinally axially aligned with each cutting scallop 336, the cutting disc has a through opening 338. More particularly, the mill element 62 is formed so that each cutting scallop 336 extends above the adjacent top surface of the element. The scallops 336 are milled to define a cutting edge 340 that forms a perimeter of the adjacent opening 338.

Impingement plate 308 is formed from material against which bone stock can be pressed without causing the fracture of the plate. In some versions of the invention, this material is 304 stainless steel. The impingement plate 308 is fitted to the mill head top shell 304 so as to be located to one

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side of the opening through feed sleeve 306 and so as to be above the mill element 62. Impingement plate 308 is further positioned so that, as the mill element 62 rotates, first an element opening 338 followed by the cutting edge-defining scallop 336 that defines the opening rotates towards and under the plate 308.

V. Operation

System 50 of this invention is used to clean and mill bone that is harvested to serve as stock from which bone chips are formed. To prepare the system for use, a lower brush 58 is fitted in lower shell 192 and an upper brush 59 is fitted in upper shell 194. The harvested bone is placed against the lower shell bristles 268. The upper shell 194 is coupled to the lower shell so that that harvested bone stock is sandwiched between the lower and upper brushes 58 and 59, respectively. To couple the shells 192 and 194 together, the upper shell pins 248 are rotating held in the lower shell slots 224. Collectively, these steps are called out as step 350 in FIG. 19A.

Once the two shells 192 and 194 are coupled together, the post 286 may be pressed downwardly to unlock the post from the ball plunger 257. This allows upper brush 59 to move longitudinally within cleaning head 56. Pins 253, which are urged against the upper brush superstrate 284 by springs 254, provide a force in addition to gravity that presses the upper brush bristles 290 against the bone to be cleaned. It should be understood that owing to the relative dimensions of the lower shell 192 and brush 58, the brush may shift laterally within the shell.

The cleaning head 56 is then mounted to base unit 52, step 351. More particularly, the cleaning head 56 is positioned so that the head lower shell 192 is positioned on pedestal top surface 76 so that pedestal teeth 84 seat in shell notches 210. Base unit retention arms 86 are set so that fingers 88 seat in pedestal notches 220. This seating of the retention arm fingers 88 in the lower shell notches 220 is what holds cleaning head 56 to the base unit 52. The seating of base unit teeth 84 in cleaning head notches 210 prevent rotation of the head 56 relative to the base unit 52.

As part of the seating of the cleaning head 56 on the base unit 52, step 351, drive spindle 108 extends through the lower shell opening 208. The spindle alignment pin 110 seats in lower brush opening 262. The seating of pin 110 in opening 262 may laterally shift the lower brush 58 in the lower shell 192 so that substrate openings 264 define a circle that is aligned over spindle drive teeth 112.

Also as part of preparing the system 50 for use, the base unit 52 is connected by a cable 67 to control console 66. The control console 66 is actuated. (Steps not illustrated.)

Once the system is actuated, the control console display controller 164, through RFID interface 168, reads the contents of the base unit memory 126, step 354 of FIG. 19A. Base on unit-identifying data in memory 126, in step 324, display controller 164 determines that the device attached to control console 66 is a bone cleaner/bone mill base unit 52, step 356. Based on this determination, display controller 164, as part of step 356, recognizes that it must determine the type of head 56 or 60 attached to the base unit before it determines the characteristics of the energization currents that should be applied to the base unit motor 54. Accordingly, in a step 358 the display controller 164, through the RFID interface 168, reads out the data in the cleaning head brush RFID 270, step 360. These data are read as a consequence of the inductive signal exchange between base unit coil 130, shell coils 232 and 236 and brush coil 271.

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It should be understood that there is always the possibility that either cleaning head **56** or mill head **60** once attached to the base unit **52** will be removed. Therefore, periodically throughout the time the base unit **52** is attached to the control console **66**, the display controller **164** will request the RFID interface to conduct a basic interrogation to determine if the head **56** or **60** previously attached to the base unit **52** is still attached. This integration process often involves the outputting of a by the RFID interface **168** of a request that any attached RFID write back some basic identifying data. These data, if written back to the display controller **164** indicate that the device that was attached to the base unit **52** is still attached. Thus, as long as the base unit is connected to the console **66**, data are written from the base unit memory **126** back to the console. The same read out of data occurs as long as the mill head **56** is attached to the base unit **52**.

The absence of responses to these write request is interpreted by the display controller **164** that the base unit **52** is no longer attached to the console or the system component, the cleaning head or the mill head, is not longer attached to the base unit. While not represented in the flow chart of FIGS. **19A** through **19C**, it should be understood that these basic interrogation processes are repetitively executed as long as the control console **66** is turned on and connected to the base unit **52**.

Based on the data the brush RFID **270**, display controller configures system **50** for operation to clean the bone, step **362**. Step **362** involves generating a sequence of instructions that indicate the speed and direction in which the motor should rotate. As part of step **362**, the display controller **164** also causes to be presented on the display **174** information about the state of the system **50**. These data include data indicating that the attached device is bone cleaning head, the speed range at which the base unit **54** should operate as well as the default operating speed. The surgical personnel can, if they want to, in a step **364**, set the system for operating based on personal preferences of the characteristics of the bone disposed in the cleaning head **56**.

The surgical personnel then actuate the cleaning head by depressing base unit button **120**, (step not illustrated). In response to the depression of button **120** display controller **164**, in step **368**, causes motor driver **142** to apply energization signals to the base unit **54** to cause the motor to turn in the directions in which the lower brush should be turned. Initially, the spindle teeth may not be seated in the brush notches **264**. However, owing to the alignment of the brush with the spindle, after less than 900 rotation of the spindle and the biasing force provided by spring **118**, spindle teeth **112** seat in the brush head notches **264**. Once the spindle teeth **112** are so seated, the continued rotation of the spindle **104** results in a like rotation of cleaning head lower brush **58**.

As discussed above, once the cleaning head shells **192** and **194** are clamped together, the harvested bone is pressed between the bristles **268** and **290** of, respectively, the lower brush **58** and upper brush **59**. The rotation of the lower brush bristles, forces the bone to rub against both sets of bristles **268** and **290**. The rubbing of the bone against the bristles strips ligaments and other debris off the bone so as to clean the bone, step **370**. It is believed that once actuated, cleaning head **56** of this invention can clean bone stock in 5 minutes or less and, in some circumstances, 3 minutes or less.

It should be understood that during steps **368** and **370**, the motor is actuated according the sequence data specified in direction field **278** in the memory **272**. Likewise, unless modified by the user, the motor is run at a speed necessary to rotate the brush according to the data specified in default speed field **276**.

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At the conclusion of the cleaning process, cleaning head **56** is removed from the base unit **52**. The cleaning head shells **192** and **194** are unlocked from each other so as to allow removal the cleaned bone from the head **56**. Both these process are called out as step **372** in FIG. **19B**.

Mill head **60** is then coupled to the base unit **52**, step **373**. To so position the mill head **60**, the mill head bottom shell is seated on the pedestal surface **76** so that pedestal teeth **84** seat in bottom shell notches **312** and catch tray **310** seats in pedestal notch **83**. Mill head **60** is releasably secured to the base unit **52** by the seating of the retaining arms **86** so that the arm fingers **88** seat in the top shell notches **328**. During this process, the spindle alignment pin **110** seats in the mill element opening **332**. This serves to alignment the mill element **62** with the drive spindle **104** so that the element openings **334** are on a circle disposed over the spindle teeth **112**. The seating of base unit teeth **84** in mill head notches **312** prevents rotation of the mill head **60** relative to the base unit **52**.

As mentioned above, as long as the display controller **164** recognizes that the base unit **52** is attached to the control console **66**, the display controller **164**, through RFID interface **168** continues to perform interrogations to determine whether or not a device is attached to the base unit. Once the mill head **60** is fitted to the base unit **52**, in response to this basic interrogation, the mill head RFID **320**, writes out data indicating that the mill head is attached to the base unit. In response to this event occurring, the display controller **164**, in step **374**, reads out all the data in the mill head RFID **320**.

Based on the data read from the mill head RFID, in a step **376**, display controller **164** configures system **50** to actuate the mill head **60**. This process involves generating instructions to operate the base unit motor **54** at the speed which will cause the mill element **64** to rotate at the appropriate speed. Also, the display controller **164** causes data to be presented on the console display **174** to indicate that a mill head **60** is attached to the base unit **52**.

The cleaned bone stock is placed in the feed sleeve **306** and the plunger **307** placed in the sleeve over the bone, step **378**. Surgical personnel start the actual milling process by depressing the base unit button **120**. The personnel also press downwardly on the plunger **307** so as to press the bone stock against the mill element **62**. These two steps are called out as step **380** in FIG. **19C**.

As a consequence of the display controller **164** detecting the depression of button **120**, (step not identified) in step **380** the display controller **164**, in step **380**, sends instructions to the motor driver **162** that causes the driver to actuate the base unit motor **54** at the speed necessary to rotate the mill element **62** at the desired speed. Again, it should be understood that based on the data in memory **320** this speed may be different than the speed at which the motor **54** is actuated to activate the cleaning head **56**. As a consequence of the bone being pressed against the mill element **62** and the mill element rotating so that the cutting edges **340** turn towards the impingement plate **308**, the bone is compressed between the mill element scallops and the impingement plate. The mill element cutting edges **340** shear the bone stock into chips. The chips fall into the catch tray **310**.

After the milling process, the catch tray **310** filled with bone chips is removed from the mill head, step **384**. The chips are then extracted for further processing and subsequent implantation into the patient.

System **50** eliminates the need to hand clean bone stock before it is ground into chips. Thus, the possibility that the individual charged with cleaning the bone stock, will, in the

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cleaning process, rip a glove so as to result in the risk of cross contamination is likewise eliminated.

Still another feature of this invention is that the most costly non-disposable components used to perform the cleaning process, the base unit **52** and the control console **66**, have two functions. These components are used to activate both the mill element **62** in the mill head **60** and the brush **58** in the cleaning head **56**. This reduces the costs of providing system **50** of this invention.

The memories (RFIDs) **270** and **320** provide configuration data about, respectively, the brush **58** and mill element **62**, with which they are associated. Control console **66**, based on these configuration data, controls actuation of the base unit motor **54** so that the motor operates at the speed for the attached brush **58** or mill element **62**. This minimizes the amount of time personnel need to configure the system **50** for operation when first the cleaning head and then the mill head are mounted to the base unit. Given that the data transfer from the cleaning head and the mill head is automatic, the likelihood the human error could result in the console receiving incorrect configuration/operational data is essentially eliminated. Thus, the control console **66** ensures that the base unit motor **54** operates at the appropriate speed regardless of which head, the cleaning head **56** or the mill head **60**, is attached to the base unit **52**.

Still another feature of system **50** of this invention is that, in some versions of the invention, the only disposable portions of the cleaning head **56** are the brushes **58** and **59**. This further reduces the costs associated with operating system **50**.

VI. First Alternative Cleaning Head

FIG. **20** illustrates the basic features of an alternative cleaning head **410** for use with system **50** of this invention. Head **410** includes lower and upper shells **412** and **414**, respectively. A lower brush **416** is disposed in the lower shell **412**. An upper brush **418** is disposed in the upper shell **414**. A drive assembly, the components of which are discussed below, connects the brushes for simultaneous rotation. In one version of the invention, the brushes are connected together so that when the lower brush **416** rotates in one direction, the upper brush **418** rotates in the opposite direction.

Lower shell **412** is formed from the same material from which lower shell **192** of cleaning head **56** is formed. The lower shell **412**, seen best in FIG. **24**, has the same basic circular shape as shell **192**. More particularly, at the disk shaped base of the shell there is a center located opening (not illustrated) through which the base unit drive spindle **104** can extend. There are notches **422** in which the pedestal teeth **84** seat when cleaning head **410** is seated on the base unit **56**. Lower shell **412** has a circular ring **424** that extends upwardly from the base. Ring **424** of shell **412** is shorter in height than ring **214** of shell **194**. Accordingly the notches **426** formed in the ring in which the base unit retention arms **88** seat extend downwardly from the exposed outwardly directed face of the ring **426**. Slots **224** are formed in the outwardly directed face of ring **424** for the same reason the slots **224** are present in shell **192**.

Cleaning head upper shell **414** is formed from the same material from which the lower shell **412** is formed. Upper shell **414**, like shell **194** of cleaning head **56**, is cylindrically shaped. As seen best in FIGS. **20** and **21**, upper shell **414** is larger in top-to-bottom height than shell **194**. A number of coaxial bores extend longitudinally through upper shell **414**. A bore **430** extends upwardly from the bottom of upper shell

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414. Bore **430** extends approximately 50% of the distance through the shell **414** and is the largest diameter bore. A bore **432** is located immediately above and is contiguous with bore **430**. Bore **432** has a diameter that is slightly less than that of bore **430**. Upper shell **414** is shaped so that the outer perimeter of bore **432** is defined by a circular ring of teeth **434** that extends inwardly from an interior surface of shell **414**.

Above bore **432** there is a bore **436**, a bore **438** and a bore **440**. Bore **436** is present for manufacturing reasons and has a diameter between that of bore **430** and bore **432**. Bore **438** is immediately above bore **436**. Bore **438** has a diameter less than that of bore **436**. Bore **440** extends between bore **438** and the top face of shell **414**. Bore **440** has a diameter less than that of bore **438**.

The same pins **248** that extend downwardly from the base of upper shell **194** extend downwardly from the base of upper shell **414**.

Lower brush **416** has a substrate **444** similar to substrate **260**. As seen in FIG. **23**, a bore **446** extends through substrate **444** along the axis around which substrate **444** rotates. Bore **446** has a square cross-sectional shape. The width across bore **446** is equal to or greater than the maximum outer diameter of the drive spindle alignment pin **110**. The notches **264** present in substrate **260** are equiangularly spaced around the bore **446** of substrate **444**.

Substrate **444** is further shown as having a closed end notch **448** on the underside of the substrate. This is to indicate that substrate **444**, like substrate **260**, holds the memory (RFID) **270** (FIG. **5**) in which data regarding the identity and operation of the cleaning head **410** are stored.

Bristles **268** extend upwardly from the top directed face of substrate **444**. Not identified is the adhesive layer between the substrate **444** and the bristles.

Upper brush **418** has a disc shaped superstrate **450** with an outer diameter at least 0.5 mm smaller than the diameter of shell bore **430**. Superstrate **450** is formed with three bores. A through bore **452** extends axially through the substrate along the longitudinal axis of the substrate. There are also two closed end bores **454** and **456**. Bore **452**, **454** and **456** are linearly aligned. Bore **456** is located immediately inward of the outer perimeter of superstrate **450**.

The drive assembly of cleaning head **410** includes a shaft **460**, now described by reference to FIGS. **24** and **25**. Shaft **460** is formed from stainless steel or other metal or, in some cases, as plastic and as a single piece unit. The shaft **460** is shaped to have a generally elongated cylindrical body **462**. Body **462** has a diameter that is less than the diameter of bore **452** of upper brush substrate **450**. Formed integrally with the body **462**, at the bottom end of the body, shaft **460** has a foot **464**. Foot **464** has a rectangular cross-sectional profile. More particularly, shaft foot **464** is shaped to closely fit in bore **446** internal to lower brush substrate **444**. While not illustrated, shaft foot **464** may have a bore that extends upwardly from the base of the foot. This bore is shaped to receive the alignment pin **110** integral with the base unit drive spindle **104**. Shaft **460** is further formed to have an elongated slot **466**. Slot **466** is located in shaft body **462** approximately 2 to 4 cm below the top of the shaft. Slot **464** extends diametrically through the shaft body **462**.

Shaft **460** is positioned so that foot **464** is seated in lower brush substrate bore **446**. The shaft body **462** extends through and above bore **452** in the upper brush substrate **450**. As seen in FIG. **21**, shaft body **462** also extends through upper shell bores **436**, **438** and **440**.

The cleaning head drive assembly also includes three gears **468**, **470**, **472** located immediately above the top of the

upper brush substrate **450** as seen best in FIGS. **20** and **24**. Gear **468** is slip fitted over the section of the shaft body **462** that extends above the superstrate **450**. A ring shaped collar **474** integral with gear **468** extends upwardly from the gear so as to also extend around the shaft body **462**. Collar **474** is formed to have diametrically opposed openings **476**, one shown. A pin **478** extends through collar openings **476** and shaft slot **466** to couple gear **468** to the shaft **460** so that the gear rotates with the shaft. Given that pin **478** extends through shaft slot **466** it should be apparent the gear **468**, like upper brush **418**, is able to move longitudinally along shaft **462**.

It should further be appreciated that when cleaning head **410** is assembled, collar **474** may extend into upper shell bore **438**. Accordingly, the upper shell **414** is formed so that bore **438** is of larger diameter than collar **474**.

Gears **470** and **472** are both rotationally mounted to the top surface of superstrate **450**. More particularly, gear **470** is disposed over the shaft of an arbor **480** seated in superstrate bore **454**. Gear **472** is disposed over the shaft of an arbor **482** seated in superstrate bore **456**. Pins **486** hold the gears **470** and **472** to, respectively, arbors **480** and **482**. A washer **484** is disposed between each gear **470** and **472** and the overlying pin head. Gear **470** interlocks with both gears **468** and **472**. Gear **472** is positioned so the teeth of the gear project beyond the perimeter of superstrate **450**.

Bristles **290** extend downwardly from superstrate **450**. Not identified is the adhesive layer that holds the bristles **290** to the superstrate.

Cleaning head **410** also includes a ring **488**. Ring **488** is dimensioned to slip fit over the shaft body **462**. For reasons apparent below, this version of the invention may include a number of different rings of different heights.

Bone stock is cleaned using cleaning head **410** by first placing the lower brush **416** in lower shell **412**. Shaft **460** is mounted to the brush **416** so the shaft foot **464** seats in substrate bore **446**. A ring **488** having a height slightly less than the height of the bone stock is slipped over the shaft body **462**. The bone is placed on the brush bristles **268**.

Upper brush **418** is disposed over the lower brush so the shaft body extends through superstrate bore **452**. It should be appreciated that the presence of ring **488** limits the extent to which the upper brush **418** is pushed down against the bone stock.

The upper shell **414** is then disposed over the lower shell **412**, the bone stock and the brushes **416** and **418**. As a consequence of the positioning of the upper shell **414** in place, the teeth of the gear **472** that extend beyond upper brush substrate **450** engage the teeth **434** internal to the upper shell **414**.

Cleaning head **410** is then releasably coupled to the base unit pedestal **74** in the same manner in which cleaning head **56** is so attached. The cleaning head **410** is actuated in the same general process in which cleaning head **56** is actuated. If the cleaning head, more particularly, the lower brush **416** is provided with a memory **270**, in which data defining the operational parameters of the cleaning head are stored, control console **66** activates the base unit motor **54** based on these data.

During the actuation of the cleaning head **410**, base unit spindle **110** rotates the lower brush **416**. This results in a like rotation of shaft **460**. The rotation of shaft **460** results in the rotation of gears **468**, **470** and **472**. The engagement of the teeth of gear **472** in upper shell teeth **434** results in the rotation of upper brush **418** around the shaft **460**. More particularly, superstrate **450** and the upper brush are rotated in a direction opposite that in which the lower brush **416**

rotates. Thus, bristles **268** and **290** simultaneously rub against opposed surfaces of the bone stock in opposed directions. This simultaneous brushing of two surfaces of the bone stock in opposed directions rolls the bone stock between the brushes during the cleaning process. This facilitates the thorough cleaning of the bone stock and potentially reduces the overall time required to clean the bone stock.

VII. Integrated Cleaning and Mill Head

In another alternative version of this invention, the system may include a single head **490** now described by reference to FIG. **26**, with components for both cleaning and milling bone stock. Head **490** includes lower and upper plates **492** and **496**, respectively. Rotatably disposed between the plates **492** and **496** is a mill element **494**. A cleaning module, including lower and upper shells **502** and **508**, respectively is moveably mounted to upper plate **496**. Lower and upper brushes **504** and **506**, respectively, are disposed inside the brush housing. Lower brush **504** is provided with features that releasably couple the brush to the mill element **494** so that the brush rotates in unison with the mill element.

The head lower plate **492**, is formed from a plastic such as a sterilizable plastic such as a polycarbonate plastic. As seen in FIG. **27**, the lower plate is formed to have cylindrical base **510**. Base **510** has a diameter that allows it to seat on the base unit pedestal top surface **76** within lip **78**. Base **510** has a height that is at least as great as that of lip **78** above surface **76**. Lower plate **492** is further formed to have a rim **512** that is integral with and extends circumferentially and radially beyond base **510**. The base is formed so as to have four notches **514** and two notches **516** (one notch **514** seen in FIG. **27**). Notches **514** are located around the bottom surface of base **510**. Notches **514** are dimensioned to receive teeth **84** when head **490** is seated on base unit **52**. Notches **516** are diametrically opposed to each other. Notches **516** are located at the top of base **510** immediately below when the rim **512** projects outwardly from the base. The notches **516** are positioned and dimensioned to receive the retention arms fingers **88** to facilitate the releasable coupling of the head **490** to the base unit **52**.

Lower plate **492** is further formed so as to have a circular void space **518** concentric with and located within rim **512**. More specifically the lower plate **492** is formed so that the void space **518** extends through rim **512** and partially through the base **510**. A circular opening **520** seen in phantom in FIG. **27**, extends concentrically from the bottom of void space **518** through the plate base **510**. Opening **520** has a diameter slightly greater than that of the drive spindle head **108**. The lower plate is also formed with a discharge port **522**. Discharge port **522** is square in cross section and extends from the bottom of the void space through the plate base **510**. Port **522** is located between opening **520** and the outer perimeter of plate base **510**.

While not seen in the Figures, it should be understood that a catch tray similar to catch tray **310** is removably mounted to the lower plate base **510**. Also not illustrated are the rails integral with the lower plate **492** that facilitate the removable mounting of the catch tray below discharge port **522**.

Upper plate **494** is formed from the same type of plastic from which lower plate **492** is formed and is generally disc-shaped. The upper plate **494** has an outer diameter equal to the outer diameter of base plate rim **512**. The upper plate **494** is formed to have a center located hole **524** (shown in phantom in FIG. **28**). Upper plate **496** is further formed to have a feed port **526**. Feed port **526** is generally square in

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shape. In the illustrated version of the invention the plate is shaped so that the interior surfaces of the plate **496** that lead into the port are inwardly tapered such that the size of the port **526** decreases progressing inwardly from the top of the plate. When head **490** is assembled, the plates **492** and **496** are oriented so that the upper plate feed port **526** is in registration with the lower plate discharge port **522**.

The upper plate **496** is further formed to have a center located tube-shaped sleeve **528**. The plate is formed so that sleeve **528** extends around and above plate center located-hole **520**.

An impingement plate **530**, seen as a phantom rectangle in FIG. **28**, is mounted to the bottom of the upper plate **496**. The impingement plate **530** is located immediately adjacent the plate feed port **526**.

Two parallel webs **534** extend upwardly from upper plate **496**. In some versions of the invention, webs **534** are formed integrally with the upper plate **496**. The webs **534** are positioned so that the plate hole **524**, feed port **526** and sleeve **528** are located between the webs. Webs **534** are spaced apart from each other a distance greater than the outer diameter of shells **502** and **508**. In the illustrated version of the invention, a rectangular rib **536** is formed integrally with each web **534**. Each rib **536** (one shown) extends outwardly from the face of the web that is directed toward sleeve **528**. Ribs **536** provide structural strength to the webs **534**.

Webs **534** are generally solid structural members. Each web **534** is further formed to have a slot **540** immediately below the top of the web that extends across the web. The slots **540**, which are in registration with each other, are each formed to have three downwardly directed indentations **542**, **544** and **546**. Two indentations, indentations **542** and **546**, are located at the opposed ends of each slot. Each indentation **544** extends downwardly from the mid-point of the slot **540** with which it is integral.

Mill element **494** includes a cutting disc **540**. Disc **540** has the same basic features of mill element **62**. Disc **540** has an outer diameter slightly less than the diameter of the lower plate void space **518**. Mill element **494** also has a post **544** that extends upwardly from the top surface of disc **540**, the surface above which the disc scallops (not identified) extend. In the illustrated version of the invention, post **544** extends upwardly from a circular base **542** welded or otherwise secured to the top of disc **540**. Post **544** has a diameter less than that of the underlying base **542**.

A circular head **546** is mounted to the top of post **544**. Head **546** is shaped to have an alignment pin **548** similar to spindle alignment pin **110**. The head **546** also has teeth **550** similar in drive teeth **112** integral with drive spindle **112**. Teeth **550** are disposed equiangularly around and are smaller in height than pin **548**. Mill element **494** is dimensioned so that when head **490** is assembled, the cutting disc **540** is disposed in the lower plate void space **518**, post **544** extends through sleeve **528** and element head **546** is spaced above the sleeve.

Lower shell **502** has the same basic shape of shell **192** of cleaning head **56**. Shell **502** does not have notches that facilitate the seating of the shell or the coupling of the shell to the base unit **52**. Shell **502** has a notch **554** that extends downwardly from the outer lip of the shell. In some versions of the invention, notch **554** extends to the base of the shell **502**. Lower shell **502** is further formed to have two diametrically opposed, linearly aligned closed end bores **556**.

Pins **558** moveably mount the lower shell **502** to webs **534** integral with upper plate **494**. Each pin **558** has an end that is seated in one of the shell bores **556**. The end of the pin **558**

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that extends outwardly of the shell **502** extends into the adjacent web slot **540**. Pins **558** are dimensioned relative to the slots **540** so that the pins can slidably move in the slots and indentations **542-546**. It should therefore be appreciated that the pins **558** allow the lower shell **502**, and the components mounted to the shell, to both move over the upper plate **496** and to pivot.

Upper shell **508** has the same basic cylindrical shape as shell **194** of cleaning head **56**. A rectangularly shaped notch **560** extends through the downwardly extend skirt of the shell **194** (skirt not identified). Collectively, shells **502** and **508** are shaped so that, when assembled together to form the cleaning module housing, upper shell notch **560** is in registration with lower shell notch **554**. It should also be understood that notches **554** and **560** share a common width. Two linearly aligned laterally extending bores **564** are also formed in upper shell **508**. Bores **564** are positioned to open into notch **560**.

Pins **248** (one shown) integral with upper shell **508** seat in slots **224** formed in the lower shell **502** to facilitate the releasable coupling of the shells together. Again, collectively, shells **502** and **508** form the housing of the cleaning module of head **490**.

Lower brush **504** has the same basic features of brush **58**. Upper brush **506** has the same basic features of brush **59**.

Head **490** of this version of the invention also includes a slide plate **568**. Plate **568** is mounted in contiguous notches **554** and **560** integral with the cleaning module housing. Plate **568** has a generally rectangular shape. The width of the plate **568** is selected to facilitate the sliding movement of the plate **568** in the notches **554** and **560**. The opposed side surfaces of the plate **568** are each formed to have longitudinally extending groove **570** (one shown). A tab **572** extends perpendicularly away from the plate **568**. In the illustrated version of the invention, the plate **568** is constructed so that when the head **490** is assembled, tab **572** extends over the top of the upper shell **508**.

Two pins **574** hold the plate **568** to the cleaning module housing. More particularly, each pin **574** is partially seated in one of the bores **564** formed in the upper shell **508**. The exposed end of the pin **574** seats in the adjacent groove **570** integral with the plate **568**. Pins **574** thus hold the plate to the cleaning module housing while allowing the plate to slide so as to selectively cover and expose notches **554** and **560**.

While not illustrated, a memory similar to memory **320** may be disposed in the lower plate base **510**. This memory includes data indicating the speeds at which both the mill element **494** and lower brush **504** should be rotated.

To use head **490**, the head is initially mounted on the base unit **52**. As a consequence of the seating of the lower plate base **510** on base unit pedestal surface **76**, the drive spindle alignment pin **110** causes mill element **494** to align with the drive spindle **104**.

To load bone stock for cleaning in the cleaning module, the module is first moved so that pins **558** seat in web slot indentations **546**. The cleaning module is thus positioned so it is spaced furthest away from the upper plate feed port **526**. The cleaning module is then pivoted so notches **554** and **560** and plate **568** are upwardly directed. Plate **568** is opened to expose notches **554** and **560**. The bone stock is passed through the notches, between the brushes **504** and **506**. To ensure sufficient clearance between the notches, the person perform this process may pull on the post integral with brush **506** to hold it away from brush **504**. Plate **568** is returned to the closed state.

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The cleaning module is then moved so that pins **554** seat in web slot indentations **544**. As a consequence of the cleaning module being so positioned, the alignment pin **548** integral with mill element **494** in the complementary opening formed in brush **504** aligns the drive openings internal to the brush with the circle in which teeth **550** are disposed. Collectively, the components of head **490** may further be constructed so that when pins **558** seat in indentations **544**, the undersurface of lower shell **502** seats against sleeve **528**.

The memory integral with head **490** contains data that informs the control console display controller **164** that the head is a combined cleaning and mill head, step **582** of FIG. **29A**. In response to this information, display controller **164** presents on display **174** buttons requesting the user to indicate if the device is to be operated in the cleaning mode or the milling mode, step **584**. Upon entry of a command, step **585**, that the device is to be operating in the cleaning mode, display controller, in step **586**, prepares the instructions to cause the motor driver **162** to actuate the motor **54** in a manner appropriate to rotate brush **504** at the appropriate speed and in the appropriate direction. Once button **120** is depressed, (step not shown) control console **66** actuates the motor **54** so as to cause the appropriate actuation of the cleaning module, step **588**.

In step **588** the actuation of the motor **54** and drive spindle **108** result in a like actuation of the mill element **494**. The cutting disc **540**, post **544** and head **546** are all rotated. If teeth **550** are not already seated in the brush openings, identical to openings **264** of FIG. **12**, they seat in these openings during the initial rotation of the head **546**. As a consequence of the engagement of teeth **550** in the brush openings, the brush undergoes a like rotation. As with the rotation of brush **58**, this action results in the cleaning of the bone stock.

Once the cleaning processes is terminated, the control console may return to step **584** to await an indication if, the next time the motor is actuated, it is actuated to actuate the mill element **494** or the brush **504**.

Once the bone is cleaned, the cleaning module is moved so that pins **554** seat in web slot indentations **542**, the indentations adjacent feed port **526**. The cleaning module is pivoted so that the module notches **554** and **560** face the feed port **526**. Plate **568** is slide to the open state. Gravity causes the cleaned bone stock to fall into the feed port **526**. To facilitate this process, it may be necessary to pull on the post integral with brush **506** so as to hold the brushes apart.

Once the bone stock is transferred to the lower plate, essentially the mill assembly, the cleaning module is moved away from the feed port **526**. This provides clearance for the subsequent insertion of a plunger into the feed port.

The user depresses the appropriate button presented on the control console display **174** to indicate that head **490** is now to be operated in the cleaning mode, step **585** is reexecuted. In response to this button being depressed, in step **562**, the display controller **164** readies the instructions to cause the base unit motor **54** to be actuated at a speed appropriate for rotating the cutting disc **540**, step **562**.

When the bone is to be milled the button **120** is depressed, (step not illustrated). This results in the control console **66** actuating the base unit motor **54** at the appropriate speed, step **564**. Simultaneously with the actuation of the motor a plunger, such as the plunger **307** of FIG. **16**, is used to force the bone stock against the rotating cutting disc **540**. The rotating cutting disc **540** and impingement plate **530** cooperate to convert the bone stock into bone chips.

One advantage of head **490** of this invention is that a single unit includes the components that first clean the bone

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stock and then mill the stock into chips. The need to change heads between these processes is eliminated. A further advantage of a system of this invention including head **490** is that the transfer of the cleaned bone from the cleaning module to the mill module occurs using gravity. The need to have an individual perform this transfer is eliminated.

It should of course be appreciated that there may procedures in which it may be appropriate to use one of the cleaning module or the milling module of this invention but not both the modules of head **490**.

VIII. Alternative Brush

FIGS. **30** and **31** illustrate an alternative brush **602** that can be employed with the system of this invention. The illustrated brush **602** is a lower brush. Versions of the brush can be configured as an upper brush. Brush **602** includes a disc substrate **606**. A number of brush heads **608** (one shown) are removably attached to the substrate **606**.

Substrate **606** is dimensioned to fit in and rotate in the head/module in which the brush **602** is seated. Not shown is the opening on the underside of the substrate for receiving the alignment pin **110** or **548**. Also not shown are the openings in the underside of substrate **606** for receiving the drive teeth **112** or **550**. Substrate **606** is formed to have four equiangularly spaced apart slots **610**. Each slot **610** starts from a location radially spaced from the center of the substrate and extends radially outwardly to the outer perimeter of the substrate. While not identified, it can be seen in the Figures that each slot **610** includes a wide width lower section and an upper section with narrower width. The slot lower section, it is observed, form the closed end base of the slot **610**. Slot **610** upper section is open to the surface of substrate **606**.

Each brush head **608** includes a rectangular base **614**. Base **614** has a width that allows the base to slide in the lower section of substrate slot **610**. Bristles **616** extend upwardly from the top surface of base **614**. Bristles **616** may be adhesively secured to the base **614**. Alternatively, the bristles may be compression packed in closed end bores (not illustrated) in the base. Regardless of how the bristles **616** are mounted to the base **614**, the bristles are extend across the base a distance approximately equal to that of the width of the upper section of slot **610**. Bristles **616** are of sufficient length that the bristles extend above and out of the upper section of slot **610**.

Brush **602** of this invention is prepared for use by sliding a brush head **608** in each one of the substrate slots **610**. Brush **602** is then mounted the cleaning head/module and used in the same manner as the previously described brushes **58**, **59**, **416**, **418**, **504** or **506**.

Once a system of this invention in which one or more brushes **602** is employed, the brush heads **608** can be removed from the substrate (or superstrate) for cleaning or disposal and replacement. The substrate (or superstrate) can be independently sterilized. Thus use of brushes **602** of this version of the system of the invention thus eliminate the need to, after each use, discard the whole of the brush.

IX. Second Alternative Cleaning Head

A second alternative cleaning head **630** of this invention is now described by initial reference to FIG. **32**. Cleaning head **630** includes a base **632**. Base **632** is dimensioned to be seated over the base unit top surface **76**. Brush **58** (or

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brush 602) is rotatably disposed within the base 632. A flexible cap 634 is fitted to the base 632 so as to extend over the brush 58.

Cleaning head base 632 can be formed from the same material from which the shells 192 and 194 of head 56 are formed. Alternatively, if cleaning head 630 is a use-once unit, base 632 may be formed from a sterilizable plastic such as a polycarbonate plastic. As best seen in FIGS. 33 and 34, base 632 is formed to have a cylindrical foot 638 dimensioned to seat on base unit surface 76. Foot 638 has an outer diameter that allows the foot to be slip fitted within the circular void space defined by the base unit lip 78. Four equiangularly spaced notches 640 extend upwardly from the bottom of the foot 638 around the outer perimeter of the foot. Notches receive pedestal teeth 84 when the cleaning head 630 is seated on the base unit 52. In the illustrated version of the invention, a groove 642 extends inwardly around the circumferential outer surface of the foot 638. Foot 638 is further formed to have two diametrically opposed notches 644 (one shown in FIG. 32) that extend inwardly from the outer circumferential surface of the foot. Notches 644 intersect groove 642. Each notch 644 is dimensioned to receive a separate one fingers 88 integral with the base unit retention arms 86. Base foot 638 also has a through hole 646 that extends top-to-bottom through the foot. Hole 646 is centered along the top to bottom longitudinal axis of the foot 638. Hole 646 is dimensioned to allow the base unit spindle head 108 freely move therein.

While not illustrated, it should be understood that RFID 270 and coil 271 (FIG. 5) are embedded in the foot 638 of cleaning head base 632. The data in RFID 270 are used by control console 66 to regulate the operation of the system base unit 52 when cleaning head 630 is attached.

Base 632 is further formed to have a multi-section ring 647 that extends upwardly from the outer perimeter of foot 638. The ring has a lower section, section 650, with a relatively thick cross sectional width. The ring has an upper section, section 652, with a narrower cross sectional width. Ring 647 is shaped so that the inner walls of the lower and upper sections 650 and 652, respectively, form a shell that defines a void space 654 above foot 638 of constant diameter. Void space 654 has a diameter that is at least 0.5 mm greater than that of brush 58. Thus, around the outer surface of the ring the ring upper section 652 is stepped inwardly from the lower section 650. The ring is of sufficient height so that void space 654 is able to receive all of the brush; both the substrate 260 and the bristles 268. The ring is further formed to have a flange 656 that projects radially outwardly from the top of the upper section 652. While not called out, it can be seen in FIG. 34 that the outer surface of the flange is outwardly tapered.

Cleaning head base 632 is further formed to have a lip 660 that extends substantially circumferentially around and above the ring. Lip 660 is formed to have a horizontal section 662 that extends radially away from the top of the ring lower section 650. Lip 660 also has a vertical section 664 integral with the outer end of the horizontal section 662. More particularly, the lip 660 is formed so that vertical section 664 extends both outwardly and upwardly away from the horizontal section. In many versions of the invention, including the illustrated version, lip 660 is formed so that the top edge of the vertical section 664 is located above the top of the ring upper section 652.

The cleaning head base 632 is further formed to have a spout 668. The spout 668 includes a finger 670 that extends radially and upwardly from an arcuate section of the ring lower section 650. The face of this finger 670 is flush with

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the adjacent arcuate step between the ring upper and lower sections 652 and 650, respectively. Finger 670 is formed with a tip 672 that extends radially outwardly beyond the portion of the finger on either side of the tip. Spout 668 also has a three-wall chute 678 that extends radially beyond finger 672. The walls that form the chute 678, two opposed side walls as well as a wall between the side walls are extensions of the lip vertical section 664 that are extend vertically upwardly more than the vertical section itself. Collectively, the outer end of the spout finger 670 and the adjacent though spaced away chute walls defining an opening 676 through the spout 668. In the illustrated version of the invention a three-sided lip 680 extends inwardly from the inner faces of the chute walls into the opening 676. Opening 676 is dimensioned to allow the seating thereon of mill head feed sleeve 306.

While not identified, it is observed from FIG. 32 that the upwardly directed portion of the cap lip 660 that leads to spout 668 is recessed relative to the adjacent surface of the lip 660.

The cap 634 of cleaning head 630 is formed from a flexible material such as a thermoplastic resin. The cap 634, shown best in FIGS. 32, 35 and 36 has a ring shaped rim 684. A concavo-convex dome 686 extends inwardly from the inner edge of the rim 684 and forms the center of the cap 634. The cap 634 is shaped so that the dome 686 extends upwardly away from the outer surface of rim, away from the outer surface of the underlying brush 58. Bristles 688, also part of cap 634, extend downwardly from the surface of the dome 686 directed towards brush 58.

A C-shaped lip 690 extends outwardly from the outer perimeter of the cap rim 684. One end of the lip 690 is contiguous with the outer edge of the rim 684. The cap 634 is shaped so that the lip 690 curves under the side of the rim that faces base 632. The cap 634 is formed so that the lip 690 can snap fit over flange 656 integral with base 632. Cap 634 is further formed to have a number of flexible tabs 692. Tabs 692 project radially and upwardly away from the outer surface of the cap lip 690 adjacent where the lip starts to curve under the rim 684. Tabs 692 serve as finger holds for stretching the cap over and removing the cap from the base ring upper section 652.

Cleaning head 630 of this invention is used by seating the brush 58 (or brush 602) in the base void space 654. The bone stock to be cleaned and milled is placed on brush bristles 268. Cap 634 is snap fitted over the base ring upper section 652. The cleaning head 630 is then fitted and secured to the base unit 52.

The base unit motor 54 is actuated as previously described in order to actuate the brush 58. During the time the brush 58 is rotated, the individual responsible for the cleaning process may press down on the cap dome 686. This action presses the caps bristles 688 against the bone. The bone is therefore compressed between both the brush bristles 268 and the cap bristles 688. The rotation of the brush 58 rotates the bone stock against both sets of bristles 268 and 688 to result in the striping of ligaments and other debris away from the bone.

Once the bone is cleaned, the cleaning head 630, possibly with cap 634 still attached, is removed from the base unit 52. Mill head 60 is attached to the base unit 52. As seen in FIG. 37, the cleaning head 630 is then positioned so that chute 678 extends on the top of the mill head feed sleeve 306. The underside of lip 680 that surrounds spout opening 676 abuts the top edge surfaces of the feed sleeve 306. This lip-to-sleeve contact prevents from the cleaning head 630 from sliding down the mill head feed sleeve 306

The cleaning cap 634 is removed from the rest of the head 630. The presence of lip 660, which flares outwardly, prevents the cleaned bone stock from falling out of the base 632. Assume the bone stock passes inspection, a hand held instrument, such as forceps, are used to guide the cleaned bone stock off the brush 58 through the spout 668 and into the feed sleeve 306. During this process, both lip 660 and the walls of chute 678 prevent the bone stock from falling out of the cleaning head base 632.

In the described version of the invention, once the bone is transferred into the feed sleeve 306, plunger 307 can then be fitted into the feed sleeve without first having to remove cleaning head 630.

Cleaning head 630 and mill head 60 of this embodiment of the system of this invention do more than, respectively, clean and mill the bone stock. These components are designed to be coupled to each between the cleaning and milling processes. In the described version of the invention this mating occurs without having to provide supplemental components such as a set screws or ball pins with moving components. This component mating minimizes the extent the cleaned bone stock needs to be handled before it is transferred to the mill head 60. This simplifies the transfer of the bone stock to the mill head and reduces the likelihood that, during the transfer process, the bone stock will be inadvertently mishandled.

Lip 660 and chute 668 of cleaning head 630 do more than function as structural members that prevent the bone stock from inadvertently fall out of the head base 632. The lip 660 and chute 668 function as features of the base 632 an individual can hold when handling the base. These features are both spaced away from the brush 58. Thus, an individual by holding onto either the base lip 660 or chute 668 can handle the base while having his/her fingers spaced from the brush 58. This reduces the likelihood that the individual may inadvertently touch the brush 58 and the possible problems caused by such contact.

X. Third Alternative Cleaning Head

Referring to FIGS. 38-47, a third alternative cleaning head 700 is shown. The cleaning head 700 comprises a base 702. A shell 704 is mounted to the base 702. The shell 704 defines a void space 706 for receiving the bone to be cleaned. A cap 707, (shown only in FIG. 39,) is removably mounted on top of the shell 704 to enclose the void space 706. The base 702, shell 704, and cap 707 can be formed from the same material from which the shells 192 and 194 of head 56 are formed. Alternatively, if cleaning head 700 is a use-once unit, base 702, shell 704, and cap 707 may be formed from a sterilizable plastic such as a polycarbonate plastic.

The base 702 is shown in FIGS. 38 and 39 as being rectangular in shape without any features for engaging the base unit 52. Accordingly, the cleaning head 700 may be a stand-alone unit for cleaning bone in which the base 702 is simply attached to a separate drive system (shown but not numbered). However, in other embodiments the base 702 has an outer diameter that allows the base 702 to be slip fitted within the circular void space defined by the base unit lip 78. As a result, the cleaning head 700 can be operated by the base unit 52. In this case, like the second alternative cleaning head 630, the base 702 is circular in shape (not shown) and four equiangularly spaced notches (not shown) extend upwardly from the bottom of the base 702 around the outer perimeter of the base 702. The notches receive the pedestal teeth 84 when the cleaning head 700 is seated on the

base unit 52. A groove (not shown), like groove 642, extends inwardly around the circumferential outer surface of the base 702. Base 702 has two diametrically opposed notches (not shown), like notches 644 (one shown in FIG. 32) that extend inwardly from the outer circumferential surface of the base 702. Each notch being dimensioned to receive a separate one of the fingers 88 integral with the base unit retention arms 86.

Referring back to the embodiment of FIG. 38, the base 702 has a through hole 708 that extends top-to-bottom through the base 702. The hole 708 is centered along the top to bottom longitudinal axis of the base 702. The hole 708 is occupied by a spindle 710. The spindle 710 forms part of a drive assembly 714 of the cleaning head 700. The spindle 710 is rotatably supported by bearings 712 mounted to the base 702. A needle bearing assembly, shown in exploded view, is disposed between base 702 and the overlying surface of the spindle 710. The needle bearing assembly absorbs the thrust load of the spindle and the attached components. In alternative embodiments in which the cleaning head 700 is operated by the base unit 52, the spindle 710 is not present and the hole 708 is dimensioned to allow the base unit spindle head 108 to freely move therein and engage the cleaning head 700.

Referring to FIGS. 38 and 39, the shell 704 includes a shell base 716 to which a plurality of shaving blocks 718 are mounted. Alternatively, the shell 704 may be formed in one-piece with the shaving blocks 718 being integrated with the shell base 716. The cap 707 is mounted to the shell base 716 by a plurality of fasteners 717, (seen only FIG. 39). Alternatively, the cap 707 may be releasably locked to the shell base 716 by locking features (not shown) or the cap 707 may be simply fitted on top of the shell 704 temporarily during bone cleaning. The shell base 716 includes a bottom 720 that is generally circular in shape. A plurality of spacers 721 space the bottom 720 from the base 702. The bottom 720 defines a through bore 722 that is generally centrally located within the bottom 720. A bushing 724 is mounted to the bottom 720 in the through bore 722 and rotatably supports an agitator 726.

The agitator 726 includes a cross-shaped first end 728 that engages a correspondingly cross-shaped grooved end 730 of the spindle 710. The ends 728 and 730 mate such that rotation of the spindle 710 results in direct rotation of the agitator 726. The agitator 726 extends from the first end 728 disposed beneath the bottom 720, through the through bore 722 and upwardly into the void space 706 to a second free end 732. When the cap 707 is attached, the agitator second end 732 is seated in closed end bore formed in cap 707. Cap 707 thus supports the agitator second end 732.

The agitator 726 is generally cylindrical in shape and is elongated between its ends 728, 732. The agitator 726 includes a stem 734 disposed about a rotational axis and a pair of fins 736 extending radially outwardly from the stem 734. The fins 736 extend helically about the rotational axis from the first end 728 toward the second end 732. When bone is placed in the void space 706 the fins 736 rotate to pick up and tumble the bone and press the bone outwardly away from the rotational axis of the agitator 726. The fins 736 may be tapered from the first end 728 toward the second end 732 such that the fins 736 increase in radially outward dimension from the stem 734 as the agitator 726 extends from the second end 732 toward the first end 728.

Cleaning elements 738 in the form of fluted screws 738 clean soft tissue from the bone placed in the void space 706. Axles 740 rotatably support the fluted screws 738 in the shaving blocks 718. The axles 740 are supported and fixed

at a first end in the base **702** and extend upwardly from the base **702** to a second end. The second end is disposed in a top opening **742** in a top plate **759** of the shaving blocks **718**.

Referring to FIGS. **40** and **41**, each of the fluted screws **738** include a sleeve **744**, a pair of couplings **746**, **748** for press-fitting into opposing ends of the sleeve **744** to rotate about the axle **740**, and a pinion gear **750** for engaging the drive assembly **714** of the cleaning head **700**. Each sleeve **744** has an axially extending through bore, (not identified). Each sleeve **744** is further shaped to have a plurality of flutes **752** that extend helically around the outer surface of the sleeve. Flutes are formed with adjacent surfaces that meet at defined edges **761** discussed below. This flute geometry facilitates gripping and tearing of soft tissue from the bone as part of the cleaning process. As shown in FIG. **39**, each fluted screw **738** is rotatably mounted to a separate one of the shaving blocks **718**. Sleeves **744** are preferably formed of stainless steel.

Referring to FIGS. **45** and **47**, the shaving blocks **718** are preferably formed of stainless steel. The shaving blocks **718** are mounted adjacent the shell base **716** to form part of a generally cylindrical inner surface **754** of the shell **704**. The inner surface **754** further defines the void space **706**. Bone is compressed against the inner surface **754** by the agitator **726**. Each shaving block **718** includes a ring shaped collar **755**. The collar **755** defines an opening **757** that receives the associated fluted screw **738**. A front post **756** extends upwardly from the collar **755** to the top plate **759**. Top plate is longitudinally supported by a thrust bearing (not illustrated). Post **756** has a front surface **758** forming part of the inner surface **754**. An elongated space **760** is defined in the front post **756** to receive the fluted screw **738**. The fluted screw **738** rotates in the elongated space **760** in the front post **756**. A portion of the sleeve **744** extends radially inwardly toward the agitator **726** a distance away from the front surface **758** to grip soft tissue attached to bone disposed in the void space **706**.

Each sleeve flute **752** is shaped to define a pair of cutting edges **761** disposed on opposing sides of the elongated space **760** (see FIG. **47**). These cutting edges **761** are sharp enough to cut soft tissue that the associated sleeve **744** grips or captures during rotation. Each sleeve **744** engages the soft tissue still attached to bone and impinge that soft tissue against the cutting edges **761** during rotation to cut the soft tissue away from the bone. The agitator **726** forces the bone with soft tissue attached thereto toward the sleeves **744** to facilitate gripping of the soft tissue by the fluted screws **738** and corresponding cutting of the soft tissue by the cutting edges **761**. Fluted sleeves **744** also separately act to cut the soft tissue from the bone although to a limited extent.

A plurality of wire brushes **762** are fixed to each shaving block **718** to clean the associated fluted screws **738**. In the embodiment shown two wire brushes **762** are attached to each shaving block **718**. Each brush **762** has a first end **764** fixed to the top plate **759** of the shaving block **718** and a second end **766** disposed adjacent the base collar **755**. A plug **763** (see FIG. **46**) is inserted into an opening **765** in the top plate **759** of the shaving block **718**. The first end **764** of the wire brush is press fit into a bore **767** in the plug **763**. The plug **763** is press fit into the opening **765** and snugly holds the first end in the opening **765** so that the brushes **762** remain fixed relative to the rotating fluted screws **738**.

Referring to FIG. **39**, each of the wire brushes **762** includes an elongated cylindrically shaped substrate **777** formed of metal or a sterilizable plastic such as a glass-filled nylon. The bristles **768** are preferably made of metal such as stainless steel. The bristles **768** clean out soft tissue trapped

in the flutes **752** of the fluted screws **738** as the fluted screws **738** are rotating in the void space **706**. Accordingly, the bristles **768** are positioned such that they penetrate into the flutes **752** of the fluted screws **738**. The bristles **768** remain fixed to the shell **704** while the rotating fluted screws **738** rotate relative to the shell **704** and the fixed bristles **768**.

As previously discussed, the hole **708** extending through the base **702** is occupied by spindle **710**. The spindle **710** extends from the first end **730** that engages the first end **728** of the agitator **726** to a second end **770** having features (such as a square shaft configuration) for mating with a drive system such as a drive motor (shown but not numbered). In the embodiment shown, the drive assembly **714** further includes a pinion gear **772** that is fixedly mounted to the spindle **710**. The gear **772** is adapted to mate with the gears **750** of the fluted screws **738** to transmit power to the screws **738**. This causes the fluted screws **738** to rotate when the drive system is actuated. Simultaneously, the spindle **710** also transfers power from the drive system to the agitator **726** so as to rotate the agitator. Alternatively, the first end **728** of the agitator **726** and the gear **772** could be configured to engage and be rotatably fixed to the base unit spindle head **108** via the alignment pin **110** and teeth **112** so as to be operated by the base unit **52**.

In the embodiment shown, the gear ratio of the gear **772** to gears **750** is 1:1. In alternative embodiments, different gear ratios could be employed such as 1:2 or 1:3 and vice versa depending on the particular desired relative rotational speeds of the agitator **726** and fluted screws **738**.

During operation, uncleaned bone is first placed in the void space **706** for cleaning and the cap **707** is then placed to cover the void space **706**. The uncleaned bone includes soft tissue attached thereto that requires removal prior to processing by the mill head **60**. The drive system or base unit **52** (if the cleaning head **700** is mounted to the base unit **52**) is then actuated to start rotation of the drive assembly and simultaneous rotation of the agitator **726** and fluted screws **738**. The agitator **726** then acts to tumble the bone and forces the bone against the fluted screws **738**. The fluted screws **738** grip soft tissue attached to the bone and cut the soft tissue away from the bone either by the nature of the flutes **752** on the fluted screws **738** or by impinging the soft tissue against the cutting edges **761** of the shaving blocks **718**. The wire brushes **762** continuously act to clean the fluted screws **738** by removing material out from the flutes **752**. Once the cleaning head **700** has sufficiently removed soft tissue from the bone, the cap **707** is removed and the cleaned bone is grabbed by forceps or other device for further processing. The cleaning head **700** may then be cleaned or discarded.

While not illustrated, it should be understood that RFID **270** and coil **271** (FIG. **5**) may be embedded in the base **702** of cleaning head **700**. The data in RFID **270** are used by control console **66** to regulate the operation of the system base unit **52** when cleaning head **700** is attached.

XI. Fourth Alternative Cleaning Head

Referring to FIGS. **48** and **49**, a fourth alternative cleaning head **800** is now described. Cleaning head **800** includes a base **802**. A shell **804** is mounted to the base **802**. The shell **804** defines a void space **806** for receiving the bone to be cleaned. A pair of shields **805** are fixed to the base **802** on opposing sides of the shell **804**. A cap **807** is mounted to each of the shields **805** above the shell **804** to cover the void space **806**. The base **802**, shell **804**, and cap **807** can be formed from the same material from which the shells **192** and **194** of head **56** are formed. Alternatively, if cleaning

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head **800** is a use-once unit, base **802**, shell **804**, and cap **807** may be formed from a sterilizable plastic such as a polycarbonate plastic.

The base **802** is shown in FIGS. **48** and **49** as being rectangular in shape without any features for engaging the base unit **52**. Accordingly, the cleaning head **800** may be a stand-alone unit for cleaning bone in which the base **802** is simply attached to a separate drive system (shown but not numbered). However, in alternative embodiments, the cleaning head **800** is operated by the base unit **52**. In these embodiments, the base **802** has an outer diameter that allows the base **802** to be slip fitted within the circular void space defined by the base unit lip **78**. In this case, like the second alternative cleaning head **630**, the base **802** is circular in shape and four equiangularly spaced notches (not shown) extend upwardly from the bottom of the base **802** around the outer perimeter of the base **802**. The notches receive the pedestal teeth **84** when the cleaning head **800** is seated on the base unit **52**. A groove (not shown), like groove **642**, extends inwardly around the circumferential outer surface of the base **802**. Base **802** is further formed with two diametrically opposed notches (not shown), like notches **644** (one shown in FIG. **32**) that extend inwardly from the outer circumferential surface of the base **802**. Each notch being dimensioned to receive a separate one of the fingers **88** integral with the base unit retention arms **86**.

Referring back to the embodiment of FIGS. **48** and **49**, the base **802** has a through hole **808** that extends top-to-bottom through the base **802**. The hole **808** is centered along the top to bottom longitudinal axis of the base **802**. The hole **808** is occupied by a spindle **810**. The spindle **810** forms part of a drive assembly of the cleaning head **800**. The spindle **810** is rotatably supported in a bearing **812** mounted to the base **802**. In the alternative embodiments in which the cleaning head **800** is operated by the base unit **52**, the spindle **810** is not present and the hole **808** is dimensioned to allow the base unit spindle head **108** to freely move therein and engage the cleaning head **800**.

Referring to FIG. **48**, the shell **804** includes an outer basket **814** and an inner basket **816** spaced from the outer basket **814** to define a gap **818** therebetween. In the embodiment shown, the outer **814** and inner **816** baskets are fixed to one another and to the spindle **810** such that rotation of the spindle **810** results in rotation of both of the baskets **814**, **816**. Outer basket **814** has a bottom **820** with an annular cavity **822** defined therein for receiving a circular plate **824** of the spindle **810**. The circular plate **824** is secured to the outer basket **814** with fasteners (not shown). The base **802** defines a pocket **807** for receiving the bottom **820** of the outer basket **814**. A pair of washers **821** with roller bearings (not shown) disposed therebetween are seated in the pocket **807** between the base **802** and the outer basket **814** to rotatably support the outer basket **814** for rotation relative to the base **802**. Shields **805** protect users from the rotating shell **804**.

The inner basket **816** has a bottom plate **826**. A tube-shaped sleeve **828** extending upwardly from plate **826**. Sleeve **828** has an open end **830** through which the bone to be cleaned is deposited in the basket **816**. Sleeve **828** includes a plurality of openings **832**. Each opening **832** is defined by a raised, inwardly directed scallop (not illustrated). Each scallop, which is similar to mill element scallop **336**, has a sharp edge, that defines the adjacent opening **832**. Thus, the opening **832**-defining scallops give the sleeve **828** a shape similar to that of a grater. Here, since the scallops are inwardly directed, towards the longitudinal axis of the basket, the inner wall of the sleeve **828** is the

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grating surface of the inner basket **816**. The openings **832** are sized and configured to be small enough to prevent the bone from falling therethrough, but large enough to grasp or at least partially capture and temporarily hold the bone and tumble the bone about an inner surface **834** of the inner basket **816**. The plate **826** may include similar openings in alternative embodiments, but is solid in the embodiment shown.

A cleaning element **840**, in the form of a brush **840**, is rotatably supported in the void space **806** for engaging the bone tumbling against the inner surface **834** of the inner basket **816**. The brush **840** has a first end **842** disposed outside of the void space **806** and a second end **844** disposed inside the void space **806**. The brush **840** includes an elongated cylindrically shaped substrate **846** formed of metal or a sterilizable plastic such as a glass-filled nylon. Bristles **848** are attached to the substrate **846** by an adhesive **847** such as an epoxy adhesive. In manufacture, adhesive **847** is initially applied over an outer cylindrical face of substrate **846**. Before the adhesive cures, the bristles **848** are planted in the adhesive **847**. The bristles **848** are preferably made of metal such as stainless steel. The bristles **848** grab and tear soft tissue from the bone while the bone is frictionally grabbed by the inner basket **816** via the openings **832**. Soft tissue that is removed by the openings **832** is captured in the gap **818** between the outer **814** and inner **816** baskets. In FIG. **48** the brush **840** is shown generally centrally disposed in the void space **806** for rotating about a central axis. In other embodiments, the brush **840** is disposed about an axis offset from the central axis of the void space **806** and the brush **840** can likewise be disposed at an acute angle to the shell **804**.

As previously discussed, the hole **808** extending through the base **802** is occupied by the spindle **810**. The spindle **810** extends from the circular plate **824** to an end **850** having features (such as a square shaft configuration) for mating with a drive system such as a drive motor (shown but not numbered). In the embodiment shown, the spindle **810** is rotatably fixed to both the outer **814** and inner **816** baskets to transmit power from the drive system to the baskets **814**, **816** thereby causing the baskets **814**, **816** to rotate when the drive system is operational. Alternatively, the circular plate **824** could be configured to engage and be rotatably fixed to the base unit spindle head **108** via the alignment pin **110** and teeth **112**. As a result, the cleaning head **800** could be operated by the base unit **52**.

In the embodiment shown, a separate drive system (shown but not numbered) engages the first end **842** of the brush **840** to rotate the brush **840** in the void space **806**. The separate drive system rotates the brush **840** in a direction opposite the direction of rotation of the baskets **814**, **816**. Alternatively, the brush **840** could rotate in the same direction as the baskets **814**, **816**, either at the same rotational speed or at different rotational speeds. This separate drive system could be a stationary head fixed to the base **802** above the void space **806** with a motor disposed in the stationary head above the void space **806**.

In other embodiments the same drive system rotating the baskets **814**, **816** also rotates the brush **840**. Referring to FIG. **49A**, in such an embodiment, the spindle **810a** could be offset from the center of the shell **804a** and include a pinion gear **860** that engages teeth **862** in the annular cavity **822a** defined in the bottom wall **820a** to rotate the outer **814a** and inner **816a** baskets. A separate pinion gear **864** is attached to the first end **842a** of the brush **840a**. Thus, when the spindle **810a** rotates in a first direction, the outer **814a** and inner **816a** baskets also rotate in the first direction, while the brush

840a rotates in a second direction, opposite the first direction. Thus, the spindle **810a**, gear **860**, teeth **862**, and gear **864** would form a drive assembly of the cleaning head **800**. The gear ratio between the gears **860**, **864** and teeth **862** are selected such that the rotational speed of the brush **840a** is substantially faster than the rotational speed of the baskets **814a**, **816a**. In some cases the rotational speed of the brush is 5,000 to 10,000 RPM and the rotational speed of the baskets **814a**, **816a** is less than 1000 RPM, often less than 500 rpms, and possible 100 rpms or less. Thus, the ratio of rotational speeds of the brush **840**, **840a** to the baskets **814**, **814a**, **816**, **816a** is from about 5:1 to about 100:1.

During operation, uncleaned bone is first placed in the void space **806** for cleaning and the cap **807** is then placed to cover the void space **806**. The uncleaned bone includes soft tissue that requires removal prior to processing by the mill head **60**. The drive system or base unit **52** (if the cleaning head **800** is mounted to the base unit **52**) is then actuated to start rotation of the spindle **810**, **810a** and subsequent rotation of the shells **814**, **814a**, **816**, **816a**. If the brush **840a** is connected to the spindle **810a**, then rotation of the brush **840a** is also actuated. Alternatively, the brush drive system is actuated simultaneously to rotate the brush **840** in a direction preferably opposite to the baskets **814**, **814a**, **816**, **816a**. The inner basket **816**, **816a** and the openings **832** in the inner basket **816**, **816a** operate to grab and tumble the bone. The bristles **848** of the brush **840**, **840a** grip and tear away soft tissue attached to the bone. Once the cleaning head **800** has sufficiently removed soft tissue from the bone, the cap **807** is removed and the cleaned bone is grabbed by forceps or other device for further processing. The cleaning head **800** may then be cleaned or discarded. In some cases only the inner basket **816**, **816a** and the brush **840**, **840a** is discarded while the remaining components are sterilized and reused.

While not illustrated, it should be understood that RFID **270** and coil **271** (FIG. 5) may be embedded in the base **802** of cleaning head **800**.

XII. Fifth Alternative Cleaning Head

Referring to FIGS. 50 and 51, a fifth alternative cleaning head **900** is now described. The cleaning head **900** comprises a base **902**. A ring-shaped section **904** of the base **902** protrudes upwardly from a bottom section **905** of the base **902**. A cavity **906** is defined radially inwardly from the protruding ring-shaped section **904**. A shell **908** is mounted to the base **902** about the ring-shaped section **904** and above the cavity **906** using a plurality of fasteners **907**. The base **902** and shell **908** can be formed from the same material from which the shells **192** and **194** of head **56** are formed. Alternatively, if cleaning head **900** is a use-once unit, base **902** and shell **908** may be formed from a sterilizable plastic such as a polycarbonate plastic.

The bottom section **905** of the base **902** is shown in FIGS. 50 and 51 as being rectangular in shape without any features for engaging the base unit **52**. Accordingly, the cleaning head **900** may be a stand-alone unit for cleaning bone in which the base **902** is simply attached to a separate drive system (shown but not numbered). However, in other embodiments, the base **902** is attached to the base unit **52** for operation by the base unit **52**. In this case, the base **902** has an outer diameter that allows the base **902** to be slip fitted within the circular void space defined by the base unit lip **78**. Like the second alternative cleaning head **630**, the base **902** is circular in shape and four equiangularly spaced notches (not shown) extend upwardly from the bottom of the base

902 around the outer perimeter of the base **902**. The notches receive the pedestal teeth **84** when the cleaning head **900** is seated on the base unit **52**. A groove (not shown), like groove **642**, extends inwardly around the circumferential outer surface of the base **902**. The base **902** is further formed to have two diametrically opposed notches (not shown), like notches **644** (one shown in FIG. 32) that extend inwardly from the outer circumferential surface of the base **902**. Each notch being dimensioned to receive a separate one of the fingers **88** integral with the base unit retention arms **86**.

Referring to FIG. 51, it can be seen that base **902** has a through hole **909** that extends top-to-bottom through the base **902**. The hole **909** is centered along the top to bottom longitudinal axis of the base **902**. The hole **909** is occupied by a spindle **910**. The spindle is rotatably supported by bearings **912** mounted to the base **902**. The spindle **910** forms part of the drive assembly of the cleaning head **900**. In embodiments in which the cleaning head **900** is attached to the base unit **52**, the spindle **910** is not present and the hole **909** is dimensioned to allow the base unit spindle head **108** to freely move therein and engage the cleaning head **900**.

Cleaning head **900** includes a rotating grater disc **914** that functions as the cleaning element. Grater disc **914** is preferably formed of stainless steel. The grater **914** is disc shaped and has openings (not identified) around the center of the disc. The openings are positioned and shaped to engage the teeth integral with spindle plate **916**, (teeth not identified). The engagement of disc **914**, along with disc **918**, to the spindle **910** causes the discs to rotate in unison with the spindle. A reinforcing disc **918** is disposed between the circular plate **916** and the grater disc **914**. Grater disc **914** includes a plurality of openings **920** configured to file away soft tissue from bone. Reinforcing disc **918** includes larger openings **922** sized to allow filed off pieces of soft tissue to fall therethrough and out of the cleaning head **900** via a chute **924** in the base **902**. Both discs **914**, **918** are rotatably fixed to the spindle **910** to rotate with the spindle **910**.

Grater disc openings **920**, seen best in FIG. 51A, are arranged in arcuate groups on the disc **914** that are angularly spaced apart from each other. Thus, grater disc **914** has a number of arcuate sections that are free of openings **920**. This increases the mechanical strength of the grater disc **914**. The sections of the grater disc **914** in which openings **920** are formed extend over the sections of reinforcing disc **918** in which larger openings **922** are located. Thus, the grated off tissue falls through both grater disc openings **920** and reinforcing disc openings **922**. The dimension(s) of the openings **920** are such that the soft tissue and debris present on the uncleaned bone is filed away and removed to clean the bone, but the bone itself is not damaged or diminished beyond a usable state. In other words, the openings **920** are sized and configured not to result in milling the bone like the mill element **62** of the mill head **60**. In one embodiment, the openings **920** are rectangular in shape with a larger length than width (see FIG. 51A). In a more specific embodiment, the openings **920** are 0.39 cm (0.1 inches) or less long by 1.3 cm (0.5 inches) or less wide. Adjacent openings **920** are spaced in parallel columns and rows with approximately 0.2 cm (0.08 inches) or less between rows and 0.1 cm (0.04 inches) or less between columns.

While not identified, it is seen from FIG. 51A that grater disc **914** is formed with a number of arcuately spaced apart and radially outwardly extending tabs. During the assembly of disc **914**, grater disc **914** is placed over reinforcing disc **918**. The grater disc tabs are bent over and against the

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exposed face of the underlying reinforcing disc **918**. The tabs hold discs **914** and **918** together as a unitary assembly.

An opening **926** is defined through the shell **908** above the grater disc **914**. A plunger housing **930** is mounted to the shell **908** around the opening **926** using a plurality of fasteners **907a**. Plunger housing **930** has a chute **932** that defines a plunger passage **934**. One end of chute **932** extends into and around the outer perimeter of shell opening **930**. A flange **936** extends outwardly from the chute above the end of the chute disposed against shell **908**. The flange **936** is mounted to the shell **908**. Chute **932** defines a void space **938** above shell opening **926** for receiving the bone stock to be cleaned. A cap **915** covers the plunger passage **934**.

A plunger **940**, seated in chute **932** presses the bone against the grater disc **914** to facilitate removal of soft tissue from the bone. The plunger **940** includes a plunger head **942** that is sized with an outer perimeter slightly smaller than the inner perimeter of the plunger passage **934** to slidably fit in the plunger passage **934**. A shaft **944** has a first end **946** fixed to the plunger head **942**. A handle **948** is fixed to a second end **950** of the shaft **944**. The shaft **944** extends from the plunger head **942** in the passage **934** to handle **948**. Shaft **944** extends through and is able to slide within cap **915**.

A spring **952** is disposed about the shaft **944** between the cap **915** and the plunger head **942**. Spring **952** urges the plunger head **942** downwardly toward the bone to press the bone against the grater disc **914**. The plunger **940** is either manually or automatically operated.

As previously discussed, the hole **909** extending through the base **902** is occupied by the spindle **910**. The spindle **910** extends from the circular plate **916** to an end **960** having features (such as a square shaft configuration) for mating with a drive system such as a drive motor (shown but not numbered). In the embodiment shown, the spindle **910** is rotatably fixed to both the discs **914**, **918** to transmit power from the drive system to the discs **914**, **918** thereby causing the discs **914**, **918** to rotate when the drive system is operational. Alternatively, in the embodiments in which the cleaning head **900** is mounted to the base unit **52**, the spindle **910** is not present and the discs **914**, **918** engage and are rotatably fixed to the base unit spindle head **108** via the alignment pin **110** and teeth **112**.

During operation, uncleaned bone is first placed in the void space **938** for cleaning and the cap **915** is then placed to cover the void space **938**. The uncleaned bone includes soft tissue attached thereto that requires removal prior to processing by the mill head **60**. The drive system or base unit **52** (if the cleaning head **900** is mounted to the base unit **52**) is then actuated to start rotation of the spindle **910** and subsequent rotation of the discs **914**, **918**. Spring **952** presses the plunger **936** downwardly so that the plunger head **942** presses the bone against the grater disc **914**. The grater disc **914** relies on the scallop edges that define the openings **920** to cut away the soft tissue from the bone.

During operation of the mill head **900**, the user periodically pulls on handle **948** in order to overcome the force of the spring **952** that holds the plunger head **952** against the bone. The bone stock then rotates with the discs **914** and **918**. Almost as soon as the bone stock rotates, the bone presses against the interior wall of the shell **908** that defines opening **926**. This abutment of the bone stock against the shell and the rotation of the underlying grating disc **914** causes the bone to tumble in the opening **926**. When manual force on the plunger **940** is released, spring **942** causes the plunger head **942** to return to its position against the bone.

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Since the bone has tumbled, at this time a different surface of the bone should be pressed against the grating disc for cleaning.

Once the cleaning head **900** has sufficiently removed soft tissue from the bone, the cap **915** is removed and the cleaned bone is grabbed by forceps or other device for further processing. The cleaning head **900** may then be cleaned or discarded. In some cases only the discs **914**, **918** and the plunger **940** is discarded while the remaining components are sterilized and reused.

Again, RFID **270** and coil **271** (FIG. 5) may be embedded in the cleaning head base **902** to perform their previously defined functions.

Also, alternative structures may be employed to sequentially and repetitively press the bone against the grating disc **914** and tumble the bone so that each surface is pressed against the disc. For example, in some versions of the invention, spring **946** may be eliminated. In these versions of the invention, when the cleaning head **900** is actuated, the technician performing the cleaning operation will repetitively depress and retract the plunger **940**. The depression of the plunger causes the plunger head **942** to press the bone against the grating disc **914**. The retraction of the plunger **940** allows the bone to tumble. A mechanical device, such as a cam assembly may be used to lift and depress the plunger **940**. This mechanical device may be used in versions of the invention that both include and do not include the spring **946** that acts against the plunger **940**.

In versions of the invention in which manual force is used to press the bone against the cleaning disc, there may be a mechanism to limit the amount of this force. This may be desirable to minimize the likelihood that, due to the bone being pressed against the disc **914**, the bone is inadvertently milled. One such force-limiting component may be a spring connected at one end to the cap and at the second end to the plunger head. This spring is sized such that, instead of forcing the plunger head against the disc, it holds the head above the disc. Manual force is then required to overcome the spring force to push the plunger head downwardly in order to press the bone against the disc. The spring thus attenuates the manual force the technician is able to apply to the plunger head during the cleaning process.

In still another version of the invention, the grating disc **940** may be formed to have one or ribs. Each rib projects upwardly from the surface of the disc against which the bone stock is depressed. As the disc rotates, the rib/ribs rotate under the bone stock. The movement of a rib under the bone stock serves to force the bone to at least partially rotate, tumble, within the shell opening **926**. Again, this version of the invention may or not include the spring **946** that acts against the plunger **940**. Should the spring be present, the spring is selected so that the spring force exerted can be overcome by the action of a disc rib pushing up against the bone.

Alternatively, the grater disc may be formed with one or more flexible tabs. A tab may, for example, be formed in an arcuate section of the disc that is free from of openings. The surface of the tab angles upwardly from the opening defining section of the disc. As the disc rotates, the bone rides over the inclined surface of the tab. The flexible nature of the tab allows the tab and bone to at least partially overcome any force holding the bone against the grating disc. As the disc continues to rotate, the bone drops of the elevated edge of the tab. This dropping off, rolling off, of the bone serves to

rotate the bone so as to present another surface of the bone against the grater disc for cleaning.

XIII. Sixth Alternative Cleaning Head

FIGS. 52 and 53 depict a sixth alternative cleaning head 1000. Cleaning head 1000 is similar to the cleaning head 900 except that the plunger 940 is replaced with an impingement plate 1052. Impingement plate 1052 compresses the bone against grater openings. The cleaning head 1000 comprises a base 1002. A ring-shaped section 1004 of the base 1002 protrudes upwardly from a bottom section 1005 of the base 1002. A cavity 1006 is defined radially inwardly from the protruding ring-shaped section 1004. A shell 1008 is mounted to the base 1002 on top of the ring-shaped section 1004 and above the cavity 1006. The base 1002 and shell 1008 can be formed from the same material from which the shells 192 and 194 of head 56 are formed. Alternatively, if cleaning head 1000 is a use-once unit, base 1002 and shell 1008 may be formed from a sterilizable plastic such as a polycarbonate plastic.

The bottom section 1005 of the base 1002 is similar to that of base 902 bottom section 905 described with respect to FIGS. 50 and 51. Accordingly, the cleaning head 1000 may be a stand-alone unit for cleaning bone in which the base 1002 is simply attached to a separate drive system (shown but not numbered). Alternatively, base 1002 may also be configured for use with the base unit 52 so that the cleaning head 1000 is operated by the base unit 52. In this embodiment, the base 1002 has an outer diameter that allows the base 1002 to be slip fitted within the circular void space defined by the base unit lip 78. Like the second alternative cleaning head 630, the base 1002 is circular in shape and four equiangularly spaced notches (not shown) extend upwardly from the bottom of the base 1002 around the outer perimeter of the base 1002. The notches receive the pedestal teeth 84 when the cleaning head 1000 is seated on the base unit 52. A groove (not shown), like groove 642, extends inwardly around the circumferential outer surface of the base 1002. The base 1002 is formed to have two diametrically opposed notches (not shown), like notches 644 (one shown in FIG. 32) that extend inwardly from the outer circumferential surface of the base 1002. Each notch being dimensioned to receive a separate one of the fingers 88 integral with the base unit retention arms 86.

Referring back to the FIGS. 52 and 53, base 1002 has a through hole (not shown, but same as the hole 909 shown in FIG. 51) that extends top-to-bottom through the base 1002. The hole is centered along the top to bottom longitudinal axis of the base 1002. The hole is occupied by a spindle (not shown, but same as the spindle 910 shown in FIG. 51). The spindle is rotatably supported in bearings (not shown, but same as the bearings 912 shown in FIG. 51) mounted to the base 1002. The spindle forms part of the drive assembly of the cleaning head 1000. In the embodiments in which the cleaning head 1000 is mounted to and operated by the base unit 52, the spindle is not present and the hole is dimensioned to allow the base unit spindle head 108 to freely move therein and engage the cleaning head 1000.

A cleaning element 1014 in the form of a rotating grater 1014 is disposed in the cavity 1006. The grater 1014 is preferably formed of stainless steel. The grater 1014 is disc shaped and has features for engaging a circular plate (not shown, but same as the circular plate 916 shown in FIG. 51) of the spindle to rotate with the spindle. An reinforcing disc 1018 is disposed between the circular plate and the grater disc 1014. The grater disc 1014 includes a plurality of

openings 1020. The perimeter of each opening 1120 is defined by a raised scallop (not identified). The edges of the scallops that define the openings file away soft tissue from bone. Reinforcing disc 1018 includes larger openings 1022 sized to allow filed off pieces of soft tissue to fall there-through and out of the cleaning head 1000 via a chute 1024 in the base 1002. Both discs 1014, 1018 are rotatably fixed to the spindle 1010 to rotate with the spindle 1010 during operation.

The openings 1020 are preferably confined to circumferentially spaced areas on the grater disc 1014 such that the entire grater disc 1014 is not formed with the openings 1020. This increases the strength of the grater disc 1014. These spaced areas coincide with the larger openings 1022 in the reinforcing disc 1018 to further facilitate movement of the filed off soft tissue through the discs 1014, 1018. The dimension(s) of the openings 1020 are such that the soft tissue and debris present on the uncleaned bone is filed away and removed to clean the bone, but the bone itself is not damaged or diminished beyond a usable state. In other words, the openings 1020 are sized and configured not to result in milling the bone like the mill element 62 of the mill head 60. In one embodiment, the openings 1020 are rectangular in shape with a larger length than width (see FIG. 51A). In a more specific embodiment, the openings 920 are 0.39 cm or less long by 1.3 cm or less wide. Adjacent openings 1020 are spaced in parallel columns and rows with approximately 0.2 cm or less between rows and 0.1 cm or less between columns.

An opening 1026 is defined through the shell 1008 above the grater disc 1014. A cap 1015 partially covers the opening 1026 to define a void space 1017 for the bone. The cap 1015 is mounted to the base via fasteners 1007 with the shell 1008 captured between the cap 1015 and the base 1002. The void space 1017 in this embodiment is defined radially inwardly of the shell 1008, below the cap 1015 and above the grater disc 1014. The cap 1015 further has an opening 1030 coinciding with a portion of the void space 1017.

An impingement mechanism is disposed in the opening 1030 above the grater disc 1014. The impingement mechanism includes a plate 1052 that presses the bone against the grater disc 1014. Plate 1052 is flexible. The impingement mechanism includes a block 1045 that is mounted to a bracket 1044 by fasteners 1043. Bracket 1044 is fixed to the base 1002 by fasteners 1046. Block 1045 is formed with an elongated slot 1048 for receiving a shaft 1050 to which plate 1052 is mounted. One end of shaft 1050 is seated in block slot 1048. Set screws 1049 hold the shaft 1050 in a fixed rotational position in slot 1048. The impingement plate 1052 is fixed fitted over the second end of the shaft 1050. Set screws 1049 allows the angular relationship between the impingement plate 1052 and the grater disc 1014 to be altered for different applications and different sizes of bone. A secondary cap 1019 extends over block 1045. The secondary cap 1019 is formed with a slot 1021 sized to receive a section of the impingement plate 1052.

As previously discussed, the hole extending through the base 1002 is occupied by the spindle. The spindle extends from the circular plate to an end (not shown, but same as the end 960 in FIG. 51) having features (such as a square shaft configuration) for mating with a drive system such as a drive motor (shown but not numbered). In the embodiment shown, the spindle is rotatably fixed to both the discs 1014, 1018 to transmit power from the drive system to the discs 1014, 1018 thereby causing the discs 1014, 1018 to rotate when the drive system is operational. Alternatively, in the embodiments in which the cleaning head 1000 is operated by the

base unit **52**, the discs **1014**, **1018** are configured to engage and be rotatably fixed to the base unit spindle head **108** via the alignment pin **110** and teeth **112**.

To clean bone stock, the bone is initially placed in the void space **1017**. Cap **1015** is then placed to cover the void space **1017**. The drive system or base unit **52** (if the cleaning head **1000** is mounted to the base unit **52**) is then actuated to start rotation of the spindle and subsequent rotation of the discs **1014**, **1018**. The bone rotates with the grater until the bone rotates below the impingement plate **1052**, the surface of the impingement plate that is directed towards the disc **1014**. Initially, the impingement plate holds the bone against the grater disc **1014**. This holding action presses the bone against the cleaning disc so the disc removes the soft tissue from the bone. However, the impingement plate is flexible. As a result, the disc is able to push the bone below and across the impingement plate **1052**. It should be appreciated that during this transit of the bone, the impingement plate flexes. As the bone travels under the impingement plate, the bone is momentarily caught by the bottom edge of the impingement plate **1052**. As the impingement plate **1052** snaps back to the static state, the plate rotates the bone over the grater disc. This presents a different surface of the bone is disposed against the disc. Once the cleaning head **1000** has sufficiently removed soft tissue from the bone, the cap **1015** is removed and the cleaned bone is grabbed by forceps or other device for further processing. The cleaning head **1000** may then be cleaned or discarded. In some cases only the discs **1014**, **1018** are discarded while the remaining components are sterilized and reused.

XIV. Seventh Alternative Cleaning Head

A seventh alternative cleaning head **1100** is now described by reference to FIGS. **54** through **56**. Cleaning head **1100** includes a base **1102**. A ring-shaped section **1104** of the base **1102** protrudes upwardly from a bottom section **1105** of the base **1102**. A cavity **1106** is defined radially inwardly from the protruding ring-shaped section **1004**. A shell **1108** is mounted to the base **1102** on top of the ring-shaped section **1104** and above the cavity **1106**. The base **1102** and shell **1108** can be formed from the same material from which the shells **192** and **194** of head **56** are formed. Alternatively, if cleaning head **1100** is a use-once unit, base **1102** and shell **1108** may be formed from a sterilizable plastic such as a polycarbonate plastic.

The bottom section **1105** of the base **1102** is shown in FIG. **54** as being rectangular in shape without any features for engaging the base unit **52**. Accordingly, the cleaning head **1100** may be a stand-alone unit for cleaning bone in which the base **1102** is simply attached to a separate drive system (shown but not numbered). However, the base **1102** may be configured to be seated in the base unit **52** and operated by the base unit **52** in other embodiments. In these embodiments, the base **1102** has an outer diameter that allows the base **1102** to be slip fitted within the circular void space defined by the base unit lip **78**. In this case, like the second alternative cleaning head **630**, the base **1102** is circular in shape and four equiangularly spaced notches (not shown) extend upwardly from the bottom of the base **1102** around the outer perimeter of the base **1102**. The notches receive the pedestal teeth **84** when the cleaning head **1100** is seated on the base unit **52**. A groove (not shown), like groove **642**, extends inwardly around the circumferential outer surface of the base **1102**. The base **1102** is further formed with two diametrically opposed notches (not shown), like notches **644** (one shown in FIG. **32**) that extend inwardly

from the outer circumferential surface of the base **1102**. Each notch being dimensioned to receive a separate one of the fingers **88** integral with the base unit retention arms **86**.

Returning to FIGS. **54-56**, base **1102** has a through hole **1109** that extends top-to-bottom through the base **1102**. The hole **1109** is centered along the top to bottom longitudinal axis of the base **1102**. The hole **1109** is occupied by a spindle **1110**. The spindle **1110** is rotatably supported in bearings **1112** mounted to the base **1102**. The spindle **1110** forms part of the drive assembly of the cleaning head **1100**. In the embodiments in which the cleaning head **1100** is mounted to the base unit **52**, the hole **1109** is dimensioned to allow the base unit spindle head **108** to freely move therein and engage the cleaning head **1100**.

A cleaning element, in the form of a rotating grater **1114** is disposed in the cavity **1106**. Grater **1114** is preferably formed from stainless steel. The grater **1114** is disc shaped and has features for engaging a circular plate **1116** of the spindle **1110** to rotate with the spindle **1110**. A reinforcing disc **1119** may be disposed between the circular plate **1116** and the grater disc **1114**. The grater disc **1114** includes a plurality of openings **1120** configured to file away soft tissue from bone. The filed off tissue thereafter falls through the openings **1120** and out of the cleaning head **1100** through an opening **1124** in the base **1102**. The disc **1114** is rotatably fixed to the spindle **1110** to rotate with the spindle **1110** during operation.

The openings **1120** are preferably confined to circumferentially spaced areas on the grater disc **1114** such that the entire grater disc **1114** is not formed with the openings **1120**. This increases the strength of the grater disc **1114**. The dimension(s) of the openings **1120** are such that the soft tissue and debris present on the uncleaned bone is filed away and removed to clean the bone, but the bone itself is not damaged or diminished beyond a usable state. In other words, the openings **1120** are sized and configured not to result in milling the bone like the mill element **62** of the mill head **60**. In one embodiment, the openings **1120** are rectangular in shape with a larger length than width (see FIG. **51A**). In a more specific embodiment, the openings **1120** are 0.39 cm or less long by 1.3 cm or less wide. Adjacent openings **1120** are spaced in parallel columns and rows with approximately 0.2 cm or less between rows and 0.1 cm or less between columns.

An opening **1126** is defined through the shell **1008** above the grater disc **1114**. A cap **1115** partially covers the opening **1126** to define a void space **1117** for the bone. The cap **1115** is mounted to the base via fasteners **1107**. The shell **1108** is captured between the cap **1115** and the base **1102**. The void space **1117** in this embodiment is defined radially inwardly of the shell **1108**, below the cap **1115** and above the grater disc **1114**. The cap **1115** further has an opening **1130** coinciding with a portion of the void space **1117**.

A plunger **1140** is moveably mounted to cleaning head **1100**. The plunger **1140** has a handle **1142** to which, at one end, a head **1144** is attached. The plunger head **1144** is sized slightly smaller and has a shape similar to that of opening **1130** to fit through the opening with minimal clearance. The plunger head **1144** is manually (shown) or mechanically (not shown) reciprocated in the opening **1130** to press bone against the grater disc **1114**. The plunger **1140** is preferably formed of stainless steel or sterilizable plastic.

A second cleaning element, a fluted screw **1150** is disposed in the void space **1117**. A shaving block **1152** rotatably supports the fluted screw **1150**. The shaving block **1152** is located through another opening **1154** in the cap **1115**. The shaving block **1152** is generally centrally located in the void

space 1117 but in other embodiments may be located about the periphery of the void space 1117 similar to the third alternative cleaning head 700. Additional fluted screws 1150 may also be employed. In the embodiment shown, the fluted screw 1150 comprises a sleeve 1160 defining a through bore 1162. The sleeve 1160 is preferably fixed to rotate with the spindle 1110. The sleeve 1160 defines a plurality of flutes 1164 that facilitate gripping and tearing of soft tissue from the bone in the void space 1117 during rotation to clean the bone. The fluted screw 1150 is preferably formed of stainless steel.

Referring to FIG. 56, the shaving block 1152 is preferably formed of stainless steel. The shaving block 1152 has a flange 1170 mounted to the cap 1115 about the opening 1154. A wall 1172 of the shaving block 1152 extends through the opening 1154 and is suspended slightly above the grater disc 1114 so as not to disrupt rotation of the grater disc 1114. The spacing between a bottom of the wall 1172 and the grater disc 1114 is too small for any bone to pass thereunder. An elongated space 1174 is defined in the wall 1172 to receive the fluted screw 1150. The fluted screw 1150 rotates in the elongated space 1174 in the wall 1172. A first end of the fluted screw 1150 is rotatably mounted to the shaving block 1152 while the opposite end of the fluted screw 1150 is fixed to the spindle 1110. A portion of the fluted screw 1150 extends a distance away from a front surface 1176 of the wall 1172 to grip soft tissue attached to bone disposed in the void space 1117.

Shaving block 1152 is shaped to define a pair of cutting edges 1178 disposed on opposing sides of the elongated space 1174 (see FIG. 56). These cutting edges 1178 are sharp enough to cut soft tissue that the fluted screw 1150 has gripped or captured during rotation. In other words, the fluted screw 1150 engages the soft tissue still attached to bone and impinges that soft tissue against the cutting edges 1178 during rotation to cut the soft tissue away from the bone. Plunger head 1144 is designed to force the bone toward the fluted screw 1150. As a result of the abutment of the bone against the screw 1150, soft tissue attached to the bone is pressed into the spaces between the flutes. As the screw 1150 rotates, the tissue entrained in the screw 1150 is rotated against the cutting edges 1178. The continued rotation of the screw with the entrained tissue results in the cutting edges separating the tissue from the bone. The sharp edges of the flutes of screw 1150 also separately cut some soft tissue from the bone although to a limited extent.

The plunger head 1144 is generally box-shaped to correspond to the box-shaped opening 1130. The shell 1108 has a flat 1180 on an inner surface thereof that corresponds to a first flat side 1184 of the plunger head 1144. Accordingly, when the plunger head 1144 is disposed in the opening 1130, the first flat side 1184 of the plunger head 1144 faces the flat 1180 on the inner surface 1182 of the shell 1108 with a small gap defined therebetween (see FIG. 55). The small gap is sized to prevent bone from entering. Opposite the flat 1180 on the other side of the plunger head 1144 is the fluted screw 1150 and shaving block 1152. A gap between a second flat side 1186 of the plunger head 1144 and the fluted screw 1150 is sized to restrictively receive small pieces of bone. As a result, all of the bone must pass under the plunger head 1144 so as to be compressed by the plunger head 1144 against the grater disc 1114 to remove soft tissue from the bone or pass between the plunger head 1144 and the fluted screw 1150 to be grabbed by the fluted screw 1150 with the associated soft tissue being grabbed by the fluted screw 1150 and cut by the cutting edges 1178.

As previously discussed, the hole 1109 extending through the base 1102 is occupied by the spindle 1110. The spindle 1110 extends from the circular plate 1116 to an end 1190 having features (such as a square shaft configuration) for mating with a drive system such as a drive motor (shown but not numbered). In the embodiment shown, the spindle 1110 is rotatably fixed to both the grater disc 1114 and the fluted screw 1150 to transmit power from the drive system to the grater disc 1114 and the fluted screw 1150 thereby causing the grater disc 1114 and the fluted screw 1150 to rotate when the drive system is operational. Alternatively, when the cleaning head 1100 is operated by the base unit 52, the grater disc 1114 and the fluted screw 1150 are configured to engage and be rotatably fixed to the base unit spindle head 108 via the alignment pin 110 and teeth 112.

During operation, uncleaned bone is first placed in the void space 1117 for cleaning and the cap 1115 is then placed to cover the void space 1117. The uncleaned bone includes soft tissue attached thereto that requires removal prior to processing by the mill head 60. The drive system or base unit 52 (if the cleaning head 1100 is mounted to the base unit 52) is then actuated to start rotation of the spindle 1110 and subsequent rotation of the grater disc 1114 and the fluted screw 1150. The grater disc 1114 relies on the scallop edges that define openings 1120 to cut away the soft tissue from the bone. The grater disc 1114 and the plunger head 1144 operate to tumble the bone. The plunger head 1144 presses the bone against the grater disc 1114 to enhance the cutting away of soft tissue from the bone by the scalloped openings 1120. The fluted screw 1150 grips soft tissue attached to the bone and cuts the soft tissue away from the bone either by the nature of the flutes 1164 on the fluted screw 1150 or by impinging the soft tissue against the cutting edges 1178 of the shaving block 1152. Once the cleaning head 1100 has sufficiently removed soft tissue from the bone, the cap 1115 is removed and the cleaned bone is grabbed by forceps or other device for further processing. The cleaning head 1100 may then be cleaned or discarded. In some cases only the grater disc 1114 and plunger 1140 are discarded while the remaining components are sterilized and reused.

XV. Alternative Embodiments

The foregoing has been directed to specific versions of system of this invention. Other versions of the system of this invention may have features different from what has been described.

For example, various features of the versions of this invention may be combined. Thus, in some versions of the invention wherein there is a single head with both cleaning and milling modules, the cleaning module may have a drive assembly that rotates both the lower and upper brushes.

Likewise the features of the invention may be different from what has been described. In some versions of the invention, the cleaning head cleaning elements and mill head mill elements may have common coupling features for engaging to the common drive spindle but different coupling features for holding the cleaning head and mill head to the base unit.

Similarly the cleaning head and mill head may have common coupling features for holding the head to the base unit. These versions of the invention may then have different coupling features for coupling to different drive members integral with the base unit. Thus, instead of having a single drive spindle, the base unit may have separate drive spindles that are driven at different speeds, a first speed for the cleaning head 56 and at a second speed for the mill head 60.

A common gear assembly connects both of these spindles to the output shaft of motor **54** (shaft not illustrated). In these versions of the invention, the complementary coupling features the cleaning element and mill element are provided may be different from each other.

Similarly, there is no requirement that in all versions of the invention the cleaning elements and mill elements be disc shaped members that rotate around their center axis. In some versions of the invention, for example, one or both of the cleaning elements or mill elements may be tube shaped. A brush so shaped may have a bristles that extend inwardly from the body of the brush. This brush is used by placing the bone stock to be cleaned inside the brush. A mill element so shaped may have cutting edges that emerge from the outer face of the body. The bone is pressed against this surface. The formed chips fall into a catch tray located within the center of the mill element. The above described brush and mill element are therefore designed to be rotated around the axis that extends through its central lumen.

Both the cleaning head and mill head of this invention may have features in addition to what has been described. For example, the cap associated with mill head **56** may be fitted to the post **286** integral with brush **59**. This allows the medical personnel to by pressing down on the cap, press down on the brush **59**. The pressing down on the brush thus increases the force of the bristles against the bone located between the brushes **58** and **59**.

The integrated cleaning and mill head of this invention may have features different from the described head **490**. For example in some versions of the device, the cleaning module may be statically mounted to the other components. In these versions of the invention, the movement of a trap establishes a path through the cleaned bone can pass into the milling module. Also, in some versions of the invention, a mechanical member may physically displace the cleaned bone so as to effect the transfer of the cleaned bone to the milling module. In some versions of this embodiment of the invention, the trap (or member) that allows (performs) the transfer of the cleaned bone from the cleaning module to the milling module may be automatically actuated. This would further reduce the amount of time the surgical personnel need to devote to performing and/or monitoring the bone cleaning and bone milling processes.

Also, in some versions of the invention, the brush bristles may not always be of constant height. In some versions of the invention it may be desirable to construct the lower brush so that the bristles adjacent the center of rotation of the brush are of lower height than the bristles spaced from the center of rotation. In these versions of the invention, it may also be desirable to construct the upper brush so its bristles close to the center of rotation are longer than the bristles spaced from this axis. It is believed that an advantage of provide brushes having these features is that the bristle arrangement reduces the extent to which centrifugal force causes the bone stock being cleaned to move to the outer perimeter of the brushes. By maintaining the bone stock adjacent the center of the brushes, the likelihood that the bone stock becomes trapped between the outer perimeter of the brushes and surrounding surfaces of the cleaning head housing is significantly reduced. Should the bone stock become so trapped, the effectiveness of the cleaning process can be adversely affected.

Likewise it should be understood that in versions of the invention with integrated cleaning and milling modules, these modules may not be removable from the base as a single piece unit. In some versions of the invention the cleaning module and milling modules may be separately

removable. After use, each module is independent sterilized, the worn parts replaced, and reattached to the base unit.

It should likewise be appreciated that versions of the system of the invention can include less than all the described components. Not all versions of the invention may include memories that describe the specific speeds at which the cleaning elements and mill elements should be driven. This is especially true for versions of the invention wherein the cleaning elements and mill elements are drive at the same speed. This may also be true for versions of the invention wherein the base unit has two drive spindles that are geared to operate at different speeds.

Some versions of the invention may not include a removable catch tray for holding the milled bone. Some cleaning heads/modules of this invention may only have a single cleaning element. In some versions of the invention, the brushes may not have bristles. Instead, each brush has an abrasive surface that when is rubbed against the bone stock, cleans the bone stock.

Further in some versions of the invention the cleaning head drive assembly that simultaneously rotates brushes **416** and **418** may rotate the brushes in the same direction. Drive assembly may also be provided that rotate the brushes or other cleaning elements at different speeds.

Cleaning head **630** may, in some versions of the invention be provided with a removable retaining ring. This ring has a rim that extends over the cap rim **684**. The ring has a skirt with features that facilitate the removable coupling of the ring to the base **632**. The ring is coupled to the rest of the head during the cleaning process in order to prevent the inadvertent lifting of the cap away from the underlying base **632** and brush **58**.

Alternatively, cleaning head **630** may be constructed so that the void space in which brush **58** is seated has a depth greater than the height of the brush.

In some versions of the invention in which the cleaning head is provided with a cap that is flexed against the brush, the cap may have a hand hold. The hand hold may include a cylindrical neck that extends upwardly from the apex of the cap dome. A head, also cylindrical in shape, disposed above the neck and that extends radially outwardly beyond the neck is also part of this hand hold. During the cleaning process, the individual charged with the cleaning grasps the hand hold to push down on the cap and also move cap so the top of the cap is turned from side to side. This turning of the cap changes the orientation of the bristles that extend downwardly from the cap. This changing of the orientation of the bristles can, in some circumstances, improve the efficiency of the cleaning process.

The cleaning head and mill head of this invention may be provided with features other than the disclosed spout **668** and sleeve **308** for facilitating the coupling of these two heads together as part of the bone transfer process. For example in some versions of the invention, the cleaning head may have a spout that is dimensioned to seat in the mill head feed port into which the cleaned bone stock is introduced into mill head.

Likewise it should be understood that while this invention is intended for use to clean autograft bone, its applications are not so limited. System **50** of this invention may also be used to clean and mill donor bone, sometimes referred to as allograft bone.

The materials from which the components of this invention are fabricated may be different from what has been described. For example, in some versions of the invention, the enter cleaning head, including the housing-forming shell components may be disposable. In these versions of the

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invention, the components forming the housing, instead of being made of metal may be made of sterilizable plastic. Likewise, there is no requirement that the brushes always include bristles formed from stainless steel. The bristles may be formed from other bendable metals not prone to breakage such as titanium or alloys of titanium.

The geometry of the components may also vary. For example, in the versions of the invention wherein the cleaning element is disc, the scallops with edges that performing the cleaning function are illustrated as being in sets of arcuately spaced apart clusters. In alternative versions of the invention, the scallops with edges, and complementary openings are formed throughout the whole of the body of the disc. There are no scallop and opening free sections of the disc.

It should further be appreciated that the processing steps executed by the system may be different from what has been described. For example, the cleaning head and mill head memories are described as having flag bits that are set once the head is used. This is to prevent reuse of an unsterilized head for a procedure on a new patient. However, there may be times when, after an initial amount is cleaned and milled, the surgeon decides that it is necessary to have additional bone chips available. Accordingly, the software may have an override that allows the surgical personnel to, after acknowledging that cleaning head or mill head was used, to reuse the cleaning head. Thus allows the personnel to reuse the heads during the same procedure on a single patient.

Also, while the invention is described as a combined system for both cleaning and milling bone, other versions of the invention may not perform both functions. The base unit and cleaning head may form a system of this invention that is used just to clean bone. The advantage of the system being that it provides a mechanized means of cleaning the bone that substantially eliminates the need for surgical personnel to hold the bone. Also, depending on the system of this invention, it may be that the cleaning elements are the only disposable portion of the system. Depending on the materials forming components from which this system is fabricated, this may limit the expenses associated with providing the system.

Further it should be clear that various ones of the cleaning elements described above as well as other cleaning elements may be combined together in a single cleaning head.

Also there is no requirement that the system of this invention be constructed so that first the bone be cleaned and then milled. In some versions of the invention, the bone may initially be milled to form chips of substantially uniform size. These chips are then processed by a cleaning head/module. The cleaning head contains cleaning elements shaped especially to clean the chips of produced by the mill head/module.

Accordingly, it is an object of the appended claims to cover all such variations and modifications as come within the true spirit and scope of the invention.

What is claimed is:

1. A system for preparing bone stock for implantation into a patient including:

- a first head for removing soft tissue from bone;
- a second head for converting bone stock into bone chips; and
- a base unit including a motor, the base unit being operable to power the first head or the second head when attached thereto; and
- a controller configured to control the motor to operate the first head and the second head at a same motor speed.

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2. The system of claim 1, wherein said first head has a cleaning element moveably mounted therein, and said controller causes said motor to be actuated in order to appropriately actuate said cleaning element when said first head is attached to said base unit.

3. The system of claim 1, wherein said second head comprises a mill element moveably disposed therein, and said controller causes said motor to be actuated in order to appropriately actuate said mill element when said second head is attached to said base unit.

4. The system of claim 1 further including a control console comprising a power supply and a motor driver that are configured to generate command signals that indicate a speed at which said motor should run, based on user entered data and memory data that indicates characteristics of energization signals that are to be applied to said motor.

5. The system of claim 4, wherein said control console further comprises a display controller configured to generate:

data images for presentation on a display; and images of command buttons for presentation on said display that receive signals when an individual presses said command buttons to generate commands to cause user-requested operation of said motor.

6. The system of claim 5, wherein said control console is attached to said base unit and said control console is operable to periodically conduct a basic interrogation to determine if said first or second head previously attached to said base unit is still attached to said base unit.

7. The system of claim 1, wherein said first head has a cleaning element moveably mounted therein and said cleaning element includes at least one coupling feature shaped to engage a complementary coupling feature connected to a spindle operatively attached to said motor of said base unit when said first head is attached to said base unit so that actuation of said motor results in actuation of said cleaning element, and wherein:

said at least one coupling feature comprises plural openings in said cleaning element which are configured to receive plural teeth on said spindle; and

said complementary coupling feature comprises said plural teeth that are located equiangularly around an axis around which said spindle rotates;

wherein said plural openings receive said plural teeth when said first head is disposed on said base unit.

8. The system of claim 1, wherein said second head has a mill element moveably disposed therein and said mill element has at least one coupling feature shaped to engage with a complementary coupling feature connected on a spindle operatively attached to said motor of said base unit when said second head is attached to said base unit so that actuation of said motor results in actuation of said mill element, wherein:

said at least one coupling feature comprises plural openings in said mill element which are configured to receive plural teeth on said spindle; and

said complementary coupling feature comprises said plural teeth that are located equiangularly around an axis around which said spindle rotates;

wherein said plural openings receive said plural teeth when said second head is disposed on said base unit.

9. The system as set forth in claim 1, wherein a spindle is connected to said motor and rotates upon actuation of said motor and is able to move longitudinally with respect to said motor; and a biasing member is disposed between said motor and said spindle to urge a head of said spindle away from said motor.

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10. The system of claim 9, wherein said spindle further includes a pin and plural teeth, wherein said pin is centered around an axis around which said spindle rotates and projects above said plural teeth on said spindle.

11. A system for preparing bone stock for implantation into a patient including:

a first head for removing soft tissue from bone;
a second head for converting bone stock into bone chips;
and

a base unit including a motor, the base unit being operable to power the first head or the second head when attached thereto;

a controller configured to:
determine a type of head attached to the base unit; and
determine characteristic of energization currents that should be applied to the motor based on the determined type of head.

12. The system of claim 11, wherein said first head has a cleaning element moveably mounted therein, and said controller causes said motor to be actuated in order to appropriately actuate said cleaning element when said first head is attached to said base unit.

13. The system of claim 11, wherein said second head comprises a mill element moveably disposed therein, and said controller causes said motor to be actuated in order to appropriately actuate said mill element when said second head is attached to said base unit.

14. The system of claim 11 further including a control console comprising a power supply and a motor driver that are configured to generate command signals that indicate a speed at which said motor should run.

15. The system of claim 14, wherein said control console further comprises a display controller configured to generate:

data images for presentation on a display; and
images of command buttons for presentation on said display that receive signals when an individual presses said command buttons to generate commands to cause user-requested operation of said motor.

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16. The system of claim 15, wherein said control console is attached to said base unit and said control console is operable to periodically conduct a basic interrogation to determine if said first or second head previously attached to said base unit is still attached to said base unit.

17. A system for preparing bone stock for implantation into a patient including:

a first head for removing soft tissue from bone;
a second head for converting bone stock into bone chips;
and

a base unit including a motor, the base unit being operable to power the first head or the second head when attached thereto;

a console coupled to the base unit, the console including a display and a controller, the controller being configured to:

determine a type of head attached to the base unit; and
display data indicating the type of head attached to the base unit based on the determined type.

18. The system of claim 17 wherein said console comprising a power supply and a motor driver that are configured to generate command signals that indicate a speed at which said motor should run, based on user entered data and memory data that indicates characteristics of energization signals that are to be applied to said motor.

19. The system of claim 18, wherein said console further comprises a display controller configured to generate:

data images for presentation on a display; and
images of command buttons for presentation on said display that receive signals when an individual presses said command buttons to generate commands to cause user-requested operation of said motor.

20. The system of claim 19, wherein said console is operable to periodically conduct a basic interrogation to determine if said first or second head previously attached to said base unit is still attached to said base unit.

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